# 11 Periradicular Surgery

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Closure of the Surgical Site
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**Summary**

Although nonsurgical endodontic treatment is a highly predictable option in most cases, surgery may be indicated for teeth with persistent periradicular pathoses that have not responded to nonsurgical approaches. Surgical root canal therapy, including root-end resection, has been practiced since at least the mid-1800s.[211] In 1906, Schamberg[473] described using radiographs to assist diagnosis and the use of surgical burs to perform a rapid osteotomy and root-end "ablation."

Perhaps the single most important development in dental practice in the early 20th century was the introduction of safe, effective local anesthesia, which allowed for more meticulous and comfortable surgical treatment. Formal recognition of endodontics as a specialty in 1963 ushered in a new era of basic and clinical research focused on the prevention and treatment of pulpal and periradicular pathosis.

Since the 1990s, periradicular surgery has continued to evolve into a precise, biologically based adjunct to nonsurgical root canal therapy. The parallel development of new instruments and materials, along with a better understanding of the biology of wound healing, has made surgical treatment a viable alternative to extraction and tooth replacement, rather than a treatment of last resort.

Periradicular surgery, when indicated, should be considered an extension of nonsurgical treatment, because the underlying etiology of the disease process and the objectives of treatment are the same: prevention or elimination of apical periodontitis. Surgical root canal treatment should not be considered as somehow separate from nonsurgical treatment, although the instruments and techniques are obviously quite different. Surgical treatment accounts for about 3% to 10% of the typical endodontic specialty practice.[1,75,374] A web-based survey found that 91% of active endodontists perform some type of root-end surgery, and almost all are using a dental operating microscope and ultrasonics.[115] Endodontists perform almost 78% of surgical root canal treatments; general dentists and other specialists perform 15.5% and 6.6%, respectively.[242] Although appropriately trained general dentists and other dental specialists may perform periradicular surgery, individuals with advanced training in endodontics have developed most of the current periradicular surgery techniques and science presented in this chapter. We believe that endodontists must continue to include periradicular surgery as a routine part of clinical endodontic practice and not abrogate this treatment option to others, who may not possess the same background, skills, or values.[343,424,439] When patient preferences and quality of life measures are considered, patients can be expected to place high value on endodontic treatment and the retention of a natural tooth.[153,180]

# Indications for Periradicular Surgery

## ETIOLOGY OF PERSISTENT PERIRADICULAR DISEASE

The first and arguably most important step in treatment decision making is attempting to determine the cause of persistent periradicular disease. Treatment then is directed at eliminating the etiology, which most often is the presence of bacteria and other microbial irritants in the root canal space.[492] Nonsurgical retreatment, when possible, is often the first choice for attempting to correct obvious deficiencies in the previous treatment.[564] However, microorganisms can survive even in apparently well-treated teeth in dentinal tubules, canal irregularities, deltas, and isthmus areas.[275,307,568] If residual microorganisms remain completely entombed in the root canal system, periradicular healing should occur. Sealing off all potential routes of microbial escape from the root canal system is the goal of both nonsurgical and surgical treatment. When microorganisms of sufficient pathogenicity and number gain access to the periradicular tissues, pathosis develops.

*Enterococcus faecalis* is commonly isolated from failing root canal–treated teeth and is known to be especially difficult to eliminate with standard instrumentation and irrigation techniques.[510] Unlike primary endodontic infections, which predominantly are associated with mixed anaerobic microbiota, it has been commonly accepted that treatment failures are more frequently associated with one or two microorganisms.[492] However, research using more sophisticated techniques (16S ribosomal RNA gene clone library analysis) has determined that multiple, previously uncultivated phylotypes can be identified in the majority of teeth with persistent apical periodontitis.[464] Fungi and viruses have also emerged as potential causes of root canal failure and may play either a primary or secondary role in persistent periradicular pathosis.[369,399,462,575]

Established extraradicular colonies of microorganisms also may be a reason for failure of some teeth to respond to nonsurgical treatment. When microorganisms are able to arrange in an extraradicular biofilm, they may be particularly resistant to elimination by host defense mechanisms and antimicrobial agents.[490] Persistent extraradicular root surface colonization cannot be diagnosed by noninvasive methods but may be suspected in well-treated cases that are refractory to nonsurgical treatment.[166,218,465,495,511] Although the presence of extraradicular microbial colonies has been somewhat controversial, studies using DNA-DNA

hybridization techniques have confirmed the persistence of microorganisms in the periradicular tissues of some root canal–treated teeth.[187,509]

Overextended filling materials may contribute to treatment failure, presumably as a result of a chronic inflammatory response.[369] Although this is possible and even likely with certain toxic materials (e.g., pastes containing formaldehyde),[376] the role of relatively inert materials such as gutta-percha and set sealer is less clear, and these materials probably become a significant contributing factor only if microorganisms are present. If the root apex is close to the buccal cortical plate, apical fenestration may occur, leading to persistent symptoms, especially tenderness to palpation over the root apex.[72] Some have suggested that overextension of filling materials may contribute to endodontic failure because certain dental materials may induce periodontal ligament (PDL) cell apoptosis.[467] This specific interaction between filling materials and periradicular tissues is not fully understood and deserves further research. The poorer prognosis often reported with overextended root canal fillings also may be related simply to lack of an adequate apical seal and subsequent egress of microorganisms from the root canal space. Regardless, minor overextension of filling material is rarely a sole indication for surgery except when symptoms or periradicular pathosis develops. Significant overextension of filling material, especially when important anatomic areas and possibly toxic materials are involved, is an indication for referral to an endodontist or oral surgeon for evaluation and, possibly, treatment.

The presence of periradicular cholesterol crystals may interfere with healing after nonsurgical root canal treatment.[365] Although relatively uncommon, true periradicular cysts (completely enclosed, epithelium-lined cavities) may not be expected to resolve after nonsurgical treatment.[364] As with other extraradicular causes of failure, surgery is indicated, because definitive diagnosis and treatment require an excisional biopsy and removal of the periradicular tissue.

Vertical root fractures are a significant cause of failure and may be difficult to diagnose in the early stages.[90] Exploratory surgery often is required to confirm a root fracture. Cone-beam computed tomography (CBCT) is a promising tool for noninvasive diagnosis of root fractures[155,165] and is discussed in greater detail later in this chapter and in Chapter 2. Although several promising approaches to the management of vertical root fractures have been proposed,[226,268,506] the prognosis generally is believed to be poor. Extraction is usually the treatment of choice, especially if appropriate tooth replacement options are available. Root amputation or hemisection may be considered in a multirooted tooth if the remaining tooth structure is uninvolved and sufficient periodontal support is present.

The relationship between systemic disease and periradicular healing is not completely understood. The possible influence of certain systemic medications on wound healing is discussed later in the chapter. Compromised host healing capacity may be a contributing factor in delayed healing and failure of some root canal–treated teeth. For example, complete healing after nonsurgical root canal treatment is less likely in diabetic patients with preoperative periradicular pathosis.[78,175] Patients undergoing immunosuppressive therapy may be at greater risk for delayed healing, treatment failure, or acute exacerbation of a subclinical infection, although two studies involving subgroups of immunocompromised patients (bone marrow transplant and acquired immunodeficiency syndrome [AIDS]) did not find these patients to be at higher risk for complications related to endodontic treatment.[197,406]

## RATIONALE FOR SURGICAL TREATMENT

Although nonsurgical retreatment generally is believed to be the preferred first approach in the management of persistent apical periodontitis,[77,492,510,564] periradicular surgery is indicated when nonsurgical retreatment is impractical or unlikely to improve on the previous result. In particular, a surgical approach may be the first choice for managing teeth with long posts or irretrievable separated instruments, non-negotiable ledges and canal blockages or transportation, hard cement filling materials, failure of previous nonsurgical retreatment, and suspected vertical root fracture, or when a biopsy is indicated (Figs. 11.1–11.4). Even when surgical treatment is the likely definitive approach, nonsurgical therapy before the procedure may be recommended to help reduce the number of microorganisms in the root canal system and ensure a more favorable long-term prognosis.[230] On the other hand, surgery may be the first choice even if a tooth can be treated nonsurgically if the risks and costs of retreatment are considered excessive. For example, disassembling a recently restored bridge abutment tooth to allow endodontic retreatment may be technically possible but not economically feasible. One study found that endodontic microsurgery may be the most cost-effective option for management of persistent periapical disease, compared to nonsurgical retreatment, extraction and placement of a fixed partial



**Fig. 11.1** Previously treated maxillary lateral incisor with persistent periradicular disease. Nonsurgical retreatment is possible but would involve disassembly of an otherwise adequate coronal restoration. Periradicular surgery is a reasonable option.

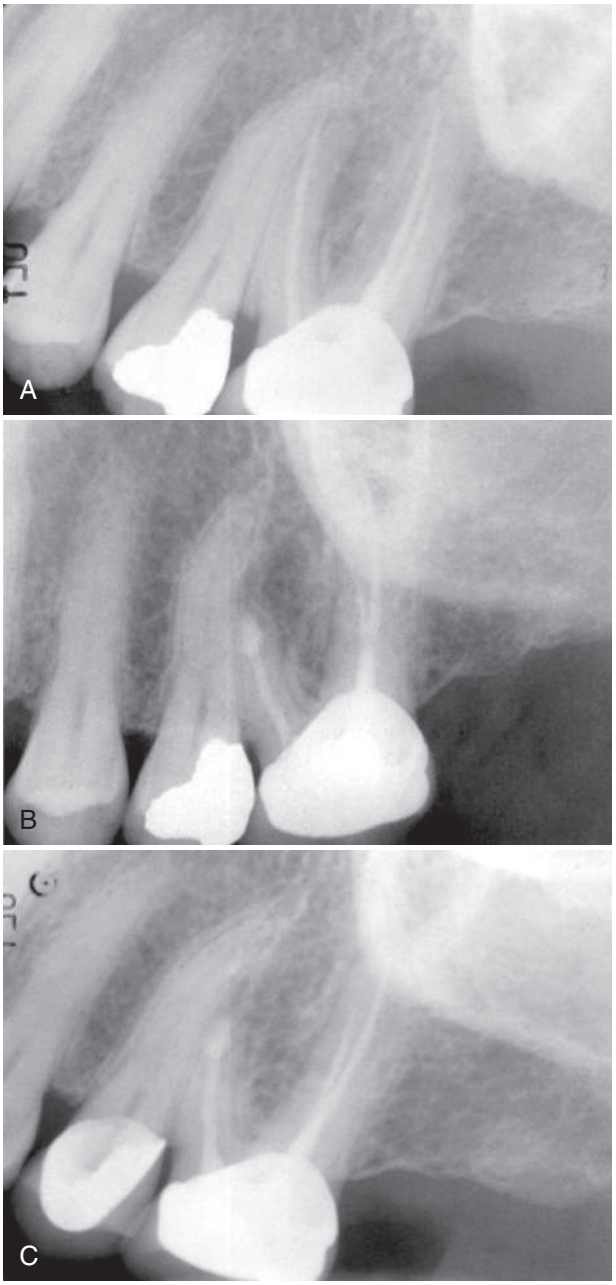



**Fig. 11.2 A,** Previously treated maxillary first molar with persistent periradicular disease and obvious transportation of the main mesiobuccal (MB-1) canal. Nonsurgical retreatment is unlikely to correct this iatrogenic condition, and surgery is the preferred choice. **B,** Immediate postsurgery radiograph. The root-end preparation and fill was extended from the MB-1 canal in a palatal direction to include the isthmus area and the second (MB-2) canal. **C,** One-year follow-up examination: the tooth is asymptomatic, and periradicular healing is apparent radiographically. Although surgery was the first-choice treatment in this case and the outcome was favorable, a good argument could be made for nonsurgical retreatment before surgery to ensure disinfection of the canal and to attempt to locate a MB-2 canal.

denture, and extraction and placement of a single tooth implant.[284] Case-specific variables and clinical judgment are key elements in the decision-making process, as current evidence supports the contention that the prognosis for surgical treatment is approximately the same as that for nonsurgical retreatment.[127,128,135,293,468,577]

## CLINICAL DECISION MAKING

Clinical decision making is a process that combines the best available evidence, clinical judgment, and patient preferences. Treatment choices are always made under conditions of at least some uncertainty. Surgical root canal treatment rarely is the only possible choice. Clinicians and patients must weigh the relative benefits, risks, and costs of two or more acceptable alternatives. Patients and clinicians can be expected to hold different attitudes about the value of potential treatment outcomes.[269] In a prognosis study of single-tooth dental implants, Gibbard and Zarb[193] reported that factors considered significant by health care professionals may not be important to patients. Even among groups of dentists and dental specialists, the threshold for treatment varies widely and treatment recommendations may depend more on personal values and experience than objective analysis of treatment costs, prognosis, risks, and alternatives.[56,343,439]

Shared decision making, in which the patient and clinician consider outcome probabilities and patient preferences and agree on the appropriate treatment, is preferable to a clinician-directed treatment decision.[46,179] Such a two-way exchange allows the clinician to provide the best available current evidence and case-specific clinical judgment while encouraging a decision that considers the patient's personal values and preferences. This decision-making model has been shown to increase patient knowledge about and satisfaction with the treatment choice.[262,386,387] In general, most patients prefer to be actively involved in the decision-making process but want to leave the specifics of the treatment to the clinician.[179] That is, the risks, benefits, and costs of treatment alternatives are important to patients, but the details of the procedure usually are not. Shared decision making is particularly relevant in light of the current trend in some areas to recommend implants as an alternative to root canal treatment.

Comparative probabilities for a successful outcome after surgical root canal treatment, nonsurgical retreatment, or extraction and replacement with a fixed prosthesis or implant are difficult to project. Many treatment variables are complex and not easily quantified, such as the affected location in the mouth, bone quality, practitioner's skill, possible influence of systemic disease on healing, periodontal support, bulk of the remaining tooth structure and resistance to fracture, quality of the coronal restoration, patient susceptibility to recurrent caries, materials used, and other factors. In addition, the definition of success varies considerably and is inconsistent from study to study.

Boioli and colleagues[66] performed a meta-analysis of implant studies and reported a 93% survival rate at 5 years. The 5- to 10-year success rate for implant retention routinely is reported to be 90% to 97%, depending on the location in the mouth and other variables.[302,309] Surgical root canal treatment was previously reported to have a lower success rate than implant placement.[230,546] However, most of these older studies would be considered weak by today's evidence-based standards and, more important, do not reflect the use of many of the newer surgical materials and techniques. With careful case selection, surgical skill, and the use of materials and techniques described later in this chapter, many studies have demonstrated success rates over 90% for surgical root canal treatment.[326,448,500,501,546,609] von Arx reported that the 5-year success for teeth receiving endodontic microsurgery

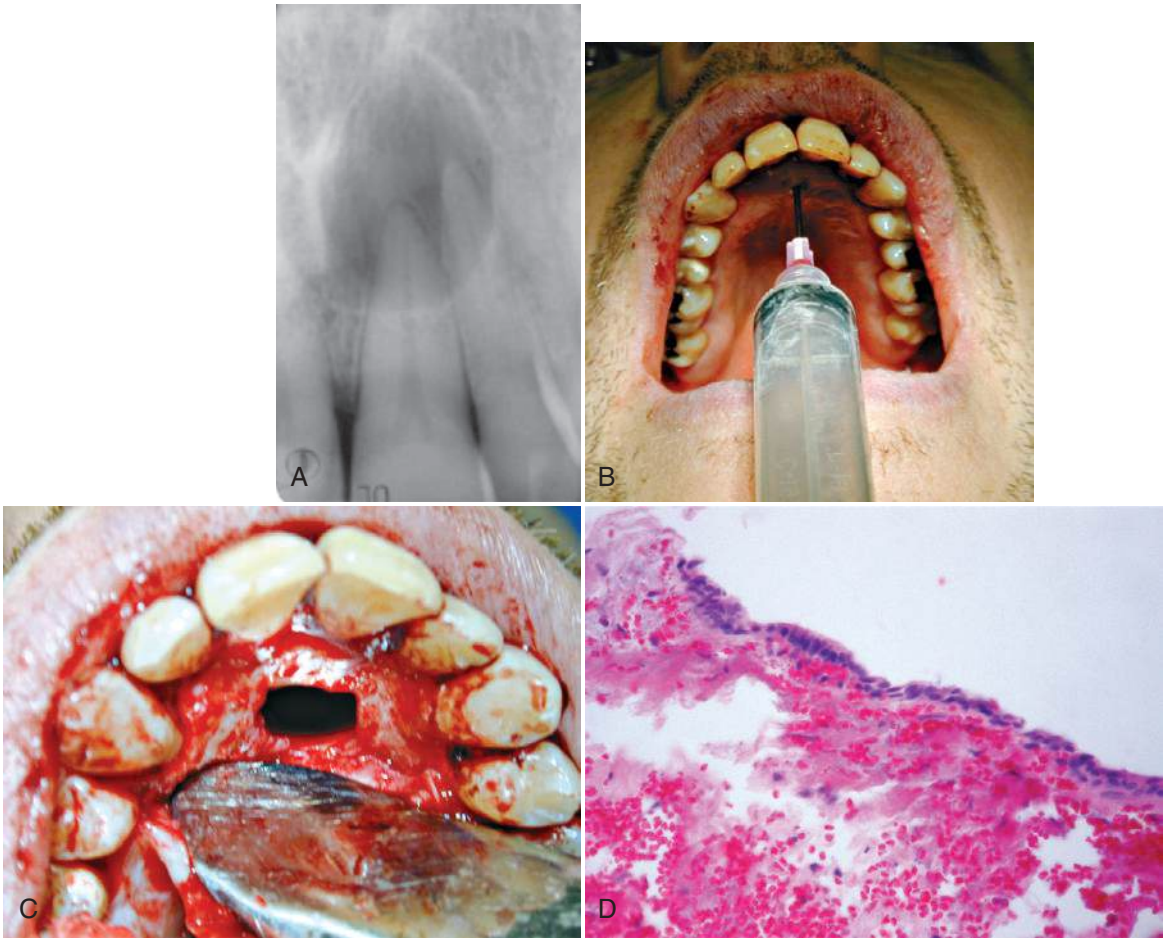



**Fig. 11.3** **A,** Indication for surgery (biopsy): large radiolucent lesion in the area of the maxillary left central and lateral incisors was detected on routine radiographic examination. All anterior teeth responded within normal limits to pulp vitality testing. **B,** After administration of a local anesthetic but before surgery, the lesion was aspirated with a large-gauge needle to rule out a vascular lesion. **C,** Buccal and palatal flaps were reflected. The lesion was directly accessible from the palate. An excisional biopsy was performed, and the sample was submitted for evaluation. **D,** Light microscopic section of the biopsy specimen; the lesion was diagnosed as a nasopalatine duct cyst (×400). (Courtesy Dr. Vince Penesis.)

was 8% less than observed at 1 year.[566] A similar study found an 85% success rate after 10 years.[500] A systematic review and meta-analysis comparing traditional root-end surgery to modern endodontic microsurgery (ultrasonic root-end preparation; root-end filling with intermediate restorative material [Denstply/Sirona, York, PA], Super EBA [ethoxybenzoic acid; Keystone Industries, Gibbstown, NJ], or mineral trioxide aggregate [MTA]; and high power [>×10] magnification and illumination) found that the outcome for endodontic microsurgery using current materials and techniques was 94%, compared to 59% success with traditional root-end surgery.[483]

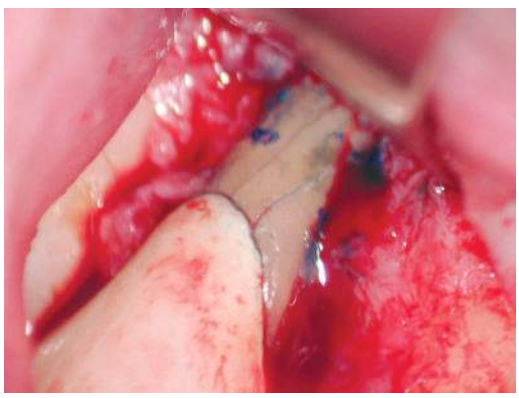


**Fig. 11.4** Exploratory surgery was indicated to rule out or confirm a root fracture. Magnification and staining with methylene blue dye confirmed the presence of multiple root fractures. The tooth was subsequently extracted. (Courtesy Dr. Martin Rogers.)

In broad terms and under ideal conditions, the prognoses for nonsurgical retreatment, surgical treatment, and implant placement should be roughly equal. The choice of treatment should be based on the best available evidence, case-specific clinical judgment, and the patient's preferences. As a stronger evidence base continues to develop, we predict that algorithms encompassing multiple patient and treatment variables will be devised to assist clinical decision making.

## General Biologic Principles of Wound Healing

Wound healing varies from region to region in the body and depends on several factors, including the type of tissue, the type of wound, and the type of healing. In periradicular surgery, the tissues include free and attached gingiva, the alveolar mucosa, periosteum, bone, the PDL, and cementum. The

wound may be intentional surgical trauma, which includes incision, blunt dissection, and excision (surgical), or pathologic or traumatic wounds. Healing occurs by primary or secondary intention.[212]

An incisional blunt dissectional wound, for example, can be considered to heal typically by primary intention, whereas a dissectional wound involving the resected root surface and osseous crypt heals by secondary intention. An important concept of the wound healing process in general is the difference between regeneration and repair. The goal of all surgical procedures should be *regeneration,* which returns the tissues to their normal microarchitecture and function, rather than *repair,* a healing outcome in which tissues do not return to normal architecture and function. Repair typically results in the formation of scar tissue.

The wounding process varies, depending on the types of tissue and injury. However, all wounds progress through three broad, overlapping phases in the process of healing; these are the inflammatory phase, the proliferative phase, and the maturation phase.[240,570] Although these phases can be identified in healing tissue, none of them has a clear beginning or end. Furthermore, in a wound such as an endodontic surgical site, which involves more than one tissue type, these phases progress at different rates in each type of tissue.

## Soft-Tissue Wound Healing

### INFLAMMATORY PHASE

Broadly speaking, the inflammatory phase of healing is similar for all tissues.[543] This phase can be broken down further into clot formation, early inflammation, and late inflammation.

#### Clot Formation

Clot formation begins with three events: (1) blood vessel contraction is initiated by platelet degranulation of serotonin, which acts on the endothelial cell and increases the permeability of the vessel, allowing a protein-rich exudate to enter the wound site; (2) a plug composed of platelets forms, primarily through intravascular platelet aggregation; and (3) both the extrinsic and intrinsic clotting mechanisms are activated. Several other events occur simultaneously, including activation of the kinin, complement, and fibrinolytic systems and the generation of plasmin.[34,247,521] These events stabilize hemostasis, begin the production of a number of mitogens and chemoattractants, and initiate the process of wound decontamination. The result is a coagulum consisting of widely spaced, haphazardly arranged fibrin strands with serum exudate, erythrocytes, tissue debris, and inflammatory cells. Compression of the surgical flap with sterile iced gauze immediately after surgery is designed to minimize the thickness of the fibrin clot and thereby accelerate optimal wound healing.

#### Early Inflammation: Polymorphonuclear Neutrophil Organization

As a result of production of chemoattractants by the various components of the clot, polymorphonuclear neutrophils (PMNs) begin to enter the wound site within 6 hours of clot stabilization. The number of PMNs increases steadily, peaking at about 24 to 48 hours after the injury. Three key steps mark PMN migration into the wound site: (1) *pavementing,* in which the red blood cells undertake intravascular agglutination, allowing the PMNs to adhere to the endothelial cells; (2) *emigration,* in which the PMNs actively pass through the vascular wall; and (3) *migration,* in which the PMNs use ameboid motion, under the influence of the various chemotactic mediators, to move into the injured tissues.[212]

The principal role of the PMNs is wound decontamination by means of phagocytosis of bacteria. The high number of PMNs in the wound site is relatively short lived; the number drops rapidly after the third day.

#### Late Inflammation: Macrophage Organization

About the time the PMN population is declining (48 to 96 hours after injury), macrophages begin to enter the wound site. They reach a peak concentration by approximately the third or fourth day. These cells, which are derived from circulating monocytes, leave the bloodstream under the influence of the chemoattractants in the wound site. The monocytes subsequently evolve into macrophages. Macrophages have a much longer life span than PMNs; they remain in the wound until healing is complete. Similar to PMNs, macrophages play a major role in wound decontamination through phagocytosis and digestion of microorganisms and tissue debris.

Macrophages are considerably more bioactive than PMNs and can secrete a vast array of cytokines. A key action of many of these bioactive substances is initiation of the proliferative phase of wound healing, which is accomplished by prompting the formation of granulation tissue. Two other major functions of macrophages are ingestion and processing of antigens for presentation to T lymphocytes, which enter the wound after the macrophages. Unlike PMNs, macrophages play an essential role in the regulation of wound healing.[109,224] A reduction in the number of macrophages in the wound site delays healing, because the wound does not progress to the next phase. For example, the age-related reduction in healing potential appears to be partly the result of loss of estrogen regulation of macrophages in healing tissues.[33]

### PROLIFERATIVE PHASE

The proliferative phase is characterized by the formation of granulation tissue in the wound. Two key cell types, fibroblasts and endothelial cells, have a primary role in the formation of granulation tissue. Granulation tissue is a fragile structure composed of an extracellular matrix of fibrin, fibronectin, glycosaminoglycans, proliferating endothelial cells, new capillaries, and fibroblasts mixed with inflammatory macrophages and lymphocytes. Epithelial cells also are active during this phase of soft-tissue healing and are responsible for initial wound closure. Guided tissue regeneration (GTR) procedures are based on control of the epithelial cell growth rate during this phase.

#### Fibroblasts: Fibroplasia

Undifferentiated mesenchymal stem cells in the perivascular tissue and fibroblasts in the adjacent connective tissue migrate into the wound site on the third day after injury

and achieve their peak numbers by approximately the seventh day. This action is stimulated by a combination of cytokines (e.g., fibroblast growth factor [FGF], insulin-like growth factor-1 [IGF-1], and 2-15 platelet-derived growth factor [PDGF]), which are produced initially by platelets and subsequently by macrophages and lymphocytes. As the number of macrophages declines and the fibroblast population increases, the tissue in the wound transforms from a *granulomatous* tissue to a *granulation* tissue.

Fibroblasts are the crucial reconstructive cell in the progression of wound healing because they produce most of the structural proteins forming the extracellular matrix (e.g., collagen). Collagen is first detected in the wound about the third day after injury. The fibroblasts produce type III collagen initially and then, as the wound matures, type I collagen. As this network of collagen fibers is laid down, endothelial and smooth muscle cells begin to migrate into the wound. Subsequently, as wound healing progresses, the collagen fibers become organized by cross-linking. Regularly aligned bundles of collagen begin to orient so as to resist stress in the healing wound.[108,286,413,516]

A focused type of fibroblast, known as a *myofibroblast,* plays a significant role in wound contraction, particularly in incisional-type wounds.[316,524] Myofibroblasts align themselves parallel with the wound surface and then contract, drawing the wound edges together. These cells are eliminated by apoptosis after wound closure.[137,138]

### Endothelial Cells: Angiogenesis

Capillary buds originate from the vessels at the periphery of the wound and extend into the wound proper. This occurs concurrently with fibroblast proliferation and can begin as early as 48 to 72 hours after injury. Without angiogenesis, the wound would not have the blood supply needed for further active healing. The capillary sprouts eventually join to form a network of capillary loops *(capillary plexuses)* throughout the wound.

In addition to a low oxygen concentration in the wound proper,[198,289] several factors have been identified as potent stimulators of angiogenesis, including vascular endothelial growth factor (VEGF), basic fibroblast growth factor (bFGF), acidic FGF (aFGF), transforming growth factor–alpha (TGF-alpha) and transforming growth factor–beta (TGF-beta), epidermal growth factor (EGF), interleukin-1 (IL-1), and tumor necrosis factor–alpha (TNF-alpha),[145] as well as lactic acid.[248,258,503] All of these have been shown to stimulate new vessel development.

### Epithelium

The first step in epithelial healing is the formation of an epithelial seal on the surface of the fibrin clot. This process begins at the edge of the wound, where the basal and suprabasal prickle cells rapidly undergo mitosis. The cells then migrate across the fibrin clot at a remarkable rate (0.5 to 1 mm/day). This monolayer of epithelial cells continues to migrate by contact guidance along the fibrin scaffold of the clot below. Migration stops as a result of contact inhibition of the epithelial cells from the opposing wound edge. Once the epithelium from both sides of the wound is in contact, an epithelial seal is achieved. In wounds healing by primary intention, formation of an epithelial seal typically takes 21 to 28 hours after reapproximation of the wound margins.[222]

### MATURATION PHASE

Under ideal conditions, maturation of the wound begins 5 to 7 days after injury. A reduction in fibroblasts, vascular channels, and extracellular fluids marks the transition to this phase of healing. During the early stages of wound maturation, the wound matrix is chiefly composed of fibronectin and hyaluronic acid. As the tensile strength of the wound increases, significant upregulation of collagen fibrogenesis occurs. Collagen remodeling ensues, with the formation of larger collagen bundles and alteration of intermolecular cross-linking. The result is a conversion of granulation tissue to fibrous connective tissue and a decreased parallelism of collagen to the plane of the wound. Aggregated collagen fibril bundles increase the tensile strength of the wound. As healing progresses in the wound, the collagen gradually reorganizes; this requires degradation and reaggregation of the collagen. The degradation of collagen is controlled by a variety of collagenase enzymes. Remodeling results in a gradual reduction in the cellularity and vascularity of the reparative tissue; the degree to which this occurs determines the extent of scar tissue formation. Active remodeling of scar tissue can continue very slowly for life.[138,247]

Maturation of the epithelial layer quickly follows formation of the epithelial seal. The monolayer of cells forming the epithelial seal differentiates and undergoes mitosis and maturation to form a definitive layer of stratified squamous epithelium. In this way an epithelial barrier is formed that protects the underlying wound from further invasion by oral microbes. The epithelial barrier typically forms by 36 to 42 hours after suturing of the wound and is characterized by a significant increase in wound strength.[222]

## Hard-Tissue Healing: Excisional Dentoalveolar Wound

The inflammatory and proliferative phases of healing for hard tissue are similar to those for soft tissue. A clot forms in the bony crypt, and an inflammatory process ensues that involves PMNs initially and macrophages subsequently. This is followed by the formation of granulation tissue with an angiogenic component. However, the maturation phase of hard-tissue healing differs markedly from that for soft tissues, primarily because of the tissues involved: cortical bone, cancellous bone, alveolar bone proper, endosteum, PDL, cementum, dentin, and inner mucoperiosteal tissue.

### OSTEOBLASTS: OSTEOGENESIS

The healing of an excisional osseous wound that is approximately 1 cm in diameter is similar to that of a fractured long bone. It progresses from hematoma to inflammation, eradication of nonvital debris, proliferation of granulation tissue, callus formation, conversion of woven bone to lamellar bone, and, finally, remodeling of the united bone ends. The coagulum that initially forms delays healing and must be removed to allow wound healing to progress.

A major difference between soft- and hard-tissue wound healing is found in the role of the osteoclast. Functionally, osteoclasts act as organizational units to debride necrotic

bone from the wound margin, much as macrophages remove tissue debris from the clot. Granulation tissue begins to proliferate from the severed PDL by 2 to 4 days after root-end resection.[223] This tissue rapidly encapsulates the root end. Simultaneously, endosteal proliferation into the coagulum occurs from the deep surface of the bony wound edge. The coagulum in the bony crypt is quickly converted into a mass of granulation tissue. In addition to those already discussed, several cell types migrate into the coagulum, including osteoprogenitor cells, preosteoblasts, and osteoblasts. These cells begin the formation of woven bone within the mass of granulation tissue. New bone formation is apparent about 6 days after surgery.[223]

Bone formation can be categorized into two types, each having several phases. The phases differ, depending on which type of formation is involved. One type of bone formation is a matrix vesicle–based process, and the other type is based on osteoid secretion. In both processes, osteoblasts produce the bone matrix. They secrete a collagen-rich ground substance that is essential for mineralization. Osteoblasts also cause calcium and phosphorus to precipitate from the blood.

In the formation of woven bone, which occurs by the matrix vesicle–based process, osteoblasts produce matrix vesicles through exocytosis (the release of substances contained in a vesicle within a cell by a process in which the membrane surrounding the vesicle unites with the membrane forming the outer wall of the cell) of their plasma membranes. As hydroxyapatite crystals accrue in the vesicles, they become enlarged and eventually rupture. This process begins with the deposition and growth of hydroxyapatite crystals in the pore regions. The crystals then amalgamate to form structures known as *spherulites.* Union of the separate spherulites results in mineralization.

The formation of lamellar bone does not require the production of matrix vesicles; rather, it proceeds by the osteoblast secretion process. Osteoblasts secrete an organic matrix composed of longitudinally arranged collagen matrix fibrils (mainly type I collagen). Mineralization occurs by mineral deposition directly along the collagen fibrils.[239,240] This stage has been associated with a rise in pH, most likely because of the enzyme alkaline phosphatase, which is secreted by osteoblasts and other cells and plays an important role in mineralization. The exact role of alkaline phosphatase during mineralization is unclear. The balance of evidence favors a positive catalytic role.[19,20] Some hypothesize[71,70,552,553] that alkaline phosphatase promotes mineralization through a combination of mechanisms involving various cells, extracellular matrix proteins, and elements. The interaction of alkaline phosphatase and phosphoproteins in both bone and dentin appears to be especially critical to the mineralization process.[71,72,557,558]

Inhibitor molecules, such as pyrophosphate and acidic noncollagenous bone proteins, regulate mineralization. Several growth factors also have been identified as key components in the production of osseous tissue. These include TGF-beta, bone morphogenic protein (BMP), PDGF, FGF, and IGF.[97,191,476,513] One clinical study demonstrated that the addition of autologous platelet concentrate to the surgical site decreased postsurgical pain and may accelerate the healing process.[134]

Three to 4 weeks after surgery, an excisional osseous wound is filled from 75% to 80% with trabeculae surrounded by intensely active osteoid and osteoblastic cells. A reforming periosteum can be seen on the outer surface of the wound; it is highly cellular and has more fibrous connective tissue oriented parallel to the plane of the former cortical plate. At 8 weeks after surgery, the trabeculae are larger and denser and the osteoblasts are less active; these cells occupy about 80% of the original wound. Also, fewer osteoid cells are associated with the maturing trabeculae. The overlying periosteum has reformed and is in contact with the newly developed bone. The osseous defect typically is filled with bone tissue by 16 weeks after surgery. However, the cortical plate has not yet totally reformed. Maturation and remodeling of the osseous tissue continues for several more months.[223]

Local healing is also influenced systemically by the endocrine system and its three general categories of hormones: polypeptide regulators (parathyroid hormone, calcitonin, insulin, and growth hormone), steroid hormones (vitamin $D_3$, glucocorticoids, and the sex hormones), and thyroid hormones.[219]

### CEMENTOBLASTS: CEMENTOGENESIS

During regeneration of the periradicular tissues, cementum forms over the surface of surgically resected root ends.[114] The exact spatial and temporal sequences of events leading to this formation of new cementum remains undefined; however, cementogenesis is important, because cementum is relatively resistant to resorption (osteoclasts have little affinity for attaching to cementum).

Cementogenesis begins 10 to 12 days after root-end resection. Cementoblasts develop at the root periphery and proceed centrally toward the root canal.[24] Considerable evidence indicates that the cells that regulate cementogenesis are derived from ectomesenchymal cells in the tooth germ proper rather than from bone or other surrounding tissues. The migration and attachment of precementoblasts to the root surface dentin is monitored by mediators from within the dentin itself.[140,210] Cementum covers the resected root end in approximately 28 days. Newly formed PDL fibers show a functional realignment that involves reorientation of fibers perpendicular to the plane of the resected root end, extending from the newly formed cementum to the woven bone trabeculae. This occurs about 8 weeks after surgery.[212,319,320]

## Systemic Medications and Wound Healing

### BISPHOSPHONATES

Bisphosphonates are commonly used for the treatment of osteopenia, osteoporosis, Paget disease of bone, multiple myeloma, and metastatic bone, breast, and prostate cancer. The potential association between bisphosphonate use and osteonecrosis of the jaw was first reported in 2003.[332,351] The current preferred term for this condition is antiresorptive agent-induced osteonecrosis of the jaw (ARONJ)[229] or medication-related osteonecrosis of the jaw (MRONJ). The change in terminology reflects the finding that, in addition to bisphosphonates, other antiresorptive agents can increase the risk for osteonecrosis of the jaw (e.g., denosumab). ARONJ can occur spontaneously but is more commonly associated with

dental procedures that involve bone trauma. Patient variables that increase the risk for ARONJ include age (over 65 years old), chronic use of corticosteroids, use of bisphosphonates for more than 2 years, smoking, diabetes, and obesity.[278,334,340,583] It is important to note that the risk for ARONJ with commonly prescribed oral bisphosphonates appears to be very low, whereas the overall benefits of this drug class for reducing morbidity and mortality related to hip, vertebral, and other bone fractures are significant.[229] The estimated incidence for ARONJ in patients taking oral bisphosphonates ranges from zero to 1 in 2260 cases, although dental extractions may quadruple the risk of developing ARONJ.[7,340] A reasonable upper estimate of the risk of developing ARONJ in a patient who does not have cancer is about 0.1%.[229] One clinical study reported a much higher 4% risk for developing ARONJ in patients taking a specific oral bisphosphonate (alendronate sodium),[479] although this was a retrospective study using an electronic patient record at a dental school and may not have adequately controlled for other relevant variables. The majority of ARONJ cases have been reported in patients taking intravenous (IV) bisphosphonates (e.g., zoledronic acid and pamidronate). According to one report, approximately 20% of patients taking IV bisphosphonates may develop ARONJ.[7,340] A systematic review found that there was a strong association between cancer and risk of ARONJ, with the prevalence ranging from 0.7% to 13.3%.[352] Interestingly, this same review found that the higher-quality studies reported the highest prevalence of ARONJ in cancer patients.

Conservative and timely management of apical periodontitis is essential to reduce the risk of ARONJ in patients taking bisphosphonates. Because periapical pathosis may exacerbate or increase the risk for ARONJ, the "no treatment" option is not a viable choice. The potential risk for ARONJ should be thoroughly discussed with all patients that are taking bisphosphonates, as well as treatment options. Nonsurgical retreatment should usually be considered the first choice, especially for patients with a history of IV bisphosphonate use or other risk factors. Even so, surgical treatment may be indicated to manage chronic or acute apical periodontitis. When the only viable treatment options for the management of persistent periradicular inflammation are surgical root canal therapy or extraction, the question of which option is more or less likely to lead to ARONJ in at-risk patients remains unanswered. In general, the procedure that could most predictably eliminate the periradicular inflammation with the least amount of surgical trauma would be preferred. Conservative surgical technique, primary tissue closure, and use of chlorhexidine mouth rinses preoperatively and during the healing stage are recommended.[229] There is some limited evidence to support the use of prophylactic antibiotics and chlorhexidine mouth rinse to decrease the risk of ARONJ for patients undergoing surgical dental treatment.[229] Because bisphosphonates have a substantial half-life when incorporated into bone, discontinuation of bisphosphonates prior to dental treatment is unlikely to provide any measurable benefit and may place the patient at greater risk for complications that the drug is intended to prevent.

The possibility of predicting which patients may be at greater risk for ARONJ based on serum levels of C-terminal cross-linking telopeptide of type I collagen (cross-linking telopeptide [CTX]) has been suggested based on clinical observations.[333] This observation may provide a useful clue for future risk assessment but requires additional refinement and validation prior to widespread acceptance. In fact, one study found that although serum CTX levels were lower in patients taking bisphosphonates (as expected), there was no association between lower CTX levels and increased risk of ARONJ.[208] This same study found that patients taking a once-a-year dose of 5 mg zoledronic acid were at no greater risk for ARONJ than a control group (1 case in 5903 patients). Once ARONJ develops, treatment options are limited. A possible treatment strategy proposed in a series of case reports is the use of teriparatide, an anabolic drug that stimulates osteoblast formation and is used for the management of osteoporosis.[505]

## GLUCOCORTICOIDS

Glucocorticoid therapy has been shown to induce rapid bone loss within the first 3 months of treatment. Even inhaled steroids have been implicated as a cause of bone loss. Bone formation is inhibited partly through a decrease in osteoblast life span and function, a reduction in the mineral apposition rate, and a prolonged mineralization lag time. Biochemical markers of bone formation (i.e., osteocalcin and bone-specific alkaline phosphatase) are suppressed. In addition to this primary suppressive effect, glucocorticoids cause accelerated bone resorption. The number and activity of osteoclasts increase during early glucocorticoid exposure. With continued use of glucocorticoids, the rapid rate of osteoclast-mediated bone resorption slows, but suppression of bone formation continues as the overriding skeletal activity.[272] Bone loss is therefore progressive because bone resorption chronically exceeds bone formation.

The adverse effects of glucocorticoids on bone are mediated by sex steroid deficiency and expression of locally produced growth factors and related proteins, such as reduced production of IGF-1 and alterations in IGF-binding proteins in osteoblasts. The direct effect on calcium metabolism through alteration of vitamin D metabolism leads to a state of secondary hyperparathyroidism.[447,581,582]

## NONSTEROIDAL ANTIINFLAMMATORY DRUGS

Bone homeostasis is regulated by many factors, including prostaglandins (PGs).[267] PGs are important to both normal and pathologic bone turnover. PGs modulate osteoblast proliferation and differentiated functions.[426,427] The levels of prostaglandin E (PGE) and prostaglandin F (PGF) are elevated in the early phase of fracture healing, and administration of $PGE_2$ has increased the rate of osseous repair in several animal studies.[272] Nonsteroidal antiinflammatory drug (NSAID) inhibition of the enzyme cyclooxygenase (COX), which is involved in the synthesis of PGs, is the same mechanism by which NSAIDs control pain. By inhibiting the COX enzymes and the subsequent production of PGs, NSAIDs accomplish the desired antiinflammatory effects—but also prevent the increased production of PGs required for bone healing. In vitro studies using animal models have shown that NSAIDs inhibit osteoblast proliferation and stimulate protein synthesis.[235] These drugs also have been shown to delay fracture healing and to affect bone formation adversely in animals and

humans.[15,192] In one study, the use of NSAIDs reduced the amount of bony ingrowth into an orthopedic implant.[220]

In other studies, NSAIDs reduced bone resorption associated with experimentally induced periapical lesions,[18,390] and systemic NSAIDs may play a positive role in maintaining bone height around endosseous titanium dental implants.[257] Additional research is needed to clarify the specific influence of NSAIDs on healing after periradicular surgery.

### CYCLOOXYGENASE-2 INHIBITORS

Although both COX-1 and COX-2 have been identified in osteoblasts, the different roles of the two COXs in bone formation remain unclear. A study using COX-1 and COX-2 knockout mice compared the roles of the two enzymes in fracture healing. COX-2 was shown to have an essential role in both endochondral and intramembranous bone formation during wound healing. COX-2 knockout mice showed a persistent delay in cartilage tissue ossification. No difference in fracture healing was observed between the COX-1 knockout mice and the wild-type control used in this study.[603] Additional studies have shown a persistent delay in healing with COX-2 NSAIDs[202,203]; however, this difference in effect was not apparent clinically in a study examining the healing of spinal fusion wounds.[182]

## Preoperative Evaluation of Medically Complex Patients

The preoperative assessment must take into account both the type of procedure planned and the type of patient (i.e., physical health and psychological status). Healthy patients clearly can be expected to tolerate surgical procedures better than medically complex patients. Clinicians should anticipate and prepare for the inevitability of treating more medically complex patients as the population ages. It is beyond the scope of this brief section to cover all possible medical considerations; rather, the most common issues that may require modification of an endodontic surgical treatment plan are presented. Relatively few absolute contraindications to periradicular surgery exist for patients well enough to seek care in an ambulatory dental office. Nonetheless, if any question arises about a patient's ability to tolerate a surgical procedure, medical consultation is advised. A thorough medical history and assessment of vital signs are required parts of the presurgical evaluation.

The American Society of Anesthesiologists (ASA) has developed a widely used system for establishing surgical risk. Patients categorized as ASA 1 are healthy and usually require no modification of the surgical treatment plan. Patients classified as ASA 4 or ASA 5 are not treated in a dental office; these individuals have significant medical problems that take priority over dental considerations. Patients considered ASA 2 or ASA 3 are commonly seen on an outpatient basis and may require medical consultation and modification of the surgical treatment plan. Patients in the ASA 2 and ASA 3 categories have mild to moderate systemic disease and often are undergoing treatment with one or more prescription medications. The ASA classification system should only be used as a general guide because, when used alone, the ASA system is not a reliable predictor of operative risk.[201] In addition, even experienced anesthesiologists exhibit differences of opinion when using the ASA system to classify patients.[227,395] The patient's psychological status and anticipated procedural stress should be considered in addition to the ASA classification of physical health. A stress reduction protocol may be helpful for patients who report moderate dental anxiety with concurrent mild to moderate systemic disease.

Surgical procedures typically require a larger amount of local anesthetic with a vasoconstrictor than a nonsurgical root canal treatment. An essential part of the presurgical evaluation is an assessment of the patient's cardiovascular status and tolerance for local anesthetics that contain epinephrine. In particular, patients with advanced cardiovascular disease, geriatric patients, and patients taking certain medications may have a reduced tolerance for local anesthetics containing a vasoconstrictor. Treatment-induced stress associated with surgical procedures can cause a significant increase in heart rate and systolic blood pressure when compared to nonsurgical root canal therapy.[189] The use of local anesthetics with vasoconstrictors was addressed in the seventh report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC 7).[231] An updated version of this report (JNC 8) was issued in 2013. In patients with cardiovascular disease, 0.036 to 0.054 mg of epinephrine (approximately two to three cartridges of local anesthetic with 1:100,000 epinephrine) should be safe for most patients except those with severe cardiovascular disease or other specific risk factors.[38,231,594] Local anesthetics with vasoconstrictors should be avoided or used with extreme caution in patients with the following cardiovascular conditions: severe or poorly controlled hypertension, arrhythmias that are refractory to treatment, myocardial infarction within the past month, stroke within the past 6 months, coronary artery bypass graft within the past 3 months, and uncontrolled congestive heart failure.[311] Patients unable to tolerate vasoconstrictors may not be good candidates for periradicular surgery, because a local anesthetic with a vasoconstrictor is essential for obtaining adequate hemostasis and visibility during this type of procedure. Surgery can be performed using only local anesthetics without vasoconstrictors, but this is not recommended.

The potential for multiple drug interactions is increasing because of the aging population and the introduction of many new drugs. Many elderly patients have diminished liver and kidney function and therefore cannot metabolize and excrete drugs as efficiently as younger, healthy patients. The potential for drug interactions and decreased metabolism and excretion of drugs should be considered, even for commonly used agents such as local anesthetics and analgesics.

A transient bacteremia is a virtual certainty with periradicular surgery; therefore patients at high risk for bacterial endocarditis should receive appropriate antibiotic prophylaxis as recommended by the American Heart Association. New guidelines for prevention of infective endocarditis were published in 2007 and represent a significant change from previous American Heart Association guidelines.[588] For example, antibiotic prophylaxis is no longer recommended for patients with a history of mitral valve prolapse

(with or without regurgitation), rheumatic heart disease, bicuspid valve disease, aortic stenosis, and certain congenital heart conditions. Antibiotic prophylaxis is now only recommended for patients with valvular disease associated with the highest risk of adverse outcomes from infective endocarditis. For patients in the highest risk category, antibiotic prophylaxis is recommended for dental procedures that involve manipulation of gingival tissues or the periapical region of teeth or perforation of the oral mucosa. For all other patients with valvular disease, the risks associated with routine antibiotic prophylaxis are greater than the potential benefits.[588] Although these guidelines represent the best available current evidence, practitioners should be aware that a recent study from England found a significant increase in cases of infective endocarditis after widespread adoption of the new guidelines in 2008. However, the authors point out that their data do not establish a causal relationship and recommend adherence to the current guidelines at this time.[130a]

New clinical practice guidelines for the management of patients with prosthetic joints were co-developed by the American Association of Orthopaedic Surgeons and the American Dental Association in December 2012.[578] The working group conducted a systematic review and presented three recommendations. Recommendation 1 states that "the practitioner might consider discontinuing the practice of routinely prescribing prophylactic antibiotics for patients with hip and knee prosthetic joint implants undergoing dental procedures." The level of evidence to support this recommendation was classified as "limited," but that was the highest level of evidence assigned to any of the three recommendations. Two case-control studies provide evidence for this recommendation.[51,498] In recommendation 2, the group was unable to recommend for or against the use of topical antimicrobial agents (grade of recommendation = inconclusive). Recommendation 3 was a consensus recommendation to support the maintenance of appropriate oral hygiene (grade of recommendation = consensus). A recent American Dental Association Council on Scientific Affairs report added further clarification by concluding that: "In general, for patients with prosthetic joint implants, prophylactic antibiotics are not recommended prior to dental procedures to prevent prosthetic joint infection.[499a]

Management of patients undergoing anticoagulant therapy depends on the type of anticoagulant, the reason for anticoagulant therapy, and the type of oral surgery planned. Warfarin (Coumadin), a commonly prescribed anticoagulant used to treat or prevent thromboembolisms, blocks the formation of prothrombin and other clotting factors. The international normalized ratio (INR) value is the accepted standard for measuring prothrombin time (PT). The desired therapeutic range for the INR usually is 2 to 3.5, depending on the underlying medical indication for anticoagulant therapy. Limited oral surgery procedures, such as simple forceps extraction of one to three teeth, may be performed safely on patients with INR values within the normal therapeutic range.[8,89,255,478] Periradicular surgery, however, may present a greater challenge for hemostasis, even for patients well maintained within the therapeutic range. The clear field visibility normally required for proper surgical management of the root end may not be possible in patients undergoing anticoagulant therapy. The patient's physician must be consulted for assistance in developing an appropriate treatment plan. Some patients may be able to tolerate discontinuation of warfarin therapy 2 days before a planned surgical procedure, which allows the INR to drift downward. In a prospective cohort study, Russo and colleagues[461] reported that suspension of warfarin 2 days before a surgical procedure resulted in no bleeding problems and no thromboembolic events. They found that the average time spent at an INR below 2 (critical value) was 28 hours and that 90% of the patients returned to the desired therapeutic INR value within 7 days. However, this strategy may place certain patients at greater risk for a thromboembolic event; therefore discontinuation of anticoagulant therapy would not be recommended.

In general, patients undergoing anticoagulant therapy should present minimal risk for significant bleeding during or after oral surgery, and local measures to control bleeding should be adequate.[88,569] In a review prepared for the American Dental Association Council on Scientific Affairs and Division of Science, Jeske and Suchko[260] recommended against routine discontinuation of anticoagulant therapy before dental procedures, including surgical procedures. Regardless of the management approach selected, consultation with the patient's physician and an INR test on the day of surgery are strongly recommended. A retrospective study of an urban dental school population found that INR values for 43% of patients taking warfarin were not within the recommended therapeutic range.[266] To underscore the challenge of maintaining anticoagulation levels within an appropriate therapeutic range, warfarin has been implicated in more emergency hospital admissions than any other medication.[83]

Hospitalization and conversion to heparin therapy may be considered in special cases, but the patient, physician, and surgeon must carefully weigh the potential risks against the expected outcome and benefits. A new category of heparin anticoagulant, low-molecular-weight heparins (LMWHs), allows patient self-administration and may be an alternative for patients who need to maintain a high level of anticoagulation but want to reduce the cost and time required for traditional heparin conversion therapy.

Novel oral anticoagulant drugs (NOACs) have recently emerged as an alternative to warfarin for most indications (e.g., previous stroke, atrial fibrillation, and deep vein thrombosis) *except* for mechanical heart valves. Examples include: apixaban (Eliquis), dabigratran (Pradaxa), edoxaban (Lixiana), and rivaroxaban (Xarelto). NOACs have fewer interactions with foods and other drugs, and there is no need for routine blood work monitoring. However, although NOACs are known to prolong PT by inhibiting factor X to Xa conversion, there is currently no reliable test to accurately measure level of anticoagulation. Medical consultation may be indicated to help evaluate the relative benefits and risks of discontinuing NOAC drugs for 1 or 2 days prior to endodontic surgery.

Low-dose aspirin therapy is known to increase bleeding time by irreversibly inhibiting platelet aggregation. A common practice has been to advise patients to discontinue aspirin therapy for 7 to 10 days before oral surgery.[28] At low-dose therapeutic levels (<100 mg/day),

aspirin may increase bleeding time and complicate surgical procedures. However, Ardekian and colleagues[28] concluded that low-dose aspirin therapy should *not* be discontinued before surgical procedures and that bleeding could be controlled by local measures. Higher-dose therapy may create a greater risk of bleeding during or after surgery.

Although a patient undergoing aspirin therapy may not be at high risk for significant bleeding during or after surgery, a concern in periradicular surgery is the visibility problems created by oozing blood. The clinician should consult the patient's physician about the medical reason for aspirin therapy and should weigh the risks and benefits of discontinuing aspirin before the proposed surgery. It should be possible to perform periradicular surgery without discontinuing aspirin therapy if necessary, but visibility during the procedure may be compromised, which may adversely affect the prognosis.

Herbs, dietary supplements, and vitamins can contribute to bleeding problems during surgery, and patients often fail to report use of these substances in the preoperative evaluation.[385] In a survey of surgical patients, approximately one third reported using a nonprescription medication that might inhibit coagulation or interact with anesthetics.[384] In particular, ginkgo biloba, ginger, garlic, ginseng, feverfew, and vitamin E inhibit platelet aggregation and can increase the risk of bleeding.[96] Ingredients in over-the-counter (OTC) weight loss products can potentiate the effect of epinephrine and increase cardiac stress, although the most obvious example of this phenomenon, ephedra, has been removed from the US market by a Food and Drug Administration (FDA) order.

Patients with inherited or acquired bleeding disorders are also at risk for excessive bleeding during and after periradicular surgery. Impaired liver function associated with past or current alcohol or drug abuse may also predispose a patient to excessive bleeding during surgery. A thorough medical history and patient interview should help identify these patients and, in consultation with the patient's physician, allow for any necessary treatment modifications.

Many other medical conditions, such as recent myocardial infarction, stroke, cardiac arrhythmias, diabetes, radiation therapy for head and neck cancer, immunocompromising conditions, seizure disorders, adrenal suppression, liver or kidney disease, and pregnancy may require modification of the treatment plan. However, these conditions usually present more general problems that are not unique to a surgical procedure. The value of a thorough medical history and patient interview cannot be overemphasized.

## Anatomic Considerations

Evaluating the access to the surgical site is one of the most important steps in case selection for periradicular surgery. Anatomic studies can provide some guidance, but individual variation is great, and there is no substitute for a complete clinical examination. A small oral opening, active facial muscles, shallow vestibule, and thick buccal alveolar bone all can significantly increase the difficulty of the procedure, even in cases that appear straightforward on radiographic examination.

### POSTERIOR MANDIBLE

The primary anatomic structure of concern for periradicular surgery in the posterior mandible is the neurovascular bundle that courses through the mandibular canal and exits through the mental foramen. The relationship among the mandibular canal, the mental foramen, and the root apices of mandibular teeth has been well studied; however, anthropometric averages have limited value in the treatment of individual patients. An understanding of the typical anatomic relationships is important, but even more important is an evaluation of the individual patient to develop a case-specific risk assessment.

The depth of the vestibular fornix generally is a good predictor of the possible difficulty involved in obtaining surgical access to the mandibular posterior teeth.[305] A shallow vestibule usually portends thicker alveolar bone and more difficult access to the root end.

The mental foramen, another key anatomic structure, is usually located between and apical to the mandibular first and second premolars[118,356]; however, this varies considerably, and the practitioner must examine each patient carefully to determine its location (Figs. 11.5 and 11.6). Vertical location of the mental foramen may vary even more than horizontal location. Moiseiwitsch[356] found that the average location was 16 mm inferior to the cementoenamel junction (CEJ) of the second premolar, although the range was 8 to 21 mm, which would place approximately 20% of the foramina at or coronal to the root apex. Fortunately, the mental foramen usually is easily visualized with standard periapical and panoramic radiographs. A vertically positioned periapical radiograph often may be more useful than a horizontally positioned one, especially in individuals with longer roots. Also, the mental foramen can usually be palpated.

When a vertical releasing incision is indicated, it typically is made at the mesial line angle of the mandibular canine. This location is always mesial to the mental foramen, because the foramen is located in the area ranging from the apex of the mandibular first premolar to slightly distal to the second premolar. The nerve bundle exits from

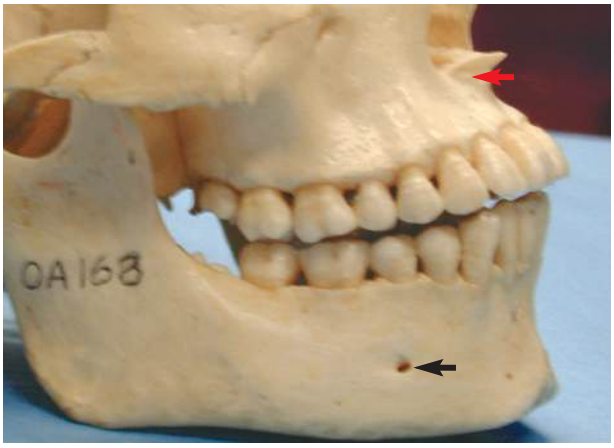



**Fig. 11.5** Lateral view of the skull showing the anterior nasal spine and the proximity of the maxillary anterior root apices to the floor of the nose *(red arrow),* as well as the typical location of the mental foramen *(black arrow).*

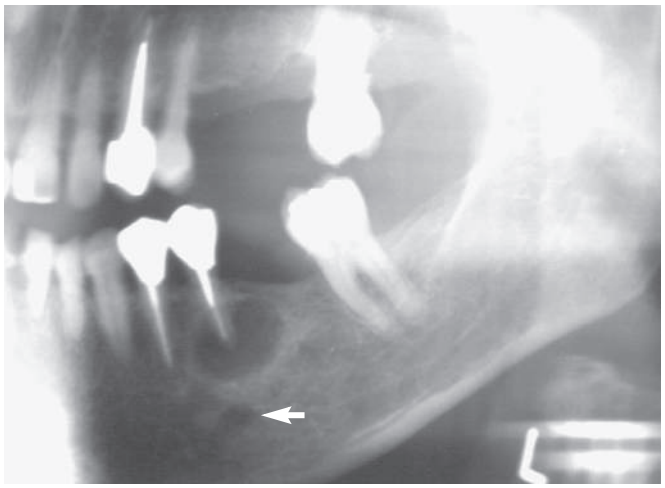


**Fig. 11.6** Preoperative evaluation of a mandibular left second premolar included a panoramic radiograph to help locate the mental foramen *(white arrow),* which was not visible on a standard periapical radiograph.

the foramen in a distal direction. An alternative technique for gaining access to mandibular posterior teeth involves a distal releasing incision between the mandibular first and second molars.[355] This approach may be especially useful for access to the mandibular second premolar and first molar. Care must be taken to avoid the facial artery as it crosses the level of the inferior vestibular fornix near the mandibular first molar. Inadvertent contact with the facial artery is unlikely if the incision is not extended beyond the depth of the vestibule.

Regardless of the approach selected, it is obviously important to avoid making an incision in the immediate vicinity of the mental foramen. The mental nerve is encased in a relatively tough sheath, and permanent damage can be avoided if careful blunt dissection is used in the area. A misguided vertical releasing incision, however, may sever the nerve, causing permanent injury. Trauma to the nerve from blunt dissection in this area or from pressure from a misplaced retractor may cause temporary paresthesia but is much less likely to do permanent injury. (Documentation and management of nerve injuries are discussed later in the chapter.)

The borders of the mandibular canal often are more difficult to visualize with conventional radiographic techniques. A parallel periapical radiograph, either horizontally or vertically positioned, usually can provide a reasonably accurate image of the relationship between the superior border of the mandibular canal and the root apices. However, the mandibular canal sometimes cannot be readily visualized. Such cases should be approached with extreme caution, because an increased risk of paresthesia secondary to nerve injury may be an unacceptable risk for many patients. CBCT imaging can be very useful for identifying the location of the mandibular canal and determining its relationship to the root apices (see Chapter 2).[68,285,556]

In the buccolingual dimension, the mandibular canal commonly follows a curving pathway from the buccal half of the mandible near the distal root of the second molar to the lingual half of the mandible near the first molar, then curving back to the buccal near the second premolar as it exits the mental foramen.[136] The average vertical distance from the superior border of the mandibular canal to the distal root apex of the mandibular second molar is approximately 3.5 mm; this increases gradually to approximately 6.2 mm for the mesial root of the mandibular first molar and to 4.7 mm for the second premolar.[136,312] This relationship typically provides a greater margin of safety for surgery on the mandibular first molar compared with the second premolar and especially the second molar. Surgery on the mandibular second molar may be further complicated by relatively thick overlying buccal bone, lingual inclination of the roots, and a more buccal location of the mandibular canal. This is not to say that periradicular surgery should not be performed on mandibular second molars; rather, it points out that the relative risks and benefits must be carefully considered. Often the most prudent choice for a mandibular second molar is an intentional replantation procedure or extraction and placement of an implant.

## POSTERIOR MAXILLA

The primary anatomic structure of concern in the posterior maxilla is the maxillary sinus. CBCT allows for a more precise preoperative three-dimensional evaluation of the relationship between the roots of maxillary posterior teeth and the sinus.[69,79] Perforation of the sinus during surgery is fairly common, with a reported incidence of about 10% to 50% of cases.[157,177,456] Even without periradicular pathosis, the distance between the root apices of the maxillary posterior teeth and the maxillary sinus sometimes is less than 1 mm.[225] An inflammatory periradicular lesion often increases the likelihood of sinus exposure during surgery. Fortunately, perforation of the maxillary sinus rarely results in long-term postoperative problems.[225] In a report of 146 cases of sinus exposure during periradicular surgery, Watzek and colleagues[579] found no difference in healing compared with similar surgical procedures without sinus exposure. The sinus membrane usually regenerates, and a thin layer of new bone often forms over the root end, although osseous regeneration is less predictable.[50,157,579] The general rule of placing a vertical releasing incision at least one tooth mesial and distal to the surgical site is especially important when sinus perforation is possible, because the exposure site should be completely covered with the mucoperiosteal flap to provide primary closure.

If the maxillary sinus is entered during surgery, special care must be taken to prevent infected root fragments and debris from entering the sinus. The most commonly used root resection technique involves grinding the root apex with a high-speed drill for approximately 3 mm in an apical to coronal direction; therefore an opening could allow infected debris into the sinus. A sinus opening can be temporarily occluded with a material such as Telfa gauze, although the gauze should be secured so as to prevent inadvertent displacement into the sinus. A suture can be placed through the packing material to prevent displacement and aid retrieval. Jerome and Hill[259] suggested securing the apical root segment by drilling a small hole in the root tip and threading a piece of suture material through the hole. The root is then resected at the appropriate level, and the root-end fragment is removed in one piece. If a root fragment or other foreign object is displaced into the sinus, it should be

removed. An orascope or endoscope may be useful for visualizing the foreign object, but referral for evaluation and surgical removal may be indicated if the fragment cannot be located and removed.

The palatal roots of maxillary molars present a special challenge for surgical access. Palatal roots may be reached from either a buccal (transantral) or palatal approach. Wallace[572] described a transantral approach in which a buccal flap is reflected, the buccal roots are resected, the osteotomy access into the sinus is enlarged to approximately 1 by 1.5 cm, and the palatal root tip is resected, ultrasonically prepared, and filled. The sinus may be packed with moist gauze to catch debris, and it should be irrigated with sterile saline upon completion of the surgery. In some cases, careful reflection and retraction of the sinus membrane may be possible, allowing access to the palatal root end without direct invasion of the sinus.[16] Many who have tried this approach find it more challenging than it might initially seem. The enhanced illumination and magnification provided by a dental operating microscope, endoscope, or orascope are essential aids in this type of surgery.[40]

A palatal approach to the palatal root of maxillary molars may seem more direct than a transantral approach, but it can present certain difficulties. Visibility in the surgical field is reduced and manipulation of instruments is more difficult than with most routine buccal approaches. Patients with a deep, vertical palatal vault are better candidates for this approach than individuals with a wide, shallow palate. A sinus tract or a large lesion on the palatal root may allow easier access and better visualization of the palatal root because only limited bone removal would be required. The position of the anterior palatine artery must be carefully considered when the incision is made and the flap reflected. This artery emerges from the greater palatine foramen distal to the maxillary second molar at the junction of the vertical section of the alveolar process and the flat portion of the palate and continues anteriorly (Fig. 11.7). A vertical releasing incision can be placed between the maxillary first premolar and canine, where the artery is relatively narrow and branches off into smaller arteries. If needed, a short distal vertical releasing incision may be made distal to the second molar, but it should not approach the junction

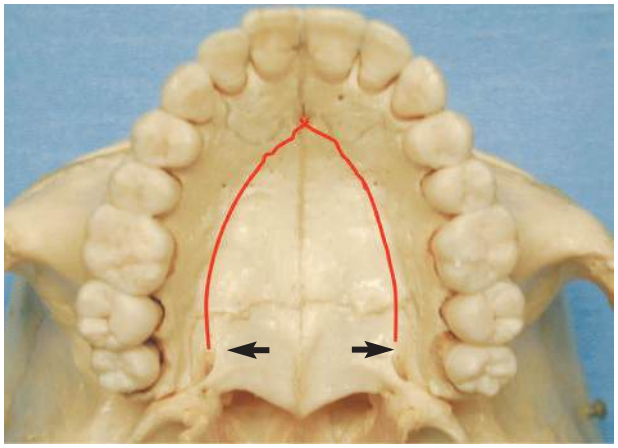


**Fig. 11.7** Palatal view showing the position of the greater palatine foramen *(arrows)*. The approximate location of the anterior palatine artery is marked in *red*.

of the alveolar process and roof of the palate. If the anterior palatine artery is severed, local clamping and pressure may not stop the hemorrhage, and ligation of the external carotid artery may be necessary. Because of the palatal vault's concave shape, repositioning the flap may be challenging. An acrylic surgical stent may be fabricated before surgery to assist repositioning of the flap and to help prevent pooling of blood under the flap.

## ANTERIOR MAXILLA AND MANDIBLE

Periradicular surgery on anterior teeth generally involves fewer anatomic hazards and potential complications than in posterior teeth. Nonetheless, access to the root apex in some patients may be unexpectedly difficult because of long roots, a shallow vestibule, or lingual inclination of the roots. As shown in Fig. 11.5, the root apices of maxillary central and lateral incisors can be quite close to the floor of the nose and the bony anterior nasal spine. Fig. 11.8 demonstrates isolation and protection of the nasopalatine neurovascular bundle during maxillary anterior surgery. The average maxillary canine is about 26 mm long and usually presents no difficulty

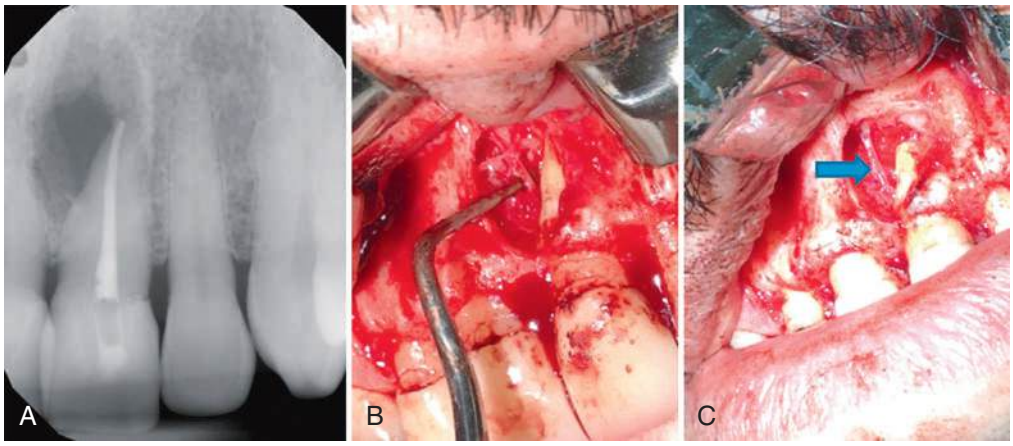



**Fig. 11.8 A,** Radiograph demonstrating periradicular radiolucency associated with previously treated tooth #9. **B–C,** After full-thickness flap reflection, the nasopalatine neurovascular bundle was identified (**B**), protected, and isolated (*arrow* in **C**) during curettage of cystic lesion. (Courtesy Dr. Tim Rogers.)

for surgical access; however, the combination of a shallow vestibule and a longer than average root length could complicate access to the root apex area. In such cases, the creation of an osteotomy apical to the root end may be impossible. An alternative approach for long-rooted teeth and root tips in the vicinity of critical anatomic structures is to enter the bone and resect the root at a level approximately 3 mm coronal to the apex. After the root tip is removed, the area apical to the root can be inspected and curetted as needed.

Periradicular surgery on mandibular incisors often is more challenging than expected. The combination of lingual root inclination, a shallow vestibule, and a prominent mental protuberance all can increase the degree of difficulty, as can proximity to adjacent roots and the need for perpendicular root-end resection and preparation to include a possible missed lingual canal.

## Cone-Beam Computed Tomography

Radiographic examination is an essential component in all aspects of endodontic treatment from diagnosis and treatment planning to assessing outcome (see also Chapter 2). Information gained from conventional films and digital periapical radiographs is limited by the fact that the three-dimensional anatomy of the area is compressed into a two-dimensional image. As a result of superimposition, periapical radiographs reveal limited aspects of the three-dimensional anatomy. In addition, there may also be geometric distortion of the anatomic structures imaged. These problems may be overcome using small volume CBCT imaging techniques, which can produce three-dimensional images of individual teeth and the surrounding tissues. CBCT may be particularly useful in diagnosis and treatment planning for periradicular surgery (Fig. 11.9).[112,163,249,318] The term *cone-beam computed tomography* is often used interchangeably with *cone-beam volumetric tomography* (CBVT).

### DIFFERENCES BETWEEN COMPUTED TOMOGRAPHY AND CONE-BEAM COMPUTED TOMOGRAPHY IMAGING

The benefits of three-dimensional medical computed tomography (CT) imaging are already well established in certain dental specialties. Current CT scanners have a linear array of multiple detectors, allowing multiple slices to be taken simultaneously, resulting in faster scan times and

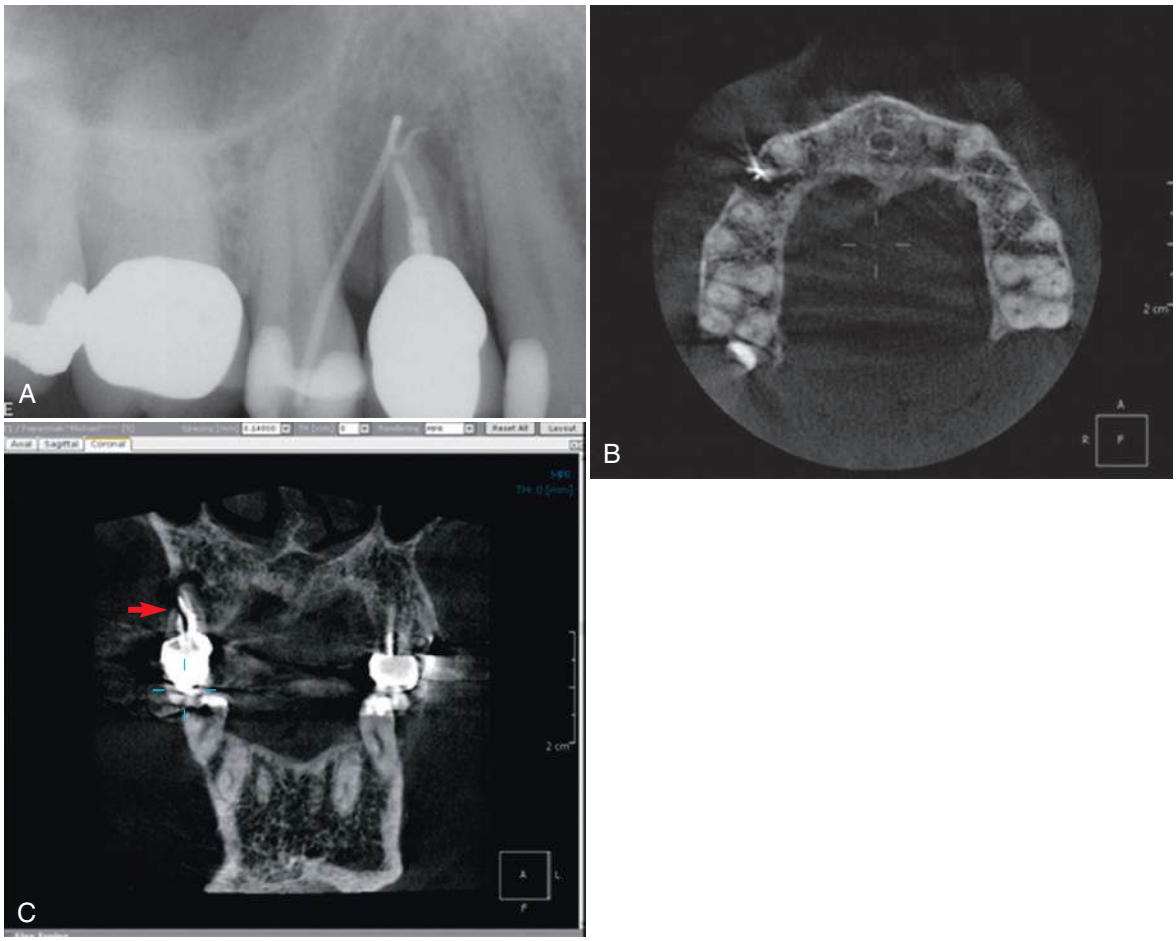



**Fig. 11.9** **A,** Periapical radiograph with gutta-percha tracing the sinus tract to tooth #5. Periodontal probings were within normal limits. This tooth was initially referred for exploratory surgery to rule out root fracture. CBCT confirmed root fracture and the treatment plan was changed to extraction with possible bone graft and guided tissue regeneration for ridge augmentation prior to implant placement. **B,** Axial CBCT view demonstrating extent of lesion. **C,** Coronal CBCT view demonstrating root fracture *(red arrow)* and perforation of buccal cortical plate in area of root fracture but intact bone in the cervical area.

often less radiation exposure to the patient.[507] The slices of data are then "stacked" and can be reformatted to obtain three-dimensional images. The high-radiation dose, cost, availability, poor resolution, and difficulty in interpretation have resulted in limited use of CT imaging in endodontics. These issues may be addressed by cone-beam innovations in CBCT technology.

In 2000, the FDA approved the first CBCT unit for dental use in the United States. Cone-beam technology uses a cone-shaped beam of radiation to acquire a volume in a single 360-degree rotation, similar to panoramic radiography. The volume of acquired images by a CBCT is composed of voxels. Essentially, a voxel is a 3D pixel. Because the data are captured in a volume as opposed to slices, all the voxels are isotropic, which enables objects within the volume to be accurately measured in different directions. Unlike the CBCT voxel, a medical CT voxel is not a perfect cube, and measurements made in multiple planes are not accurate. In addition to increased accuracy and higher resolution, CBCT offers significant scan-time reduction, radiation dose reduction, and reduced cost for the patient.[472,595,606] CBCT systems can be classified into two categories: limited (dental or regional) CBCT or full (ortho or facial) CBCT. The field of view (FOV) of limited CBCT ranges in diameter from 40 to 100 mm, whereas the FOV of full CBCT ranges from 100 to 200 mm.[112] Another difference between the limited CBCT and full CBCT is that a voxel is generally smaller for the limited version (0.1 to 0.2 mm vs. 0.3 to 0.4 mm). Thus limited CBCT systems offer higher resolution and are better suited for endodontic applications. Limited volume CBCT scanners capture small volumes of data that can include just two or three individual teeth. The most important and clinically useful feature of CBCT technology is the highly sophisticated software that allows the huge volume of data collected to be reconstructed. Tomographic slices, as thin as one voxel, may be displayed in a number of ways. One option, for example, is for the images to be displayed in the three orthogonal planes—axial, sagittal and coronal—simultaneously. The axial and proximal (sagittal in the anterior, coronal in the posterior) views are of particular value because they are generally not seen with conventional periapical radiography. The ability to reduce or eliminate superimposition of the surrounding structures makes CBCT superior to conventional periapical radiography.[314]

Scan times are typically 10 to 40 seconds, although the actual exposure time is significantly less (2 to 5 seconds) as scans involve a number (up to 360) of separate, small, individual exposures rather than one continuous exposure. With medical CT scanners, the scanning and exposure times for the skull can be significantly longer. Most CBCT scanners are much smaller than medical CT scanners, taking up about the same space as a dental panoramic machine. They are also significantly less expensive than medical CT scanners. With the help of viewer software, the clinician is able to scroll through the entire volume and simultaneously view axial, coronal, and sagittal 2D sections that range from 0.125 to 2 mm thick. Comparing the radiation dose of different CBCT scanners with medical CT scanners may be confusing because different units of radiation dose are often used. There are three basic measurement units in radiation dosimetry: the radiation absorbed dose (D), the equivalent dose (H), and the effective dose (E). The *radiation absorbed dose* is defined as the measure of the amount of energy absorbed from the radiation beam per unit mass of tissue and is measured in joules per kilogram. The *equivalent dose* is defined as a measure that indicates the radiobiologic effectiveness of different types of radiation and thus provides a common unit. The *effective dose* is calculated by multiplying the equivalent dose by different tissue weighting factors, which converts all doses to an equivalent whole-body dose and allows doses from different investigations of different parts of the body to be compared. The unit remains as the Sievert (Sv) and can be used to estimate the damage from radiation to an exposed population. The reported effective doses from CBCT scanners vary, but it can be almost as low as those from panoramic dental x-ray units and can be considerably less than that from medical CT scanners.[398] The higher effective doses from particular models of CBCT scanners is in part due to the larger size of the FOV used as well as the type of image receptor employed. The small volume scanners capture information from a small region of the jaw, approximately the same area as a periapical radiograph. The effective dose has been reported to be in the same range as two to three standard periapical radiograph exposures, whereas the effective dose for a full mouth series of periapical radiographs has been reported to be similar to the effective dose of a large volume CBCT scan.[124,194] If multiple teeth in different quadrants require endodontic evaluation or treatment, a large volume CBCT scan with a large FOV may be more appropriate. If endodontic information is required from multiple teeth in one jaw, a large volume CBCT scan with the FOV only limited to the jaw of interest could be the investigation of choice. This has the advantage of reducing the effective dosage from the full large volume CBCT scan by up to 65%.[321]

## POTENTIAL APPLICATIONS OF CONE-BEAM COMPUTED TOMOGRAPHY IN THE MANAGEMENT OF ENDODONTIC POSTTREATMENT DISEASE

Clinical radiographic examinations are usually limited to two-dimensional views captured using radiographic film or digital sensors. Crucial information related to the true three-dimensional anatomy of teeth and adjacent structures is obscured. Even with paralleling techniques, distortion and superimposition of dental structures in periapical views is unavoidable. A major reported advantage of CBCT is the accuracy of measurements in all dimensions.[331] The ability to view thin sagittal, coronal, and axial slices effectively eliminates the problem of superimposition of anatomic structures. For example, roots of maxillary posterior teeth and surrounding tissues can be viewed without superimposition of the zygomatic buttress, alveolar bone, maxillary sinus, and other roots (Fig. 11.10). CBCT enables clinicians to detect changes in apical bone density at an earlier stage when compared to conventional periapical radiographs and therefore has the potential to detect previously undiagnosed periradicular pathoses.[163,314,371] CBCT may also prove to be a useful tool for noninvasive differentiation of periapical cysts and granulomas,[489] a previously unattainable goal using conventional radiographic imaging.

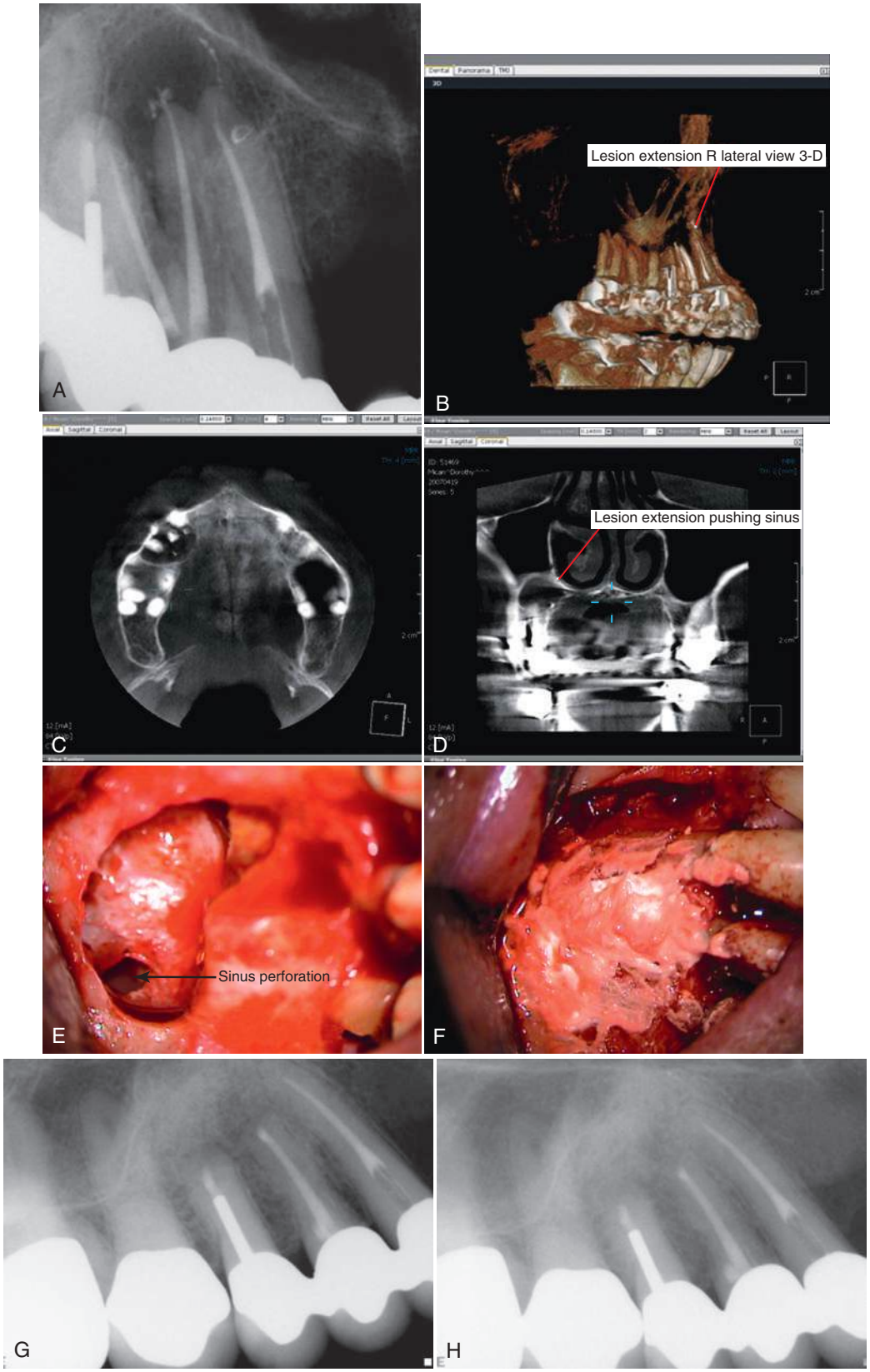



**Fig. 11.10** **A,** Preoperative periapical radiograph showing large radiolucency associated with teeth #4, 5, and 6. **B,** CBCT reconstruction *(lateral view)*. Rotation of the 3D volume on a computer monitor demonstrated that intact bone was present around the apex of tooth #6 and the lesion was associated only with #4 and 5 (this could not be detected with multiple angled periapical radiographs). **C,** Axial CBCT view demonstrating facial-palatal extent of the lesion. **D,** Coronal CBCT view showing displacement of the sinus membrane (a finding that suggests a sinus membrane perforation will be detected during surgery). **E,** Clinical image after surgical access removal of granulation tissue, and root-end resection (note sinus perforation [*arrow*]). **F,** Placement of bone graft and Capset. **G,** Immediate postoperative radiograph. **H,** Two-year follow-up radiograph demonstrating good healing.

Three-dimensional imaging allows for clear identification of the anatomic relationship of the root apices to important neighboring anatomic structures such as the mandibular canal, mental foramen, and maxillary sinus (Fig. 11.11). Velvart and colleagues reported that the relationship of the inferior alveolar canal to the root apices could be determined in every case when utilizing medical CT, but in less than 40% of cases when using conventional radiography.[556] It is likely that similar results could be achieved with CBCT using considerably less radiation. Rigolone and coworkers concluded that CBCT may play an important role in periapical microsurgery of palatal roots of maxillary first molars.[442] The distance between the cortical plate and the palatal root apex could be measured using CBCT and the presence of the maxillary sinus between the roots could be assessed. Additional information such as the thickness of the cortical plate, cancellous bone pattern, fenestrations, and inclination of the roots can be obtained prior to surgical entry.[371] Root morphology (shape, size, curvature, and number of canals) can be visualized in three dimensions. Unidentified (and untreated) canals in root-filled teeth can be routinely visualized in axial slices. CBCT has been used to determine the location and extent of invasive external root resorptive defects.[398]

One of the most useful potential applications of CBCT may be the assessment of endodontic treatment outcomes, both nonsurgical and surgical. When compared to two-dimensional film and digital images, CBCT is dimensionally precise and can detect smaller changes in bone density.[331]

It is worth remembering that CBCT still uses ionizing radiation and is not without risk. It is essential that patient exposure is kept as low as reasonably practical and that justifiable selection criteria for CBCT use are developed. CBCT may be indicated when it is determined that the additional information obtained is likely to result in a more accurate diagnosis and enhanced patient safety.

## Patient Preparation for Surgery

### INFORMED-CONSENT ISSUES SPECIFIC TO SURGERY

The general principles of informed consent discussed in Chapter 27 form the basis for informed consent for periradicular surgery. The patient must be thoroughly advised of the benefits, risks, and other treatment options and must be given an opportunity to ask questions. The main consent issues specific to surgical procedures are closely related to the anatomic considerations discussed in the previous section. That is, major neurovascular bundles may be traumatized, and a sinus exposure may occur. Paresthesia after mandibular posterior surgery is uncommon but should be discussed with the patient because this potential complication is a risk that some patients may be unwilling to assume. Postoperative swelling, bruising, bleeding, and infections are possible complications that typically are self-limiting or readily manageable. Although the incidence of

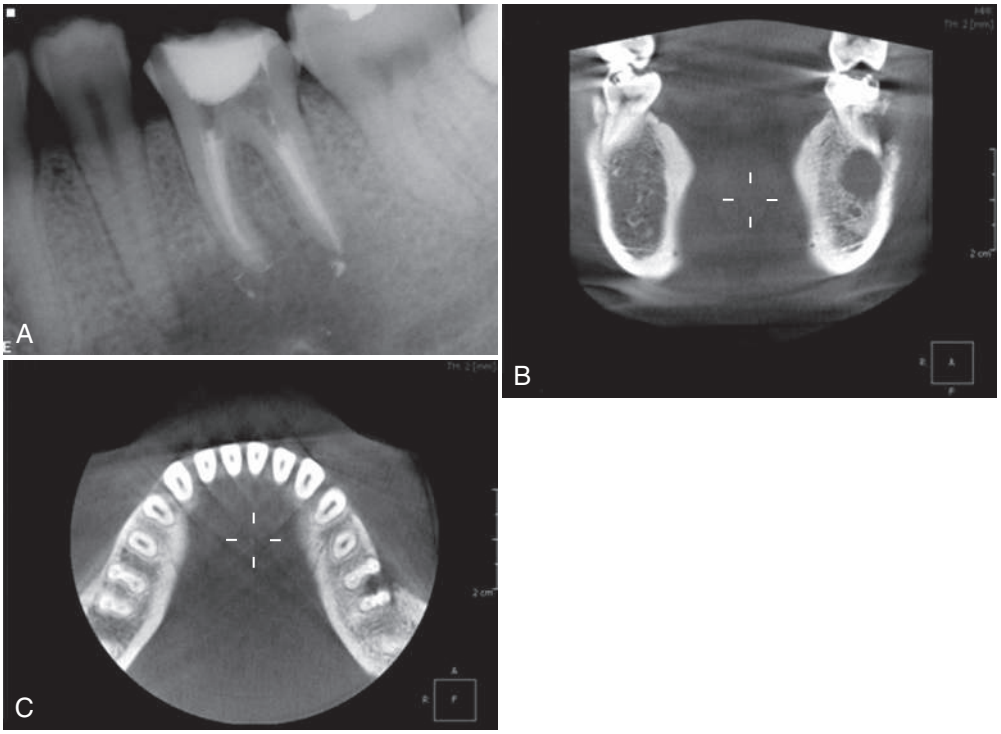



**Fig. 11.11** **A,** Periapical radiograph demonstrating persistent periradicular pathosis following nonsurgical retreatment of tooth #19. The extension of the lesion in the vicinity of the mandibular canal and possible involvement of tooth #20 was uncertain based on interpretation of periapical and panoramic radiographs. Tooth #20 responded WNL to pulp vitality test. **B,** Coronal CBCT demonstrates that the periradicular lesion does not extend apical or lingual to the mandibular canal. **C,** Axial CBCT view demonstrating perforation of the buccal cortical plate and extension into, but not through, the furcation of #19 and presence of intact bone around #20.

serious complications related to surgical procedures is very low, patients should be advised of any risks unique to their situation. Prompt attention to any surgical complications and thorough follow-up are essential from a medical-legal standpoint.

### PREMEDICATION: NONSTEROIDAL ANTIINFLAMMATORY DRUGS, ANTIBIOTICS, CHLORHEXIDINE, AND CONSCIOUS SEDATION

Administration of an NSAID, either before or up to 30 minutes after surgery, enhances postoperative analgesia.[494] NSAIDs generally have proved more effective in the management of postoperative oral surgery pain than placebo or acetaminophen and codeine combinations.[9,41,143] The combination of preoperative administration of an NSAID and use of a long-acting local anesthetic may be particularly helpful for reducing postoperative pain.[144] Many types of NSAIDs are available, but ibuprofen remains the usual standard for comparison. Ibuprofen (400 mg) provides analgesia approximately equal to that obtained with morphine (10 mg) and significantly greater than that from codeine (60 mg), tramadol (100 mg), or acetaminophen (1000 mg).[346] The analgesic effectiveness of ibuprofen tends to level off at about the 400-mg level *(ceiling effect)*, although a slight increase in analgesic potential may be expected in doses up to 800 mg.

The value of antibiotic prophylaxis before or after oral surgery is controversial, and the current best available evidence does not support the routine use of prophylactic antibiotics for periradicular surgery.[11] For most patients, the risks of indiscriminate antibiotic therapy are believed to be greater than the potential benefits.[526] The incidence of infection after oral surgery in healthy patients is very low. Peterson reported that only 1% of patients developed infections after third molar extractions.[355] A systematic review of the use of antibiotics to prevent complications after placement of dental implants found that evidence was lacking either to recommend or discourage the use of antibiotics for this purpose.[162] However, the use of prophylactic antibiotics for more invasive procedures, such as orthognathic surgery, has significantly reduced the risk of postoperative infection and complications.[146] Although routine use of prophylactic antibiotics for periradicular surgery is not currently recommended, clinical judgment is important in determining exceptions to the general rule. For example, immunocompromised patients may be good candidates for prophylactic antibiotic coverage. Certain categories of medically complex patients also may benefit from antibiotic coverage. Diabetic patients have shown impaired healing capacity after nonsurgical root canal treatment,[78,175] and a like pattern of delayed or impaired healing may emerge in studies of surgical outcomes. The global problems associated with overprescription of antibiotics are significant and should prompt due caution in the decision on whether to use these drugs prophylactically.

Chlorhexidine gluconate (0.12%) often is recommended as a mouth rinse to reduce the number of surface microorganisms in the surgical field, and its use may be continued during the postoperative healing stage.[7,281,545] Although no solid evidence supports this practice in periradicular surgery, the use of chlorhexidine follows from the general surgical principle of surface disinfection before incision and opening into a body cavity. In addition, chlorhexidine has proved to be a safe, effective adjunct in the treatment of periodontitis, and short-term use (i.e., several days) poses little or no risk. Chlorhexidine may be useful for reducing the risk of postoperative infection after oral surgery,[67,598] although the evidence in this area is conflicting. Postoperative use of chlorhexidine mouth rinse can reduce bacterial growth on sutures and wound margins,[389] but it may interfere with fibroblast reattachment to the root surface.[14] A useful empirical regimen is to have the patient rinse for 30 seconds twice a day beginning 1 or 2 days before surgery and continuing until the sutures are removed.

Conscious sedation, either by an orally administered sedative or by nitrous oxide/oxygen inhalation analgesia, may be useful for patients who are anxious about the surgical procedure or dental treatment in general. Benzodiazepines with a short half-life are particularly useful because they generally have a wide margin of safety, good absorption after oral administration, and limited residual sedative effects. When these drugs are used in sedative-hypnotic doses, the blood pressure, pulse, and respiration must be monitored, and states differ in their requirement for additional training or certification of staff members who do this monitoring. With oral conscious sedation, pulse oximetry should be used to monitor the pulse and blood oxygen saturation during the surgery.[84] As with all orally administered drugs, the dosage cannot be closely titrated; therefore the effect of the agent varies somewhat. A typical protocol is a single dose at bedtime the evening before the procedure and a second dose 1 hour before the start of surgery. The patient should not drive to or from the office and should have a responsible adult for assistance as needed. In appropriate doses, benzodiazepines and similar drugs may allow for a more relaxed patient and thus a less stressful surgical experience for both patient and surgeon.

## Instruments and Operatory Setup

The development of microsurgical techniques and new materials has changed the typical surgical instrument tray dramatically. Instruments have been designed to take full advantage of the increased visibility obtained with dental operating microscopes, endoscopes, and orascopes. Better visualization of the surgical site would have limited value without microsurgical instruments, such as ultrasonic tips for root-end preparation and micromirrors for inspecting the root end. Fig. 11.12 shows a typical basic surgery tray arrangement. This setup is not a definitive guide to surgical armamentarium but rather an adequate, efficient starting point for most periradicular surgical procedures. Although the number of instruments can easily be doubled or even tripled, the ease of locating a specific instrument is inversely proportional to the number of instruments on a tray. Specialized instruments can be kept readily available in separate sterilized bags or trays and opened as needed. A skilled surgeon can use a wide variety of instruments (Figs. 11.12–11.24) to achieve excellent results.

Periradicular surgery can be performed without the benefit of enhanced magnification and illumination; however, those who use microscopes, endoscopes, and orascopes

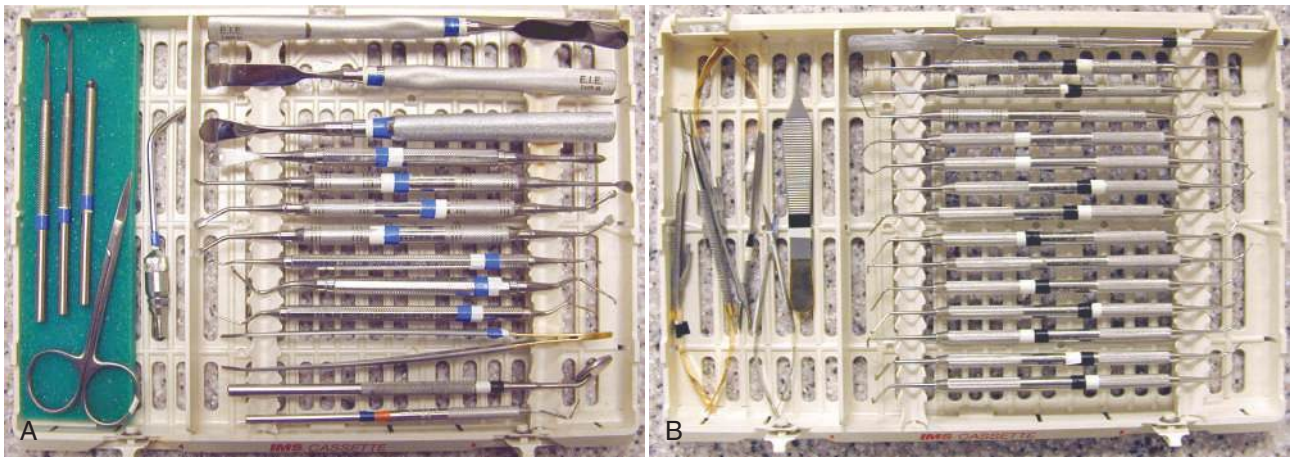


**Fig. 11.12 A,** Basic tray setup for initial surgical access. Surgical instruments shown are distributed by Hu-Friedy ([HF] Chicago), CK Dental Specialities ([CKDS] Orange, CA), EIE ([EIE]), and G. Hartzell & Son ([GHS]). *Left to right* (left section of tray): Small round micromirror (CKDS); medium oval micromirror (CKDS); handle for microscalpel (CKDS); scissors (S18 [HF]); surgical suction tip (GHS). *Top to bottom* (main section of tray): Carr #1 retractor (EIE); Carr #2 retractor (EIE); TRH-1 retractor (HF); periosteal elevator (HF); Ruddle R elevator (EIE); Ruddle L elevator (EIE); Jacquette curette (SJ 34/35 [HF]); spoon curette (CL 84 [HF]); scaler (7/8 [HF]); surgical forceps (TP 5061 [HF]); mouth mirror (HF); periodontal probe (HF). **B,** Instrument tray for root-end filling and suturing. *Left to right* (left section of tray): Two Castroviejo needle holders (Roydent Dental Products, Rochester Hills, MI); Castroviejo scissors (S31 [HF]); micro–tissue forceps (TP 5042 [HF]). *Top to bottom* (main section of tray): Cement spatula (HF); Feinstein super plugger (F1L); microexplorer (CX-1 [EIE]); endoexplorer (DG-16 [EIE]); right Super-EBA Placing & Plugging instrument (MRFR [HF]); left Super-EBA Placing & Plugging instrument (MRFL [HF]); small anterior microburnisher and plugger (HF); small left microburnisher and plugger (HF); small right microburnisher and plugger (HF); medium anterior microburnisher and plugger (HF); medium left microburnisher and plugger (HF); medium right microburnisher and plugger (HF); large anterior microburnisher and plugger (HF); large left microburnisher and plugger (HF); large right microburnisher and plugger (HF).



**Fig. 11.13** Comparison of microsurgical scalpel *(top)* to #15C surgical blade. Microsurgical scalpels are particularly useful for the intrasulcular incision and for delicate dissection of the interproximal papillae.

report dramatically improved visualization and control of the surgical site.[39,92,280,350] Studies reporting the highest surgical success rates and those cited previously in this chapter all involved some form of enhanced magnification and illumination as part of the standard operating protocol. A systematic review and meta-analysis found that the outcome for endodontic surgery performed with high magnification and illumination (microscope or endoscope) was significantly higher than surgery performed with loupes only or no magnification.[482]

## Local Anesthesia for Surgery

Local anesthesia for surgical root canal procedures differs from that for nonsurgical root canal treatment, primarily in the need for localized hemostasis in addition to profound local anesthesia. In fact, use of a local anesthetic with a vasoconstrictor may be the single most important local measure to help control hemorrhage and provide a clear surgical field. Otherwise, the same regional block and local infiltration techniques used for nonsurgical treatment are used for surgical root canal procedures. Infiltration of the surgical site with a local anesthetic containing 1:50,000 epinephrine is the technique of choice to obtain vasoconstriction and hemostasis.[280] The local anesthetic is first slowly deposited in the buccal root apex area of the alveolar mucosa at the surgical site and extended two or three teeth on either side of the site. Usually palatal or lingual infiltration is also required, although this requires a much smaller amount of local anesthetic than the primary buccal infiltration. After the injections for anesthesia, the surgeon should wait at least 10 minutes before making the first incision.

Long-acting local anesthetics (e.g., 0.5% bupivacaine with 1:200,000 epinephrine) have been shown to reduce postoperative pain and analgesic use after surgical removal of impacted third molars.[125,204] However, use of a local anesthetic with 1:200,000 epinephrine may result in greater blood loss during surgery.[234,493] To maximize postoperative analgesia and minimize intraoperative bleeding, a local anesthetic can be used with higher epinephrine concentrations (1:100,000 or 1:50,000) for the primary surgical anesthesia and supplemented with one cartridge of long-acting local anesthetic immediately after surgery. Long-lasting local anesthetics are particularly beneficial in mandibular surgery but much less so for surgery in the maxillary arch.

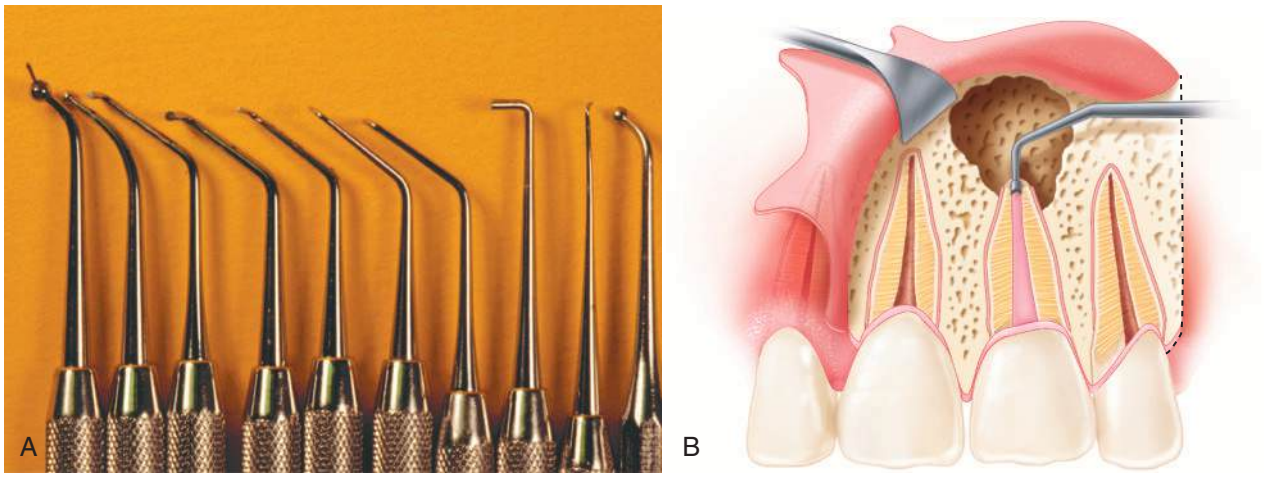


**Fig. 11.14** **A,** Microcondensers in assorted shapes and sizes for root-end filling. **B,** The microcondenser should be selected to fit the root-end preparation.



**Fig. 11.15** Comparison of standard #5 mouth mirror *(top)* to diamond-coated micromirrors (CK Dental Specialties).



**Fig. 11.17** Retractors used in periradicular surgery. *Top to bottom,* EHR-1, ER-2, and ER-1 (equivalent to Carr #2 and #1 retractors) (CK Dental Specialties).



**Fig. 11.16** Micromirror used to inspect resected mesial root of a mandibular first molar.

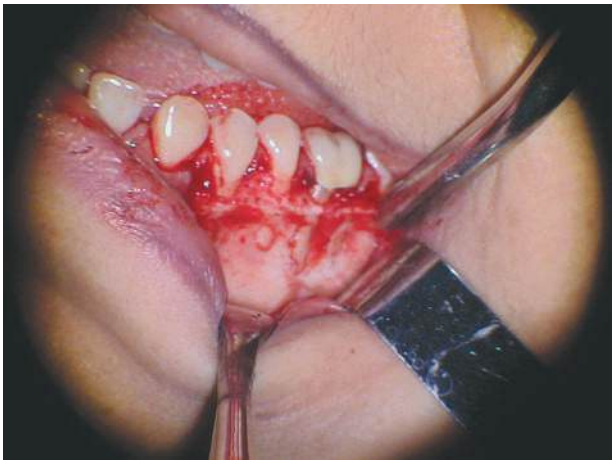


**Fig. 11.18** Retractors positioned to expose the surgical site and protect adjacent soft tissues from injury. Care must be taken to rest the retractors only on bone, not on the reflected soft-tissue flap or on the neurovascular bundle as it exits the mental foramen.



**Fig. 11.19** Teflon sleeve and plugger specially designed for placement of mineral trioxide aggregate (DENTSPLY Tulsa Dental Specialties).

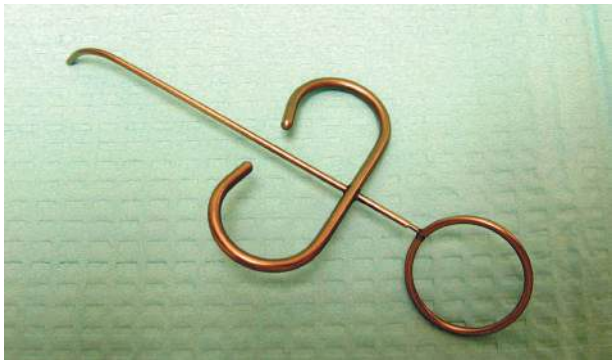


**Fig. 11.20** Messing gun–type syringe (CK Dental Specialties) can be used for placement of various root-end filling materials.

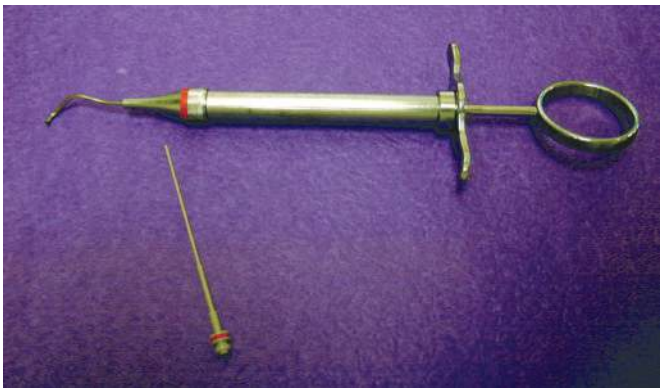


**Fig. 11.21** Another delivery system designed specifically for mineral trioxide aggregate placement (Roydent). Kit includes a variety of tips for use in different areas of the mouth and a single-use Teflon plunger.

Every effort must be made to ensure profound local anesthesia before surgery begins. Usually a minimum of 10 to 20 minutes is required from the time of injection to the start of surgery to ensure both profound local anesthesia and adequate vasoconstriction for hemostasis. The patient should be asked about the usual signs of soft-tissue anesthesia, and a sharp explorer can be used to test the surgical area for sensation. Even when the surgeon pays careful attention to local anesthetic technique, a patient sometimes has inadequate anesthesia or loss of anesthesia during the

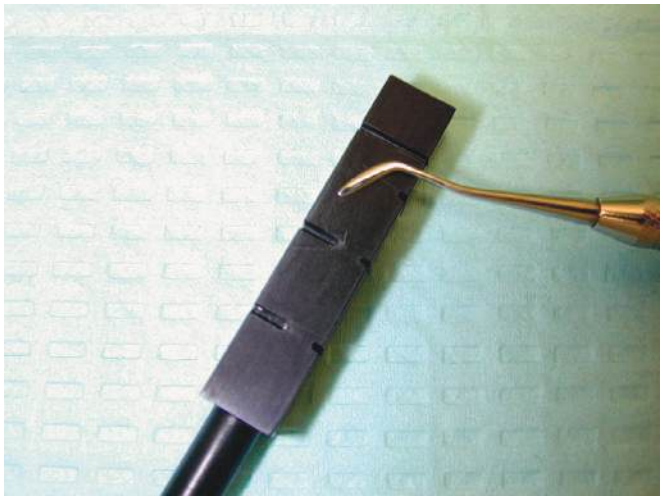


**Fig. 11.22** Hard plastic block with notches of varying shapes and sizes (G. Hartzell & Son). MTA is mixed on a glass slab to the consistency of wet sand and then packed into a notch. The applicator instrument is used to transfer the preformed plug of MTA from the block to the root end.

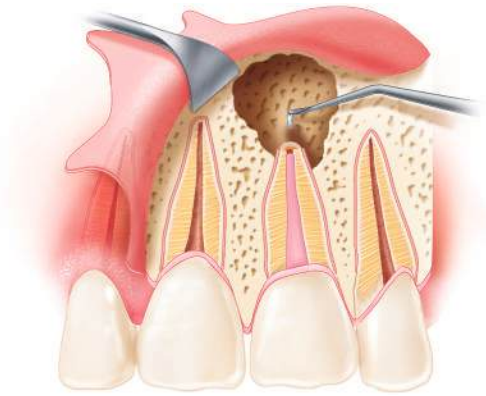


**Fig. 11.23** Placement of a root-end filling.



**Fig. 11.24** Surgeon, assistant, and patient positioned for initiation of surgery. The patient should be given tinted goggles or some other form of eye protection before the procedure begins.

surgical procedure. Providing supplemental infiltration anesthesia is difficult after a full-thickness flap has been reflected. A supplemental block injection may be useful for mandibular teeth and maxillary posterior teeth. In the maxillary anterior area, a palatal approach to the anterior middle superior nerve may be helpful. The key to this approach is slow injection of approximately 1 mL of local anesthetic in the area of the first and second maxillary premolars midway between the gingival crest and the palatal midline. An intraosseous injection also may be used to regain lost anesthesia, but even when it is effective, the area of local anesthesia often is smaller than desired for a surgical procedure. As a last resort, the procedure can be terminated short of completion and the patient can be rescheduled for surgery under sedation or general anesthesia.

## Surgical Access

The goals of periradicular surgery are to access the affected area, remove the diseased tissue, evaluate the root circumference and root canal system, and place a biocompatible seal in the form of a root-end filling that can stimulate regeneration of the periodontium. The formation of new cementum on the surgically exposed root surface and on the root-end filling material is essential to regeneration of the periodontium.

Successful endodontic surgery requires the surgeon to use several conceptual elements in planning the procedure. A vision of the immediate postoperative surgical end point (i.e., replacement of the reflected tissues) is essential for designing each phase of the surgery. Visualization of a three-dimensional image of the surgical procedure allows the surgeon to anticipate and prepare for unusual circumstances. In surgical root canal treatment, once a procedure is started, it must be completed uninterrupted within a limited time; this is a paramount difference between nonsurgical and surgical root canal treatment. For this reason, it is absolutely essential that the surgeon plan the procedure thoroughly and include alternative plans of action that anticipate unusual findings during the process of discovery.

Several general principles are important for designing the access to a diseased region: (1) the surgeon must have a thorough knowledge of the anatomic structures in relation to each other, including tooth anatomy; (2) the surgeon must be able to visualize the three-dimensional nature of the structures in the soft and hard tissue (this reduces unnecessary tissue damage); (3) the trauma of the surgical procedure itself must be minimized, which includes the preservation of the tooth and supporting structures; and (4) the tissue and instruments must be manipulated within a limited space with the aim of removing diseased tissues and retaining healthy tissues.

### SOFT-TISSUE ACCESS

In designing the soft-tissue access window to the diseased tissue, the surgeon must take into consideration various anatomic features, such as frenum-muscle attachments, the width of attached gingiva, papillary height and width, bone eminence, and crown margins. The supraperiosteal blood vessels of the attached gingiva extend from the alveolar mucosa and run parallel to the long axis of the teeth, lying in the reticular layer superficial to the periosteum.[172] A vertical, rather than an angled, releasing incision severs fewer vessels,[325] reducing the possibility of hemorrhage. Also, the blood supply to the tissue coronal to the incision is not compromised,[477] which prevents localized ischemia and sloughing of these tissues. Ultimately, the result is less bleeding during the procedure and enhanced healing. For these reasons, an angled releasing incision is contraindicated in periradicular surgery.

### VERTICAL INCISION

The general principles for placement of a vertical relieving incision are as follows:

1. The incision should be made parallel to the supraperiosteal vessels in the attached gingiva and submucosa (Fig. 11.25).
2. No cuts should be made across frenum and muscle attachments.
3. Frenum and muscle attachments should not be located in the reflected tissue, if possible.
4. The incision should be placed directly over healthy bone.
5. The incision should not be placed superior to a bony eminence.
6. The dental papilla should be included or excluded but not dissected.
7. The incision should extend from the depth of the vestibular sulcus to the midpoint between the dental papilla and the horizontal aspect of the buccal gingival sulcus.

### HORIZONTAL INCISION

Three types of horizontal incisions can be used to gain access to a surgical site in hard tissue:

- *An intrasulcular incision that includes the dental papilla.* This incision extends from the gingival sulcus through the PDL fibers and terminates at the crestal bone of the alveolar bone proper. The incision then passes in a buccolingual direction adjacent to each tooth of the dental papilla and includes the midcol region of each dental papilla. Thus the entire dental papilla is completely mobilized.
- *An intrasulcular incision that excludes the dental papilla (papillary-based incision,* Fig. 11.26). This technique consists of a shallow first incision at the base of the papilla and a second incision directed to the crestal bone.

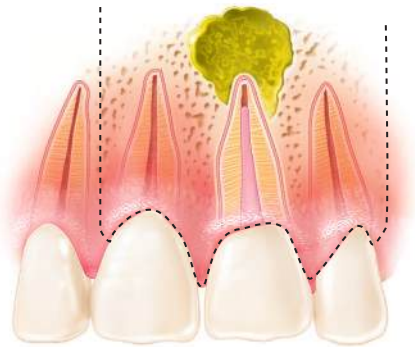


**Fig. 11.25** Intrasulcular incision with two vertical releasing incisions (rectangular flap).

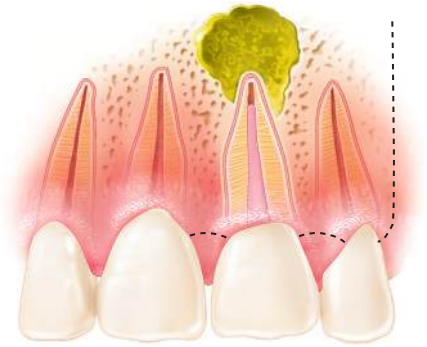


**Fig. 11.26** Papillary-based incision with one vertical releasing incision.

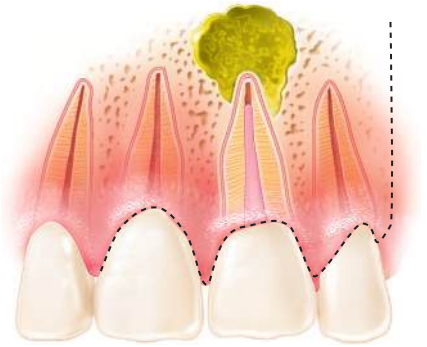


**Fig. 11.27** Intrasulcular incision with one vertical releasing incision (triangular flap).

- *An incision made in the attached gingiva (a submarginal or Ochsenbein-Luebke flap).*[322] With this technique, at least 2 mm of attached gingiva must be retained to prevent mucogingival degeneration.[296] Consequently, the incision must be placed at least 2 mm from the depth of the gingival sulcus. Extensive periodontal probing should be done to establish the depth of the gingival sulcus before the incision is made. The average width of the attached gingiva is 2.1 to 5.1 mm in the maxilla and 1.8 to 3.8 mm in the mandible.[10,199,341,563] It is widest over the central and lateral incisors, narrows over the canine and the first premolar, and then widens over the second premolar and the first molar. These variances were similar for both the maxilla and mandible.[517] Overall, this incision technique has a narrow margin of safety. It generally is recommended for use in the maxilla, especially where the aesthetics of existing crown margins are a concern.

Studies have compared the incision techniques that include and exclude the dental papilla in patients with healthy marginal periodontal conditions.[554,555,557] These researchers found that the papilla base incision resulted in rapid, recession-free healing. In contrast, complete mobilization of the papilla led to a marked loss of papilla height. The authors suggested that use of the papilla base incision in aesthetically sensitive regions could help prevent papilla recession and surgical cleft, or double papilla.

## FLAP DESIGN

Combinations of vertical and horizontal incisions are used to achieve various flap designs (Figs. 11.27 and 11.28). The full mucoperiosteal and limited mucoperiosteal are the two major categories of flap design used during periradicular surgery, the main differentiating feature being the position of the horizontal incision. In each case the entire body of soft tissue is reflected as one unit and includes the alveolar mucosa, the gingival tissues, and the periosteum. The number and position of the vertical relaxing incisions therefore governs the major variation in design:

1. Full mucoperiosteal (intrasulcular incision including the dental papilla or papillary based) (see Figs. 11.27 and 11.28):
   a. Triangular: one vertical relieving incision
   b. Rectangular: two vertical relieving incisions
   c. Trapezoidal: two angled vertical relieving incisions
   d. Horizontal: no vertical relieving incision



**Fig. 11.28** Reflection of the triangular flap to expose the root-end area.

2. Limited mucoperiosteal (Figs. 11.29 and 11.30):
   a. Curve submarginal (semilunar)
   b. Freeform rectilinear submarginal (Ochsenbein-Luebke)

## TISSUE REFLECTION

Elevation and reflection of the entire mucoperiosteal complex, maintaining the microvasculature in the body of the tissue flap, increases hemostatic control during surgery. Tissue reflection should begin from the vertical releasing incision at the junction of the submucosa and the attached gingiva (Figs. 11.31 and 11.32). By initiating the reflection process at this point, damage to the delicate supracrestal root-attached fibers is avoided. Using a flap

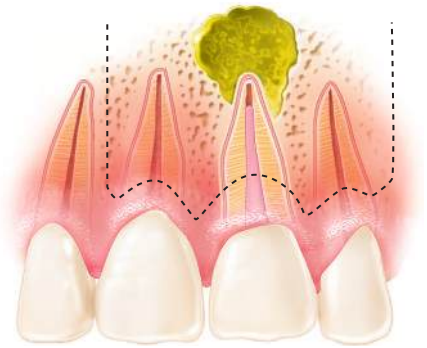


**Fig. 11.29** Submarginal (Ochsenbein-Luebke) flap.

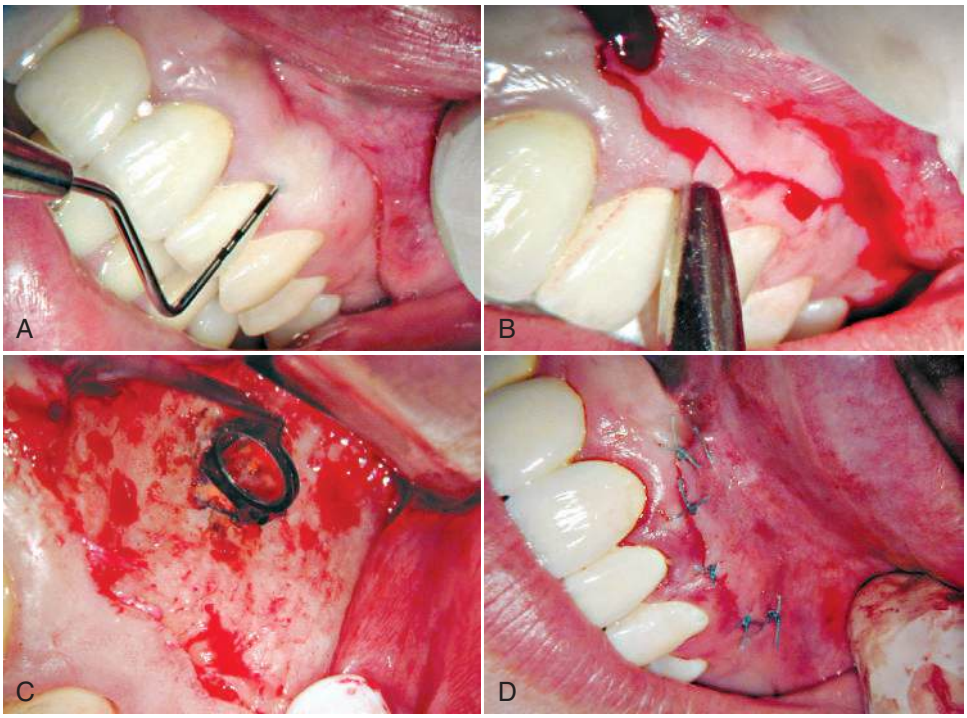



**Fig. 11.30** Clinical case involving a submarginal incision and flap. **A,** Periodontal probing of the entire area was performed before this type of flap was selected and the incision started. A submarginal incision often is used in aesthetic anterior areas of the mouth where postoperative gingival recession might expose the crown margins. **B,** The incision should be at least 2 mm apical to the depth of the sulcus. **C,** With the flap reflected, osteotomy and root-end resection were performed. Methylene blue dye was placed to mark the outline of the root end and to help identify cracks or fractures before root-end cavity preparation and filling. **D,** The flap was repositioned and sutured with 5-0 Tevdek. (Courtesy Dr. Martin Rogers.)

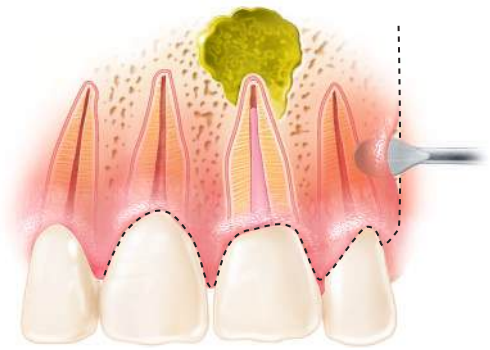


**Fig. 11.31** Elevator placed in the vertical incision for the first step in undermining flap reflection.

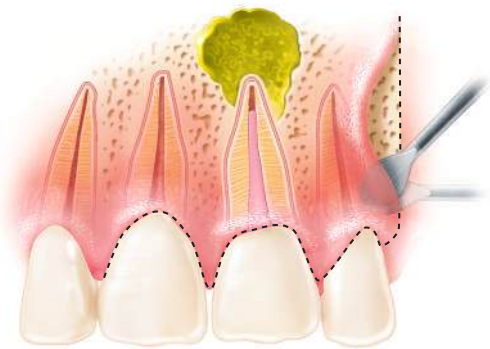


**Fig. 11.32** Continuation of reflection of full-thickness flap.

reflection technique that reduces reflective tissue forces in the intrasulcular incisional wound and avoids curettage of the root surface conserves the root-attached tissues and helps prevent apical down-growth of epithelium and loss of soft-tissue attachment.[222] Force should be applied such that the periosteum and superficial tissues are reflected as a complete unit. Using a gentle rocking motion, the surgeon initially should reflect the tissue in a horizontal direction.[213] The underlying bone of the cortical plate is irregular, and it is critical to avoid damaging the fragile tissues during elevation.

The operator should take great care not to slip during the tissue reflection process by using an appropriate instrument stabilized with adequate finger support. Slipping can result not only in a puncture of the immediate overlying tissue but also in damage to the surrounding structures.

As space permits, the elevator should be directed coronally, undermining the attached gingiva. As the interdental papilla is approached, a narrower instrument may be required to undermine and gently elevate the tissue in this region, to avoid crushing the delicate free gingival tissues. This process should be continued gradually until the osseous tissues overlying the diseased tooth structure have been adequately exposed. Generally, elevation of the flap 0.75 cm apical to the estimated apex of the root should allow adequate space to perform the surgical procedure.

No single instrument is essential for the flap elevation procedure, because every instrument has both advantages and disadvantages. Surgeons should familiarize themselves with the various instruments available.

### TISSUE RETRACTION

After the tissue is reflected, it must be retracted to provide adequate access for bone removal and root-end procedures. The main goals of tissue retraction are to provide a clear view of the bony surgical site and to prevent further soft-tissue trauma. Accidental crushing of the soft tissues leads to more postoperative swelling and ecchymoses.[222,223]

The general principles of retraction are as follows: (1) retractors should rest on solid cortical bone; (2) firm but light pressure should be used; (3) tearing, puncturing, and crushing of the soft tissue should be avoided; (4) sterile physiologic saline should be used periodically to maintain hydration of the reflected tissue; and (5) the retractor should be large enough to protect the retracted soft tissue during surgical treatment (e.g., prevent it from becoming entangled in the bone bur). No one retractor suffices for all surgical procedures, therefore the surgeon should have a selection of retractors available for the various situations that arise during surgery. If difficulty is encountered in stabilizing the retractor, a small groove can be cut into the cortical plate to support it.

### HARD-TISSUE ACCESS

Two biologic principles govern the removal of bone for hard-tissue access to diseased root ends: (1) healthy hard tissue must be preserved, and (2) heat generation during the process must be minimized.

Temperature increases above normal body temperature in osseous tissues are detrimental. Heating osseous tissue to 117° to 122°F (47° to 50°C) for 1 minute significantly reduces bone formation and is associated with irreversible cellular damage and fatty cell infiltration.[161,338] Two critical factors determine the degree of injury: how high the temperature increases and how long it remains elevated. As the temperature rises above 104°F (40°C), blood flow initially increases. It stagnates at 198°F (46°C) applied for 2 minutes. Heating osseous tissue to 133°F (56°C) deactivates alkaline phosphatase.[422,423] Studies using animal bone have shown that at temperatures above 109°F (42.5°C), for every 1°C elevation in temperature, the exposure time for the same biologic effect decreases by a factor of approximately 2.[158-161] Temperatures above 117°F (47°C) maintained for 1 minute produce effects similar to those at 118°F (48°C) applied for 30 seconds. This correlation means that the decisive exposure time declines quickly as the temperature increases. Temperatures above 127°F (53°C) applied for less than 1 second can adversely affect osteogenesis.[158-161]

Several factors determine the amount of heat generated during bone removal, including the shape and composition of the bur, the rotational speed, the use of coolant, and the pressure applied during cutting.

The round bur has the best shape for removing osseous tissue, and it should be used with a gentle brushstroke action.[519] This type of bur also readily allows access of coolant to the actual cutting surfaces. Studies comparing the heat generated with round and fissure burs found more favorable results with the round burs.[85,111,216,338,360] Cutting with round burs produced a wound site with less inflammation, which is more favorable for rapid wound healing. Although fissure-type burs cut efficiently on the sides, the tip of the bur is very inefficient because it allows no coolant access. The net result is increased inflammation and a reduced healing response.

Use of a diamond bur to remove osseous tissue is inefficient and slows ultimate wound healing. Because of its larger surface area, more of a diamond bur is in contact with the bone tissue. As a result, less coolant reaches the cutting surface, and the bur has a greater tendency to become clogged with residual bone fragments. The net effect is greater heat generation, increased inflammation, and reduced healing.[85,586]

Use of a coolant during bone cutting is essential. If an appropriate irrigant is not used, temperatures can exceed those known to impair bone healing[274]; histologically, healing can be delayed up to 3 weeks.[169] It also is critical that the coolant reach the cutting surface. Temperatures can rise above 212°F (100°C) when excess pressure is applied during cutting; this burrows the bur into the bone, where little or no irrigant can reach the cutting tip,[519] hence the recommendation for a gentle brush-stroke technique.[213] Favorable results are obtained with these provided the surgeon follows the basic tenet of minimizing heat generation: using a round fluted bur with coolant and a brushstroke technique. A high-speed handpiece that exhausts air from the base rather than the cutting end is recommended to reduce the risk of air embolism (Fig. 11.33).

## Periradicular Curettage and Biopsy

Most periradicular lesions originate in the pulp and can be classified histopathologically as granulomas or cysts.[53,58,292,295,297,359,367,382,488] Histologically, such lesions consist mainly of granulation tissue associated with angiogenesis, fibroblasts, connective tissue fibers, and inflammatory cells. Foreign material, cholesterol clefts,[365,370] and stimulated strands of epithelium also may be present. The stimulated epithelium can form into a stratified, squamous epithelium–lined cystic cavity.[366] These periradicular lesions (granulomas and cysts) are inflammatory lesions that



**Fig. 11.33** Surgical handpiece with 45-degree angle head and rear air exhaust (Impact Air 45).

develop in response to irritation caused by intraradicular and extraradicular microorganisms associated with the root canal system[363,368] or by foreign materials forced into the periradicular tissues.[600]

An important aspect of periradicular surgery is the removal of diseased tissue associated with the root apex. Because a large portion of this tissue is reactionary, the focus of surgical root canal treatment is removal of the irritant or diseased tissues. Histologically, an inflammatory periradicular lesion is similar to healing granulation tissue. If the irritant can be readily identified and successfully eliminated, it is not always necessary to completely curette all the inflamed periradicular tissues during surgery.[306] This is especially true when complete removal might result in injury to neural or vascular tissues. In addition to removing diseased tissue, periradicular curettage provides visibility and accessibility to facilitate treatment of the apical root canal system or the removal of foreign materials in the periradicular tissues.

The need for a histopathologic assessment of all tissues removed from the body cannot be overstated. Although only a small percentage of periradicular lesions are associated with pathoses other than a periradicular cyst or granuloma, all lesions must be diagnosed definitively because of the potential gravity of the few rare diseases associated with periradicular lesions.[12,121,185,383,409]

The technical aspect of removing soft tissue from the bony crypt varies among surgeons and clinical settings. Various bone and periodontal curettes are available for this purpose, and no one instrument suffices for all cases. Regardless of the instrument selected, the basic principles are the same. A sharp instrument is always preferable to a blunt instrument. The soft-tissue lesion first should be peeled away from the osseous crypt, starting at the lateral borders. This can be accomplished efficiently by using the curette with the concave surface facing the internal wall of the osseous crypt. Once the soft-tissue lesion has been separated from the osseous crypt to the point where the crypt changes its convexity, the curette can be used in a scraping manner to remove the remainder of the lesion from the medial wall of the osseous defect.

## Localized Hemostasis

Localized hemostasis during periradicular surgery is essential to successful management of the resected root end. Appropriate hemostasis during surgery minimizes surgical time, surgical blood loss, and postoperative hemorrhage and swelling.[213] The hemostatic agents used during endodontic surgery are intended to control bleeding from small blood vessels or capillaries. Localized hemorrhage control not only enhances the visibility and assessment of the root structure but also ensures the appropriate environment for placement of the current root-end filling materials and minimizes root-end filling contamination.

Many hemostatic agents have been advocated for use during surgery, and the action of these agents, their ability to control bleeding, and their effect on healing vary considerably. They generally aid coagulation by inducing rapid development of an occlusive clot, either by exerting a physical tamponade action or by enhancing the clotting mechanism and vasoconstriction (or both). No one local hemostatic agent is ideal; each has disadvantages. Therefore further investigation is indicated to find the ideal local hemostatic agent.

### PREOPERATIVE CONSIDERATIONS

A thorough review of the patient's body systems and medical history increases the likelihood of detecting an undiagnosed condition that might affect hemostasis during periradicular surgery. Review of the patient's medications, both prescribed and OTC drugs, is essential. Many OTC drugs can affect the clotting mechanism. The patient's vital signs (i.e., blood pressure, heart rate, and respiratory rate) should be assessed. Vital signs also can be used to monitor anxious patients. An increase in blood pressure and heart rate above a patient's known normal values indicates increased stress or poorly controlled hypertension. Easing the patient's anxiety before surgery reduces the possible hemostatic-potentiating effect of elevated cardiac output during surgery.[156] Anxiety and stress can be alleviated with planning, sedation, and profound local anesthesia.

### LOCAL HEMOSTATIC AGENTS

#### Collagen-Based Materials

Various collagen-based hemostatic agents are available for use as local hemostatic agents. The principal differences are in the microstructure and density of the collagen. Collagen can act as a mild allergen, but the problem of allergenization and unwanted tissue reaction does not occur when highly purified animal collagen is used.[49] The mechanisms by which collagen products help achieve hemostasis involves stimulation of platelet adhesion, platelet aggregation and release reaction,[270,271] activation of factor XII (Hageman factor),[335,336] and mechanical tamponade by the structure that forms at the collagen-blood wound interface. Collagen shows minimal interference in the wound healing process, with a limited foreign body reaction.[215] It does not increase the incidence of infection and only slightly delays early bone repair.[246] Osseous regeneration in the presence of collagen typically proceeds uneventfully, without a foreign body reaction.[168]

Collagen-based materials can be difficult to apply to the bony crypt because they adhere to wet surfaces, particularly instruments and gloves.[466] Several collagen-based products are commercially available. They include CollaCote (Integra Life Sciences, Plainsboro, NJ) (Fig. 11.34, *A*), CollaStat (American Medical Products Corp, Eatontown, NJ), Hemocollagene (Septodont, United Kingdom) and Instat (Ethicon, Somerville, NJ). These materials act in essentially similar ways, and the surgical area undergoes a similar healing pattern.[485,504] Overall, studies of wound healing with collagen-based hemostatic agents have shown favorable results.

#### Surgicel

Surgicel (Ethicon) is a chemically sterilized material prepared through oxidation of regenerated alpha cellulose (oxycellulose). The basic element of Surgicel is polyanhydroglucuronic acid, which is spun into threads and then




**Fig. 11.34 A,** Absorbable collagen (CollaCote) is a convenient, biocompatible packing material for localized hemostasis. **B,** Cotton pellets impregnated with racemic epinephrine (Racellet) also may be used for localized hemostasis.

woven into gauze. Surgicel has a pH of 3. If the material is maintained in the wound for up to 120 days, a pH this low could slow healing.[408] It is primarily a physical hemostatic agent, which acts as a barrier to blood and then becomes a sticky mass that serves as an artificial coagulum. It does not enhance the clotting cascade through adhesion or aggregation of platelets. Surgicel is retained in the surgical wound,[373] and healing is retarded, with little evidence of resorption of the material at 120 days.[57] Use of Surgicel in extraction sockets resulted in greater postoperative pain compared with a control in a split mouth–designed study.[408]

### Gelfoam

Gelfoam (Pharmacia, Peapack, NJ) is a gelatin-based sponge that is water insoluble and biologically resorbable. It stimulates the intrinsic clotting pathway by promoting platelet disintegration and the subsequent release of thromboplastin and thrombin.[164] The initial reaction to Gelfoam in the surgical site is a decrease in the rate of healing. Extraction sockets containing Gelfoam showed a greater inflammatory cell infiltrate, marked reduction in bone ingrowth, and a foreign body reaction at 8 days.[74] However, these effects were transitory and did not impair long-term bone healing.[392]

### Bone Wax

Historically, bone wax has been advocated for controlling both hemostasis and debris in the bony crypt during periradicular surgery.[480] It is a nonabsorbable product composed of 88% beeswax and 12% isopropyl palmitate. Healing with bone wax is best described as poor. The bony crypt typically contains fibrous connective tissue and has no bony or hematopoietic tissue. Bone wax retards bone healing and predisposes the surgical site to infection[117,378] by producing a chronic inflammatory foreign body reaction[35] and impairing the clearance of bacteria.[263] The use of bone wax can no longer be recommended because it impairs healing and several good alternatives are available.[589]

### Ferric Sulfate

Ferric sulfate (Cut-Trol, Ichthys Enterprises, Mobile, AL), a necrotizing agent with an extremely low pH, is one of the few products investigated for use in periradicular surgery. In studies using a rabbit model, Lemon and colleagues[301] and Jeansonne and colleagues[256] reported hemostatic control for 5 minutes and near normal healing with only a mild foreign body reaction, provided the surgical wound was adequately curetted and irrigated with saline. Failure to remove ferric sulfate from the surgical wound site resulted in severely impaired healing, a foreign body-type reaction, and, in some cases, abscess formation. The possibility of acute inflammation and necrosis of the surrounding soft tissue with careless use of this solution should not be underestimated.[298] A similar product, Monsel solution (ferric subsulfate), has been used to control local hemostasis in dermatologic procedures. However, the popularity of this solution has declined because application to wound sites has resulted in tissue necrosis for up to 2 weeks,[130] differences in the degree of epidermal maturation, and tattoo formation.[536]

### Calcium Sulfate

Calcium sulfate has been used as a substitute bone graft material to fill bone defects since the late 1800s. The presence of calcium sulfate in an osseous wound does not inhibit bone formation.[574] It is gradually removed from the site of implantation regardless of whether new bone has formed.[110] Use of calcium sulfate during periradicular surgery does not significantly affect healing, and deposition of cementum and osseous healing proceed normally.[27] As a hemostatic agent, calcium sulfate acts as a physical barrier. The material is placed in the bony crypt, allowed to set, and then partly carved away to allow access to the root end.[283] The remaining material lines the crypt walls, preventing bleeding. When the root-end filling has been placed and all extraneous root-end filling material removed, the residual calcium sulfate can be removed or left in situ.

### Epinephrine Pellets

Epinephrine, a sympathomimetic-amine vasoconstrictor, is frequently used to control hemorrhage during oral surgery.[82,283] All granulation tissue should be removed from the root apex area prior to placement of the epinephrine pellet to assure direct contact with bone.[282] Vasoconstrictive amines exert their effects by binding to and interacting with adrenergic receptors in various body tissues. When epinephrine is bound to alpha$_1$- and alpha$_2$-adrenergic receptors, a powerful vasoconstricting effect results. Racemic epinephrine cotton pellets (Racellet #3; Pascal Co., Bellevue, WA) contain an average of 0.55 mg of racemic epinephrine hydrochloride per pellet, half of which is the pharmacologically active L-form. Racellets (see Fig. 11.31, *B*) provide good localized hemostasis in periradicular surgery.[558]

Two concerns arise with the use of Racellets in the surgical site: the cardiovascular impact of the additional epinephrine and the retention of cotton fibers in the wound, resulting in impaired wound healing.[213,265] A study by Vickers and coworkers[558] examined the cardiovascular effects of epinephrine pellets and concluded that no evidence existed of cardiovascular changes (blood pressure and pulse) compared with saline-saturated pellet controls. These authors hypothesized that the vasoconstrictive effect on the capillaries is localized and immediate and that little or no systemic uptake of epinephrine occurred.

Although the cardiovascular effect appears to be of little concern, retention of cotton fibers in the surgical site could result in inflammation and impaired wound healing. Therefore due diligence on the part of the surgeon is paramount when epinephrine pellets are used. Each pellet applied during surgery must be accounted for, and the bony crypt should be lightly curetted to remove any embedded cotton fibers before the wound is closed. The retention of cotton fibers in the crypt can be eliminated by substituting CollaCote saturated with 10 drops of 2.25% racepinephrine inhalation solution.[567] As previously noted, CollaCote is biocompatible and does not interfere with wound healing.

### CAUTERY/ELECTROSURGERY

Cautery stops the flow of blood through coagulation of blood and tissue protein, leaving an eschar that the body attempts to slough.[537] The effect of cautery in the bony crypt during periradicular surgery has not been studied to date. However, the effect of electrosurgery on alveolar bone has been studied in periodontal surgery. Tissue destruction was greater in areas exposed to electrosurgery and healing was delayed compared with surgical sites not exposed to electrosurgery. Twelve hours after surgery, a more extensive inflammatory reaction and greater destruction of periosteum were noted with electrosurgery.[36] At 24 hours, many empty lacunae were observed in the bone associated with electrosurgery, and this necrosis was even more extensive by 48 hours. At 96 hours the electrosurgical connective tissue wounds were still lined by coagulum, whereas the scalpel wound was beginning to repair.[381] The detrimental effect of applying heat to bone is proportional to both temperature and the duration of application.

## Management of the Root End

Management of the resected root end during periradicular surgery is critical to the overall success of a case. The aim of surgery should be to create an environment conducive to regeneration of the periodontium—that is, healing and regeneration of the alveolar bone, PDL, and cementum overlying the root end and root-end filling material. The key to regeneration is the presence of appropriate inducible cell types, growth factors, and specific substances necessary for mineralization. Failure to create an environment conducive to this process results in tissue repair rather than regeneration and possibly less-than-ideal healing.

### DETERMINATION OF THE NEED FOR ROOT-END RESECTION AND FILLING

The basis for periradicular surgery is twofold. The first objective is to remove the etiologic factor; the second is to prevent recontamination of the periradicular tissues once the etiologic agent has been removed.

Etiologic factors typically can be categorized as intraradicular or extraradicular bacteria,[254,287,367,491,540,541,559,580] intraradicular or extraradicular chemical substances, or extraradicular physical factors.[370,496,497] The etiology cannot always be determined with complete certainty; frequently, a number of factors are involved.[98] However, most cases involve some form of bacterial participation (e.g., bacteria within apical ramifications). The only definitive means of eradicating such an irritant is physical removal through root-end resection. The rationale for root-end resection in such cases is to establish access to and remove the diseased tissues. This ensures that the optimum environment for wound healing is established.

As mentioned, the second objective of periradicular surgery is to prevent recontamination of the tissues after removal of the etiologic agent. It follows, then, that if the remainder of the canal system cannot be verified to be irritant free, then a root-end filling should be placed to seal any remaining irritants within the canal system, thus preventing recontamination of the periradicular tissues.

### ROOT-END RESECTION

Two main principles dictate the extent of the root-end resection. First, the cause (or causes) of an ongoing disease process must be removed; this includes removal of the diseased tissue and, when indicated, reduction of an apically fenestrated root. Second, adequate room must be provided for inspection and management of the root end.

The anatomy of each tooth root is complex. The surgeon must understand the anatomy of the apical third of the root to determine the extent of a root-end resection. Approximately 75% of teeth have canal aberrations (e.g., accessory or lateral canals) in the apical 3 mm of the tooth.[131,481] An apical resection of approximately 3 mm should include most accessory and lateral canals and thus eliminate most residual microorganisms and irritants (Fig. 11.35). When roots with more than one main canal are resected, isthmus tissue may be present, and the preparation should be modified to include the isthmus area (Fig. 11.36).

If the root apex is close to the buccal cortical plate, apical fenestration can occur, leading to persistent symptoms, especially tenderness to palpation over the root apex.[72] Reduction of an apically fenestrated root apex below the level of the surrounding cortical bone allows remodeling of the bone over the tooth structure. The buccal root of the maxillary first premolar often is closest to the buccal cortical plate.

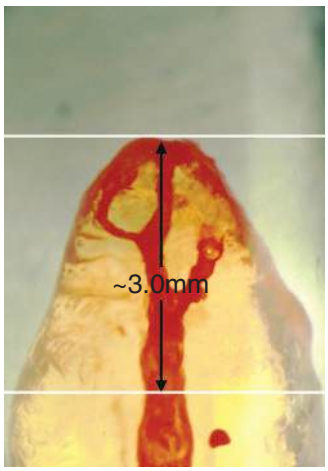



**Fig. 11.35** Cleared section of a typical single-canal root that was injected with dye to demonstrate apical accessory canals. Most of the apical ramifications can be eliminated with a 3-mm resection.

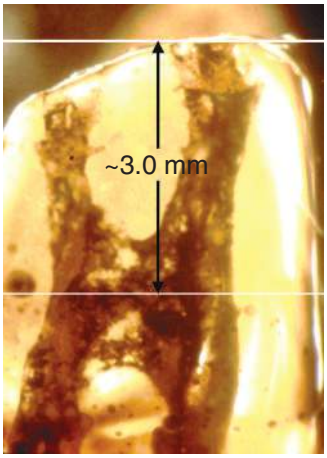



**Fig. 11.36** Cleared section of a mesiobuccal (MB) root in a maxillary first molar. Root resection at the recommended 3-mm level exposes isthmus tissue connecting the MB-1 and MB-2 canals.

The surgeon's convenience as a rationale for root-end resection depends on the individual case and the surgeon's abilities. The basic principle of the dental surgeon's convenience should be modified by the desire to minimize the trauma of the surgical procedure itself, including the preservation of tooth and supporting structures. Access to and visibility of the periradicular root structures historically has determined the extent of root-end resection. The surgeon must be able to inspect the resected root end, prepare a root-end cavity, and place a root-end filling. Enhanced visualization equipment (e.g., microscopes, endoscopes, and orascopes) have reduced the need to resect large amounts of the root to gain adequate visualization and access.[91,94,228,280,440,441,448,486] In some cases part of the root must be resected to gain access to the entire soft-tissue lesion, an additional palatally positioned root (e.g., maxillary premolars), or foreign material in the periradicular tissues.

A major consideration in determining the extent of root-end resection is the presence of anatomic structures such as the mental foramen or mandibular canal.[355,410-412] The surgeon should position the resection of the root to avoid possible damage to these structures.

## ANGLE OF RESECTION

Enhanced magnification and illumination techniques have eliminated the need to create a beveled root surface in most cases.[94,441] From a biologic perspective, the most appropriate angle of root-end resection is perpendicular to the long axis of the tooth (Figs. 11.37 and 11.38). The rationale for a perpendicular root-end resection is based on several anatomic parameters. First, a perpendicular resection approximately 3 mm from the anatomic apex is more likely to include all the apical ramifications in that region of the tooth.[339] Second, as the angle of resection increases, the number of dentinal tubules that communicate with the periradicular region and the root canal system increases significantly; therefore the probability that irritants from within the canal system will gain access to the healing tissues also rises as the resection angle increases.[181,523] Third,

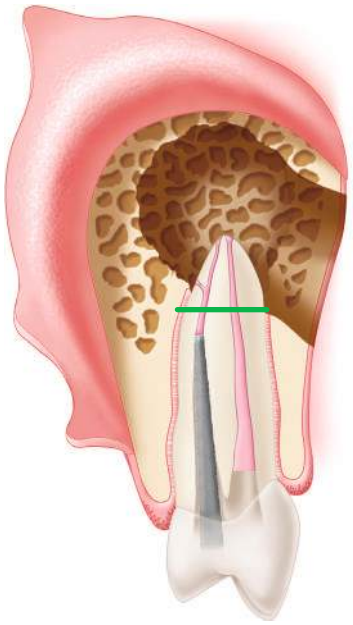


**Fig. 11.38** Perpendicular or near perpendicular root-end resection *(green line)* can be achieved with the use of microsurgical instruments and enhanced magnification and illumination.

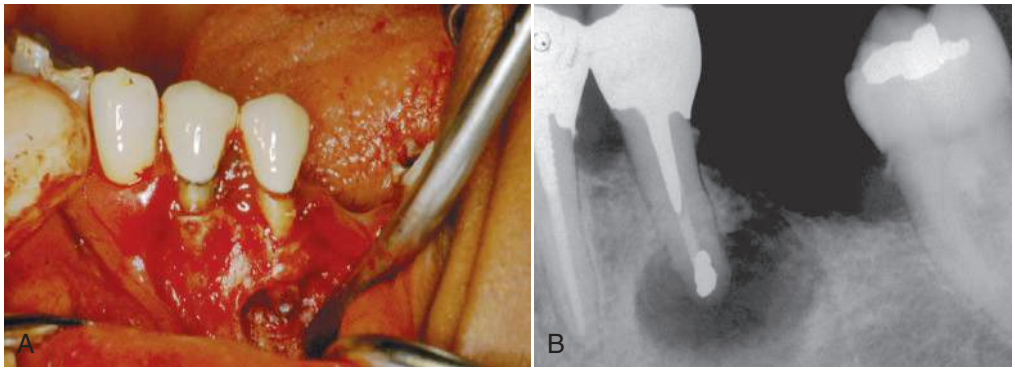



**Fig. 11.37** Periradicular surgery circa 1990. **A,** The root end was prepared with a 45-degree bevel and rotary bur microhandpiece. Amalgam was a commonly used root-end filling material at this time. **B,** Immediate postoperative radiograph of a mandibular second premolar with amalgam root-end filling. Although many teeth treated this way healed successfully, newer materials and techniques described in this chapter are currently recommended.

extending the root-end cavity preparation beyond the coronal extent of the root surface is simpler if the root-end resection is perpendicular to the long axis of the tooth.[195] Finally, with a perpendicular root-end resection, the stress forces exerted in the apical region are more evenly distributed; this may reduce the propagation of apical fractures and provide a better environment for apical healing.[470]

## ROOT-END SURFACE PREPARATION

As with all phases of endodontic surgery, the goal is to produce a resected root end with optimum conditions for the growth of cementum and subsequent regeneration of the PDL across the resected root end. Two important aspects of this process are the surface topography of the resected root end and chemical treatment of the resected root end. Healthy cementum on the root end is required for successful regeneration of periodontal tissues.[22] A number of substances found in cementum stimulate the migration, growth, and attachment of periodontal fibroblasts. Cementum extracts also activate fibroblast, protein, and collagen synthesis, which is necessary to reestablish a functional PDL.[52,471,562]

## RESECTED ROOT-END SURFACE TOPOGRAPHY

The conventional axiom for surface preparation of the resected root end has been to produce a smooth, flat root surface without sharp edges or spurs of root structure that might serve as irritants during the healing process. However, little information exists on whether smooth root ends heal differently or more quickly than rough root ends after root-end resection. A study investigating the effect of the surface topography of resected root ends on human PDL fibroblast attachment found no significant difference in fibroblast attachment to the root ends prepared with various instruments.[584] However, with a smooth resected root-end surface, the surgeon is better able to detect surface cracks and anatomic variations.[357] Given that human PDL fibroblast attachment to a smooth surface was not impaired, it would seem appropriate to produce as smooth a surface as possible to facilitate inspection of the resected root end. A smooth root end therefore should be considered advantageous.

Different types of burs tend to produce different patterns on the resected root surface.[212] Several studies have compared the root-end surface after resection.[357,377,585] Generally, crosscut fissure burs in both high-speed and low-speed handpieces produced the roughest and most irregular surfaces. Morgan and Marshall[357] compared the surface topography of root-end resection with a #57 straight fissure bur (Midwest Dental Products, Des Plains, IL), a Lindeman bone bur (Brasseler USA, Savannah, GA), and the Multi-Purpose bur (Dentsply Maillefer, Ballaigues, Switzerland), then finished with a either a multifluted carbide finishing bur (Brasseler USA) or an ultrafine diamond finishing bur (Brasseler USA). The Multi-Purpose bur produced the smoothest, most uniplanar resected root-end surface with the least amount of shattering. Regardless of the type of bur used, smearing and shredding of the gutta-percha across the root face occurred only when the handpiece was moved across the root face in reverse direction in relation to the bur's direction of rotation.[585] Burs that produce a smooth surface also tend to cut with less vibration and chatter, resulting in greater patient comfort.

## ROOT-END CONDITIONING

Root surface conditioning removes the smear layer and provides a surface conducive to mechanical adhesion and cellular mechanisms for growth and attachment. It exposes the collagenous matrix of dentin and retains biologically active substances, such as growth factors, in the dentin proper. Experimental studies have shown that demineralized dentin can induce the development of bonelike mineralized tissue.[42,43,241,550,597] Some contend that root surface conditioning produces a biocompatible surface conducive to periodontal cell colonization without compromising the vitality of the adjacent periodontium.

Three solutions have been advocated for root surface modification: citric acid, tetracycline, and ethylenediamine tetra-acetic acid (EDTA). All three solutions have enhanced fibroblast attachment to the root surface in vitro. However, citric acid is the only solution tested in an endodontic surgical application.

Citric acid traditionally has been the solution of choice. Periodontists have used an aqueous solution of citric acid (pH 1) for 2 to 3 minutes to etch diseased root surfaces to facilitate formation, new attachment, and cementogenesis.[434-438] Craig and Harrison[114] examined the effect of citric acid demineralization of resected root ends on periradicular healing. They found that 1- or 2-minute applications of 50% citric acid (pH 1) resulted in demineralized root ends and earlier complete healing than in the nondemineralized root ends. However, the periodontal literature has questioned the benefit of etching dentin surfaces with low pH agents. At a low pH, adjacent vital periodontal tissues may be compromised. Also, extended application (3 minutes) has been shown to discourage alveolar bone growth.[60,62]

EDTA, a solution with a neutral pH that endodontists have used as a canal irrigant, has been shown to be equally effective at exposing collagen fibers on dentin surfaces.[64] Unlike the lower pH solution, EDTA does not adversely affect the surrounding tissues.[63]

A series of studies that examined the effect of EDTA and citric and phosphoric acids in a periodontal application showed that application of 15% to 24% EDTA for approximately 2 minutes produces the optimum root surface.[59,61,64] These researchers concluded that EDTA, at neutral pH, was able to selectively remove mineral from a dentin surface, exposing a collagenous matrix. Citric and phosphoric acids, which have a low pH, appeared not only to remove the mineral component but also to denature the collagenous matrix.

Tetracycline has been shown to remove the dentin smear layer, leaving clean, open tubules, with application times as short as 30 seconds.[327] A histologic evaluation of new attachment in periodontally diseased human roots treated with tetracycline hydrochloride showed a trend for greater connective tissue attachment after tetracycline treatment of roots.[13] Studies comparing the effect of a 3-minute application of either EDTA (pH 7.3) or tetracycline HCl (pH 1.8) showed no significant difference in the treated tooth surfaces.[37] However, EDTA has been shown to be more favorable to human PDL cell attachment.[601]

Although the root surface conditioning effects of citric acid, EDTA, and tetracycline are well documented in the periodontal literature, this treatment has not translated into significant gains in periodontal attachment in periodontally diseased teeth.[330] Currently, only citric acid has been assessed as a root-end conditioning agent. In an animal model, citric acid was shown to enhance periradicular healing. Nonetheless, its effect and the effect of other root-end conditioning agents on the outcome of human periradicular surgery have not been established. Based on periodontal research, it would appear that if a root surface conditioning agent were to be used during periradicular surgery, EDTA might be the most appropriate solution. However, the manufacturer (personnel communication, Dr. Torabinejad) has advised against the use of EDTA when MTA is used as a root-end filling material, because it may interfere with the hard tissue–producing effect of MTA.

## Root-End Cavity Preparation

The root-end cavity preparation is a crucial step in the establishment of an apical seal. The goal is to make a cavity in the resected root end that is dimensionally sufficient for placement of a root-end filling material and at the same time avoid unnecessary damage to the root-end structures. The ideal preparation is a class I cavity prepared along the long axis of the tooth to a depth of at least 3 mm (Figs. 11.39 and 11.40). The surgical procedure is most likely to be successful if the remaining canal system has been thoroughly cleaned and shaped to eliminate microorganisms and irritants.[178,449] Traditionally, a microhandpiece with a rotating bur has been used for this purpose. However, with the advent of ultrasonic tips designed specifically for this purpose (Fig. 11.41), root-end preparations now are most often performed with the ultrasonic technique.[91] Clinical evidence is emerging to support the benefit of an ultrasonic root-end preparation when compared to traditional bur preparation, especially in molar surgery.[132]

Ultrasonic root-end preparation techniques have several advantages over the microhandpiece method. Less osseous tissue must be removed to gain proper access to the resected root end. Also, the surgeon is better able to produce a more conservative preparation that follows the long axis of the tooth and remains centered in the canal. The risk of root-end perforation is reduced, partly because of enhanced manipulation of the instrument. In addition, ultrasonic root-end techniques produce a more consistent, deeper cavity preparation that requires less beveling of the root.[95,304,347,592] Ultrasonic apical preparation generates significantly less smear layer compared with burs alone[205]; root-end preparation with a bur produces a heavy smear layer at all levels of preparation.[214]



**Fig. 11.39** Diagram of a perpendicular root-end preparation and 3-mm deep cavity preparation along the long axis of the root.



**Fig. 11.40** Error in root-end cavity preparation: ultrasonic preparation did not follow the long axis of the mesial root and therefore did not allow for proper sealing of this root. Healing is unlikely.

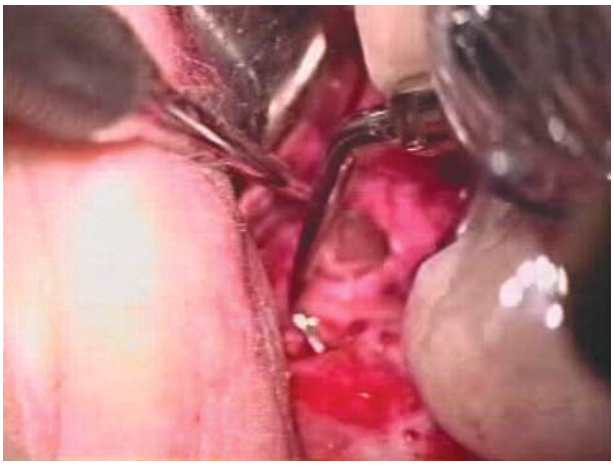


**Fig. 11.41** Ultrasonic tip in use. Preparation along the long axis of the root is possible using tips designed for each area of the mouth. In this case, the tip is properly positioned for root-end cavity preparation of a maxillary first premolar but is dangerously close to the lip because of improper retraction. Heat generated by an ultrasonic tip can cause a thermal burn, which may result in scar tissue formation.

The major concern with ultrasonic root-end preparation is the potential for creating root fractures as a result of the ultrasonic vibration (Fig. 11.42).

### ULTRASONIC ROOT-END PREPARATION AND APICAL FRACTURES

Several studies have investigated the fracture-inducing potential of ultrasonic root-end preparation techniques. This is possibly the most controversial question that arises in relation to the use of ultrasonic root-end preparation tips.

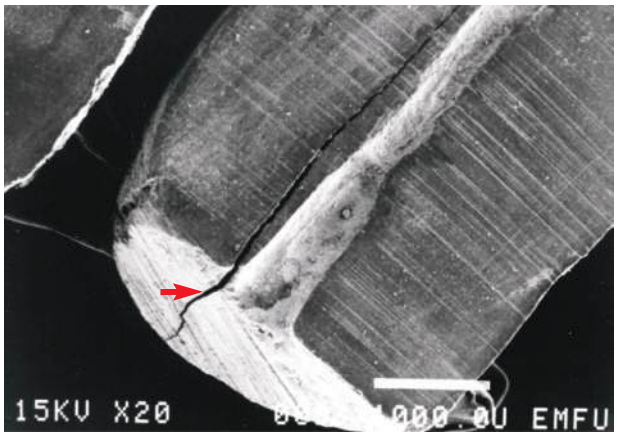



**Fig. 11.42** Scanning electron microscope image of a root end prepared in vitro with an ultrasonic device at high-power setting. A distinct fracture line can be seen *(red arrow)*. Root-end preparation with ultrasonic devices should be done at low power and with water coolant.



**Fig. 11.43** Ultrasonic tips (Obtura Spartan) are available in a wide variety of configurations for use in different areas of the mouth. Most new tips have a special coating (zirconium nitride or diamond) that improves cutting efficiency.

Three types of root-end fractures have been described: intracanal fractures (originating from the root canal system and extending into the dentin), extracanal fractures (originating on the root surface and extending into the dentin), and communicating fractures (extending from the root surface to the root canal system).[425] This classification system is not universally used in studies of root-end cavity preparation; therefore it is difficult to establish the significance, if any, of one type of fracture over another. When a strain gauge was used to measure root deformation, ultrasonic root-end preparation was shown to produce significantly greater strain, on average, than that generated by a microhandpiece. However, this did not translate into an increase in cracks observed on the resected surface of roots after root-end cavity preparation.[48,303,375,425] Several other in vitro studies that used different models to assess root-end fractures conversely concluded that ultrasonic root-end preparation does induce apical fractures.[2,184,299,354,469] Thus the degree to which apical fractures are induced during ultrasonic root-end cavity preparation is difficult to determine from in vitro studies.

On the other hand, in an in vivo study[358] and cadaver-based studies[86,207] designed to replicate the clinical scenario, root fractures were not attributed to ultrasonic use. In these studies, ultrasonic root-end preparation did not induce a significant number of root-end fractures. Several explanations may account for the differences observed in these studies. The surrounding tissues in cadaver and clinical subjects may disperse the ultrasonic energy away from the root tip. Thermal energy produced during ultrasonic preparation may have been controlled more adequately in some studies than in others. The power setting used on the ultrasonic unit may have been in the low range; a low power setting has been shown to produce fewer fractures in an in vitro setting[176,299] and is therefore recommended for clinical use.

## SIGNIFICANCE OF ULTRASONIC TIP DESIGN

Different types of ultrasonic tips are available for root-end preparation (Figs. 11.43 and 11.44), including tips of varying lengths and diameters constructed of stainless steel. These tips



**Fig. 11.44** Ultrasonic tip with diamond coating and irrigation port (DENTSPLY Tulsa Dental Specialties).

are left uncoated or are coated with diamond or zirconium nitride. Tips with a curvature of 70 degrees or greater are more susceptible to fracture under continuous loading, and fracture typically occurs at the bend.[573] Coating of ultrasonic tips undoubtedly improves the cutting efficiency compared with uncoated stainless steel tips; this translates into significantly less time required to prepare a root-end cavity.[200,405] In the coated tips, the diamond coating appears to be the most aggressive and requires the least amount of time to produce a root-end cavity preparation.[253] Furthermore, the type of tip (i.e., stainless steel, diamond coated, or zirconium nitride coated) appears to have little effect on the number or types of fractures that can be induced in the root end during root-end preparation.[76,200,253,375,425] The cavity walls of root-end preparations formed by stainless steel tips typically have cleaner canal walls than those formed by coated root-end preparation tips. Stainless steel tips appear to produce less superficial debris and smear layer. Coated root-end preparation instruments typically produce a heavily abraded, debris-covered cavity wall surface.[76,610] However, overall the quality of the coated-tip preparation has been suggested to be superior.[405]

### TEMPERATURE CHANGES INDUCED BY ULTRASONIC INSTRUMENTS

The importance of heat generation and temperature changes has already been discussed. All ultrasonic surgical tips should have an irrigation port. Use of an ultrasonic instrument in the periradicular tissues without adequate irrigation results in an extreme temperature increase in the tissues, although this specific effect has not been demonstrated during root-end preparation. Scaling without irrigation can increase the temperature in dentin as much as 95°F (35°C) above the baseline temperature[290]; such an increase can injure pulpal and periodontal tissues.[380]

### BONDED ROOT-END FILLINGS

Root-end cavity preparation for bonded root-end filling materials requires a change in the standard root-end cavity preparation technique. A shallow, scalloped preparation of the entire root surface should be made using a round or oval bur; the preparation should be at least 1 mm at the deepest concavity.[26,453] An ultrasonic preparation can be made into the root canal system, but this may not be necessary. The root-end filling material is placed in a dome fashion and bonded to the entirety of the resected root end (Fig. 11.45).

## Root-End Filling Materials

The ideal root-end filling material seals the contents of the root canal system within the canal, preventing egress of any bacteria, bacterial by-products, or toxic material into the surrounding periradicular tissues. The material should be nonresorbable, biocompatible, and dimensionally stable over time. It should be able to induce regeneration of the PDL complex, specifically cementogenesis over the root-end filling itself. Finally, the handling properties and working time should be such that the endodontic surgeon can place a root-end filling with sufficient ease.

Many materials have been used as root-end fillings, including gutta-percha, polycarboxylate cements, silver cones, amalgam, Cavit (3M ESPE, St. Paul, MN) zinc phosphate cement, gold foil, and titanium screws. However, this section focuses on root-end filling materials discussed in the literature within the past 10 years that are in common clinical use. These materials are zinc oxide–eugenol (ZOE) cements (IRM and Super-EBA), glass ionomer cement, Diaket, composite resins (Retroplast), resin–glass ionomer hybrids (Geristore), and mineral trioxide aggregate (ProRoot-MTA), and bioceramics.

### ZINC OXIDE–EUGENOL CEMENTS

Zinc oxide powder and eugenol liquid can be mixed to form a paste that is compacted into a cavity preparation. Use of this material dates back to the 1870s. Eugenol is released from ZOE mixtures, although this declines exponentially with time and is directly proportional to the liquid-powder ratio.[243] When ZOE comes in contact with water, it undergoes surface hydrolysis, producing zinc hydroxide and eugenol. This reaction continues until all the ZOE in contact with the free water is converted to zinc hydroxide.[243-245] Eugenol can have a number of effects on mammalian cells, depending on the concentration and length of exposure. These effects include cell respiration depression, macrophage and fibroblast cytotoxicity, depressed vasoconstrictor response, inhibition of PG, and suppressing or enhancing effects on the immune response.[139,349,561] Other materials have been added to the basic ZOE mixture in an effort to increase the strength and radiopacity and reduce the solubility of the final material. Commercially available ZOE materials include intermediate restorative material (IRM; Dentsply Caulk, Milford, DE) and Super-EBA (Bosworth Company, Skokie, IL).

#### Intermediate Restorative Material

IRM consists of a powder containing greater than 75% zinc oxide and approximately 20% polymethacrylate mixed in

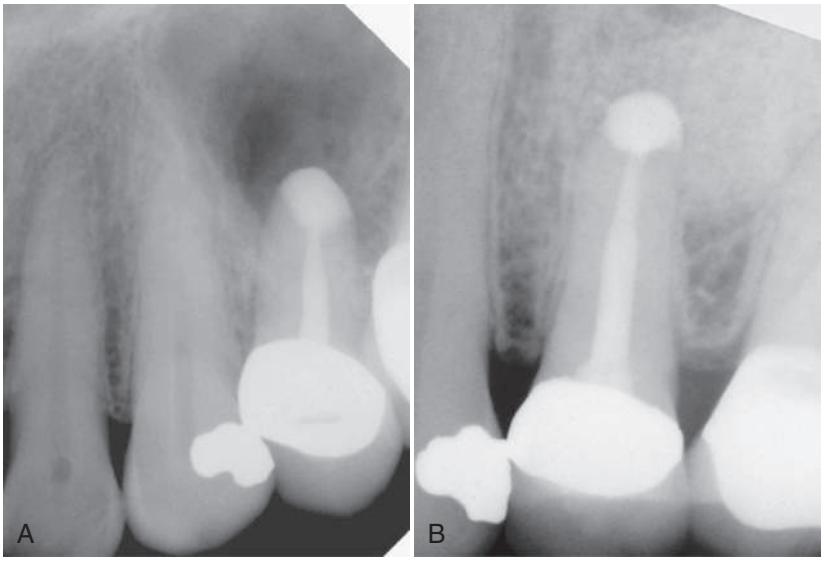



**Fig. 11.45** **A,** Immediate postoperative radiograph of a bonded root-end filling in a maxillary first premolar. **B,** Follow-up radiograph at 20 months showing good periradicular healing.

equal parts with a liquid that contains greater than 99% eugenol and less than 1% acetic acid. IRM seals better than amalgam and is not affected by the liquid-powder ratio or root-end conditioning agents.[116,407] IRM appears to be tolerated in the periradicular tissue, but it has no dental hard-tissue regenerative capacity. The response is similar to that seen with other ZOE-based materials.[221,329,417-420] In vitro, IRM prevents adherence of enamel matrix proteins.[463]

### Super-EBA

Super-EBA consists of a powder containing 60% zinc oxide, 34% aluminum oxide, and 6% natural resins. It is mixed in equal parts with a liquid that contains 37.5% eugenol and 62.5% o-methoxybenzoic acid. Super-EBA is available in two forms, fast set and regular set. Other than the setting time, the properties of the two forms appear to be the same.[593] Super-EBA has radiopacity[484] and sealing effects similar to those of IRM and is less leaky than amalgam.[232,388] The leakage pattern of Super-EBA does not appear to be affected by root-end conditioning or finishing techniques.[174,469] When Super-EBA and IRM were finished with a carbide finishing bur in a high-speed handpiece, marginal adaptation was better than with ball burnishing, which was equal to burnishing with a moistened cotton pellet.[170] The environment of the periradicular wound may affect the long-term stability of Super-EBA, which has been shown to disintegrate over time in an acid pH environment.[30]

Biologically, Super-EBA is well tolerated in the periradicular tissues when used as a root-end filling. However, it has no capacity to regenerate cementum. Bone healing has been demonstrated at 12 weeks, with some fibrous tissue persisting. Super-EBA root-end fillings show a basophilic-stained line adjacent to the filling material, which may indicate hard-tissue formation.[404,418,419,542] Collagen fibers appear to grow into the cracks of the material,[396] but the significance of this is unknown. Super-EBA has limited antibacterial effect.[101] The cytotoxicity of Super-EBA is similar to that of amalgam and IRM.[102,605] The incidence of persistent disease after endodontic surgery in which Super-EBA was used as a root-end filling material ranges from 4% to 20%. In comparative studies in which amalgam was also used as a root-end filling material, the use of Super-EBA always resulted in less persistent disease. The follow-up period for these studies ranged from 0.5 to 4.6 years.[147,326,397,448,518,565]

## GLASS-IONOMER CEMENTS

Glass-ionomer cement (GIC) consists of aqueous polymeric acids, such as polyacrylic acid, plus basic glass powders, such as calcium aluminosilicate. GIC sets by a neutralization reaction of aluminosilicate, which is chelated with carboxylate groups to cross-link the polyacids; a substantial amount of the glass remains unreacted and acts as reinforcing filler. GICs can be either light or chemically cured. Silver has been incorporated into GIC to improve the physical properties, including compressive and tensile strength and creep resistance. Both forms of GIC have been suggested as an alternative root-end filling material.[44,415,416]

The seal and marginal adaptation of light-cured GIC are superior to those with chemical-cured GIC. The seal achieved with GIC generally is better than that with amalgam and similar to that with IRM.[105,106,252,444,520,591] Long-term surface changes can occur in silver-GIC that may affect the stability of silver-GIC in the periradicular tissues.[55] GICs are susceptible to attack by moisture during the initial setting period, resulting in increased solubility and decreased bond strength.[188,508,596] Contamination with moisture and blood adversely affected the outcome when GIC was used as a root-end filling material; this occurred significantly more often in unsuccessful cases.[602] The cytotoxicity of chemical- and light-cured GIC does not differ significantly from that of Super-EBA or amalgam.[102,394] The tissue response to GIC is considerably more favorable than to amalgam and similar to that with ZOE-based materials.[100,104,133,416] In a comparative clinical study using amalgam or GIC for root-end filling, healing was evaluated clinically and radiographically after 1 and 5 years.[261,602] No difference was seen in healing capacity between the two materials. The overall success rate in both groups was 90% at 1 year and 85% at 5 years. This study showed that the 5-year follow-up result can be predicted in more than 95% of the cases at the 1-year follow-up. These authors concluded that GIC is a valid alternative to amalgam for use as an apical sealant after root-end resection and that it provides equally good long-term clinical results.[261,602]

## DIAKET

Diaket (ESPE GmbH, Seefeld, Germany) a polyvinyl resin initially intended for use as a root canal sealer, has been advocated for use as a root-end filling material.[587] It is a powder consisting of approximately 98% zinc oxide and 2% bismuth phosphate mixed with a liquid consisting of 2.2-dihydroxy-5.5 dichlorodiphenylmethane, propionylacetophenone, triethanolamine, caproic acid copolymers of vinyl acetate, and vinyl chloride vinyl isobutylether. Leakage studies comparing Diaket to other commonly used root-end filling materials have shown it to have a superior sealing ability.[190,264,313,571] Its sealing ability has not been directly compared to that of MTA. When Diaket was used as a root canal sealer, biocompatibility studies showed that it was cytotoxic in cell culture[276] and generated long-term chronic inflammation in osseous[502] and subcutaneous tissues.[393] However, when mixed at the thicker consistency advocated for use as a root-end filling material, Diaket has shown good biocompatibility with osseous tissues.[379,587] Histologically, a unique tissue barrier has been observed to form across the Diaket at the resected root end, the nature of which is unknown. This tissue resembles an osteoid or cementoid type of matrix with a close approximation of periodontal tissue fibers, suggesting a regenerative response to the root-end filling material.[587] In animal studies Diaket has shown a better healing response than resected gutta-percha in uninfected teeth[590] and a healing response similar to that with MTA. However, no cementum formation was evident.[433] This material is no longer available in the United States.

## COMPOSITE RESINS AND RESIN-IONOMER HYBRIDS

Composite resin materials have some desirable properties and may be considered for use as root-end filling materials. Generally, when assessed for sealability, composite resins perform well in vitro studies. Composite resins also tend to

leak less than amalgam, Super-EBA, IRM, and GICs.[126,344,345,520] However, blood contamination during the bonding process reduces bond strength and increases leakage.[353,560] Marginal adaptation varies, depending on the conditions and the bonding agents.[17] Certain components of composite resins and dentin-bonding agents can have a cytotoxic effect on cells; this effect varies, depending on the agent and its concentration.[80,217,428,429,514] Studies have shown that once the composite resin sets, cells can grow on its surface.[328,362,402,604] The healing response of the periradicular tissues to composite resins in general appears to be very diverse, ranging from poor to good[25,542]; this may depend on the type of material used. Two composite resin–based materials, Retroplast (Retroplast Trading, Rørvig, Denmark) and Geristore (Den-Mat, Santa Maria, CA), have been advocated for use as root-end filling materials.

### Retroplast

Retroplast is a dentin-bonding composite resin system developed in 1984 specifically for use as a root-end filling material. The formulation was changed in 1990, when the silver was replaced with ytterbium trifluoride and ferric oxide. Retroplast is a two-paste system that forms a dual-cure composite resin when mixed. Paste A is composed of Bis-GMA/TEGDMA 1:1, benzoyl peroxide N,N-di-(2-hyydroxyethyl)-p-toluidine, and butylated hydroxytoluene (BHT). This is mixed in equal parts with paste B, which is composed of resin ytterbium trifluoride aerosil ferric oxide. A Gluma-based dentin bonding agent is used to adhere the material to the root-end surface. The working time is 1½ to 2 minutes, and the radiopacity (due to the ytterbium trifluoride content) is equivalent to 6 mm of aluminum.

Only limited information is available on the physical and chemical properties of Retroplast, although a number of human clinical studies have been published.[22,25,361,451-455,457-460] In all cases the material appeared to be well tolerated and promoted a good healing response. There is evidence that Retroplast promotes hard-tissue formation at the root apex, and some have suggested that this is a form of cementum. In a limited number of case reports, Retroplast root-end fillings have demonstrated regeneration of the periodontium with a cementum layer over the root-end restoration.[22,453,454] The healing response in these cases showed deposition of minimal cementum and the insertion of new Sharpey's fibers. The PDL fibers also entered the newly formed adjacent alveolar bone, indicating that tissue regeneration, including cementogenesis, may occur on composite material, consequently forming a biologic closure of the root canal.[25] In an investigation of 388 cases comparing root-end fillings of Retroplast or amalgam, radiographic healing after 1 year was as follows: with Retroplast, 74% showed complete healing, 4% showed fibrous healing, 15% were uncertain, and 7% were failures; with amalgam, 59% showed complete healing, 3% showed fibrous healing, 30% were uncertain, and 8% were failures.[453] Complete healing occurred significantly more often after root-end filling with Retroplast. The number of immediate postoperative complications did not differ significantly between the composite and the amalgam groups. A more recent clinical study of 351 cases reported a complete healing rate of 80% to 89%.[459] A 10-year follow-up of 34 of these cases showed complete healing in 33 of the cases.[458]

### Resin-Ionomer Suspension (Geristore) and Compomer (Dyract)

The resin-ionomer suspension and compomer group of materials attempts to combine the various properties of composite resins and glass ionomers. Geristore and Dyract (Dentsply, Tulsa, OK) have been investigated for use as root-end filling materials, although the available published literature on both is limited. These two materials require light activation and resin-dentin bonding agents to attach to the tooth.

Geristore has been recommended both as a root-end filling material[93] and for use in restoring subgingival surface defects such as root surface caries, external root resorption lesions, iatrogenic root perforations, and subgingival oblique fractured roots. Clinical evaluation of Geristore as a restorative material for root caries and cervical erosions showed it to be an acceptable material.[183,372,475,548] When it is used for surgical repair of root perforations and as an adjunct to GTR, the results have been favorable in isolated case reports.[3,4,47,440,487] Geristore's dual-curing paste/paste formulation is a hydrophilic Bis-GMA with long-term fluoride release. Light activation for 40 seconds cures the material to approximately 4 mm. However, the top layer is harder until the material achieves uniform hardness at 1 day after activation.[512] In vitro leakage assessment of Geristore and Dyract indicates that the materials leak less than root-end fillings made of IRM, amalgam, or Super-EBA.[65,209] Geristore has a leakage pattern similar to that of MTA.[474] An acid pH significantly reduces dye leakage of Geristore.[446] These materials are less sensitive to moisture than conventional GIC; however, dry environments produce stronger bonds.[99] The effect of blood contamination during the bonding phase in a clinical scenario is unknown. Geristore appears to have the potential to allow regeneration of the periradicular tissue. In one study, PDL and gingival fibroblasts attached to Geristore, and the attachment improved with time and cell proliferation.[87] Studies investigating epithelial and connective tissue adherence to Geristore found clinical and histologic evidence of cellular attachment when the material was placed in subgingival cavities.[150,151,475] However, the healing response in the periradicular region is best described as unpredictable. In a study in dogs, 10 of the 18 root end–filled teeth developed abscesses. The author attributed this to the technical difficulty of placing Geristore root-end filling. However, a small number of specimens developed cementum on the root-end fillings. The cemental covering was never greater than 25% of the root-end filling surface, which was considerably less than the amount of cementum developed on both white and gray MTA.[315]

## MINERAL TRIOXIDE AGGREGATE

MTA (ProRoot MTA; DENTSPLY Tulsa Dental Specialties), a material developed specifically as a root-end filling,[529] has undergone numerous in vitro and in vivo investigations comparing its various properties to Super-EBA, IRM, and amalgam. In vitro sealing ability and biocompatibility studies comparing root-end filling materials have shown MTA to be superior to other commonly used

materials.[277,300,527,530,531,533-535] When various in vitro leakage models were used, MTA prevented leakage as well as composite resin and GIC.[6,129,171,591] However, the setting and subsequent leakage of MTA are not affected by the presence of blood.[527] Torabinejad and colleagues[535] developed the original product (gray MTA). The main constituents of this material are calcium silicate ($CaSiO_4$), bismuth oxide ($Bi_2O_3$), calcium carbonate ($CaCO_3$), calcium sulfate ($CaSO_4$), and calcium aluminate ($CaAl_2O_4$). Hydration of the powder produces a colloidal gel that solidifies into a hard structure consisting of discrete crystals in an amorphous matrix. The crystals are composed of calcium oxide, and the amorphous region is composed of 33% calcium, 49% phosphate, 2% carbon, 3% chloride, and 6% silica.[529] In a study comparing the setting time, compressive strength, radiopacity, and solubility of MTA to those of amalgam, Super-EBA, and IRM, MTA was found to be less radiopaque than amalgam but more radiopaque than Super-EBA and IRM.[529] MTA had the longest setting time (2 hours, 45 minutes) and the lowest compressive strength at 24 hours after mixing (40 MPa), although compressive strength increased to 67 MPa at 21 days after mixing. The solubility of MTA after setting was similar to that of amalgam and Super-EBA. Initially MTA has a pH of 10.2, which rises to 12.5 3 hours after mixing.[529] The pH has been reported to be approximately 9.5 at 168 hours after mixing.[152] MTA is less cytotoxic than amalgam, Super-EBA, or IRM root-end fillings.[531] Endodontic surgery studies in dogs and monkeys have reported less periradicular inflammation and cementum deposition immediately adjacent to the root-end filling material.[27,173,236,528,532] Holland and colleagues[236,237] theorized that the tricalcium oxide in MTA reacts with tissue fluids to form calcium hydroxide, resulting in hard-tissue formation.

The importance of the presence of cementum-like tissue adjacent to MTA cannot be understated. Cementum deposition is essential to regeneration of the periodontal apparatus.[310] Augmentation of new cementum across the root end and root-end restoration is essential for ideal healing of the periodontium. A layer would also enhance the integrity of the apical barrier, making it more resistant to penetration by microorganisms; in effect, establishing a biologic barrier.[22] This is seen most frequently in sections where MTA was used as the filling material. MTA appears to be able to induce cementoblastic cells to produce hard tissue. Cementogenesis in the presence of MTA has been evaluated by assessment of the expression of osteocalcin (OCN), cell growth, and the morphology of cementoblast-like cells.[522] Scanning electron microscope (SEM) analysis indicated that cementoblasts could attach to and grow on MTA. In addition, strong expression of the OCN gene was seen after application of MTA. MTA can also increase the production of both proinflammatory and antiinflammatory cytokines from osteoblasts. The clinical significance of this reaction is not known. The effect of MTA on periradicular tissues is probably partly due to these reactions.

In a human outcomes assessment study comparing ProRoot MTA to IRM, the rate of persistent disease with MTA was 16% at 12 months and 8% at 24 months.[103] The rate of persistent disease with IRM was 24% at 12 months and 13% at 24 months. These authors concluded that the use of MTA as a root-end filling material resulted in a high success rate that was not significantly better than that obtained with IRM. A prospective clinical trial using MTA as a root-end filling material along with current microsurgical techniques reported 89% clinical success, with follow-up time ranging from 4 to 72 months.[468]

A variation of the original formula of gray MTA has been introduced. This material, which is a cream white color, is often called white MTA. The chemical composition of white MTA is very similar to that of the original. White and gray ProRoot-MTA materials differ by less than 6% in any one component. Both are fine powders with a mean particle size of approximately 10 μm (the range in particle size is approximately 0.1 to 100 μm). The radiopacity of both materials is equivalent to approximately 3.04 mm of aluminum.[45] When white MTA was implanted in the subcutaneous connective tissue of rats, the results were similar to those reported for gray MTA.[238] One study compared the tissue reaction evoked by the two materials when used as root-end fillings in canines.[315] The only statistically significant difference observed was for the presence of macrophages or multinucleated giant cells adjacent to material. Gray MTA had more samples with mild to moderate infiltration of macrophages or multinucleated giant cells, and white MTA had more samples with no macrophages and/or multinucleated giant cells adjacent to the material. All other parameters assessed were essentially the same.

## BIOCERAMICS

Bioceramics are a relatively new and potentially promising addition to the group of materials available for root-end filling. In vitro testing of EndoSequence Root Repair Material (ERRM; Brasseler USA) demonstrates biocompatibility and antimicrobial activity that is similar to MTA.[107,122,317,323] ERRM is composed of calcium silicates, monobasic calcium phosphate, and zirconium oxide.[107] The material is hydrophilic, radiopaque, and has high pH. ERRM is available as a putty and a syringable paste. Because this is a relatively new material, long-term clinical studies are not yet available.

## OVERVIEW OF ROOT-END FILLING MATERIALS

Many different materials have been advocated for use as root-end filling materials, and each has specific advantages and disadvantages. However, from the biologic perspective of regeneration of the periradicular tissues, MTA, followed by Retroplast, appears to have a clear advantage over the other available materials. Bioceramic materials may join this group but require more clinical testing. Retroplast and other composite resin–based filling materials require meticulous hemostasis and a dry surgical field for optimum results. The most commonly cited disadvantage of MTA is its handling properties. Even when properly prepared, MTA is more difficult to place in the root-end cavity than most other materials. Several devices have been modified or developed specifically for use with MTA (Fig. 11.46; also see Figs. 11.19–11.23). Typical clinical cases showing the surgical procedures described in the previous sections are presented in Figs. 11.47–11.49.

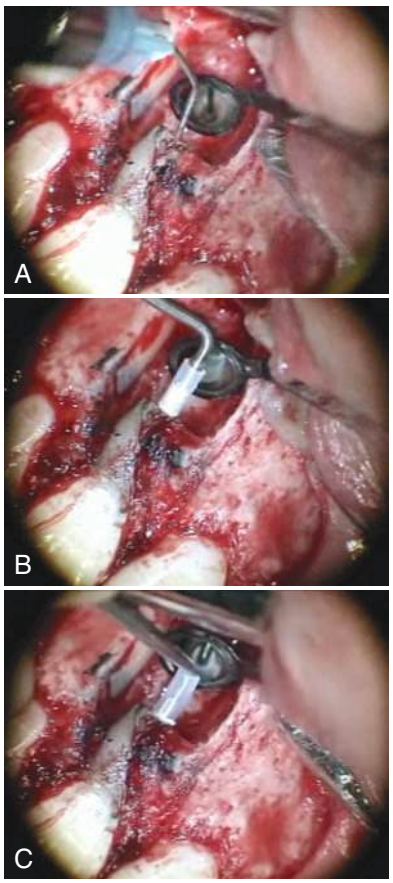



**Fig. 11.46** **A,** Stropko syringe used to dry a root-end preparation before placement of the root-end filling material. **B,** Clinical use of an MTA delivery system (Dentsply Tulsa Dental). The device is loaded with MTA and placed over the root-end preparation. **C,** Pressing the plugger into the sleeve delivers the filling material to the root-end cavity preparation. The filling material then is compacted with microcondensers, and additional filling material is placed as needed.

## Closure of the Surgical Site and Selection of Suture Material

### CLOSURE OF THE SURGICAL SITE

The surgical site should be closed only after careful visual and radiographic inspection of the area. Before suturing, a radiograph should be taken with the flap held loosely in place to detect any foreign objects in the crypt or adhering to the flap. This image is also important for confirming the depth and density of the root-end filling. The osteotomy site then is gently curetted and irrigated with sterile saline or water to remove any remnants of hemostatic agents and packing materials. Some bleeding is encouraged at this point, because the blood clot forms the initial scaffold for subsequent healing and repair. If indicated, grafting materials or barriers may be placed at this time. Slight undermining of the unreflected soft tissue adjacent to the flap facilitates the placement of sutures. The flap is then repositioned and gently compressed with a piece of chilled, sterile moist cotton gauze to express excess blood and tissue fluids.

For the common flap designs discussed in this chapter, the corners are first identified and sutured in place with a single interrupted suture. Interrupted sutures are initially passed through the free portion of the flap approximately 2 to 3 mm from the edge and then connected to the attached tissue. The suture is secured with a simple surgeon's knot, which is positioned away from the incision line. The center of the flap then is located and sutured with either an interrupted or a sling suture. A continuous locking suture technique may be used to close a submarginal (Ochsenbein-Luebke) flap.[288] The primary advantage of a continuous suture technique is the ease of suture removal compared with multiple interrupted sutures. The disadvantages are possible difficulty with precise control of tension in each area, and the fact that the entire suture may loosen if one suture pulls through the flap. A sling suture is commonly used for the central tooth in the surgical site to close a full-thickness intrasulcular (rectangular or triangular) flap. The tension on this type of suture can be varied slightly to allow some control of the apicocoronal positioning of the flap. Interrupted sutures then are placed as needed.

When suturing is complete, chilled, sterile moist cotton gauze is again placed over the flap and pressure is applied for 5 minutes. Pressure to the area provides stability for the initial fibrin stage of clot formation and reduces the possibility of excessive postoperative bleeding and hematoma formation under the flap. The iced gauze also supports hemostasis. Final inspection of the area should confirm that all soft-tissue margins have been closely approximated and bleeding has been controlled. An additional injection of long-acting local anesthetic may be administered at this time, although care must be taken not to inject it directly under the newly repositioned flap. The patient is given a cold compress and instructed to hold it on the face in the surgical area, on for 20 minutes and then off for 20 minutes, for the rest of the day. The patient is also given verbal and written postoperative instructions, including after-hours contact information (Fig. 11.50). The patient should sit in an upright position for approximately 15 minutes and the surgical site should be inspected one more time before the patient is discharged.

### SELECTION OF THE SUTURE MATERIAL

The properties of an ideal suture material for periradicular surgery include pliability for ease of handling and knot tying, a smooth surface that discourages bacterial growth and wicking of oral fluids, and a reasonable cost. Suture material in size 5-0 is most commonly used, although some clinicians prefer slightly larger (4-0) or smaller (6-0) suture. Sutures smaller than 6-0 tend to cut through the relatively fragile oral tissues when tied with the tension required to approximate the wound margins. Silk suture material has commonly been used in dental surgery for decades and is both inexpensive and easy to handle. However, silk tends to support bacterial growth and allows for a wicking effect around the sutures. For these reasons, other materials are preferable to silk.[92]

Resorbable suture materials (plain gut and chromic gut) are not routinely used for periradicular surgery, although this material may be indicated if the patient will be unavailable for the regular suture removal appointment (48 to 96 hours after surgery) or if the suture will be used in areas of the mouth where access is difficult. The primary problem with resorbable suture materials is the variable rate of resorption—that is, sutures may weaken and dissolve too

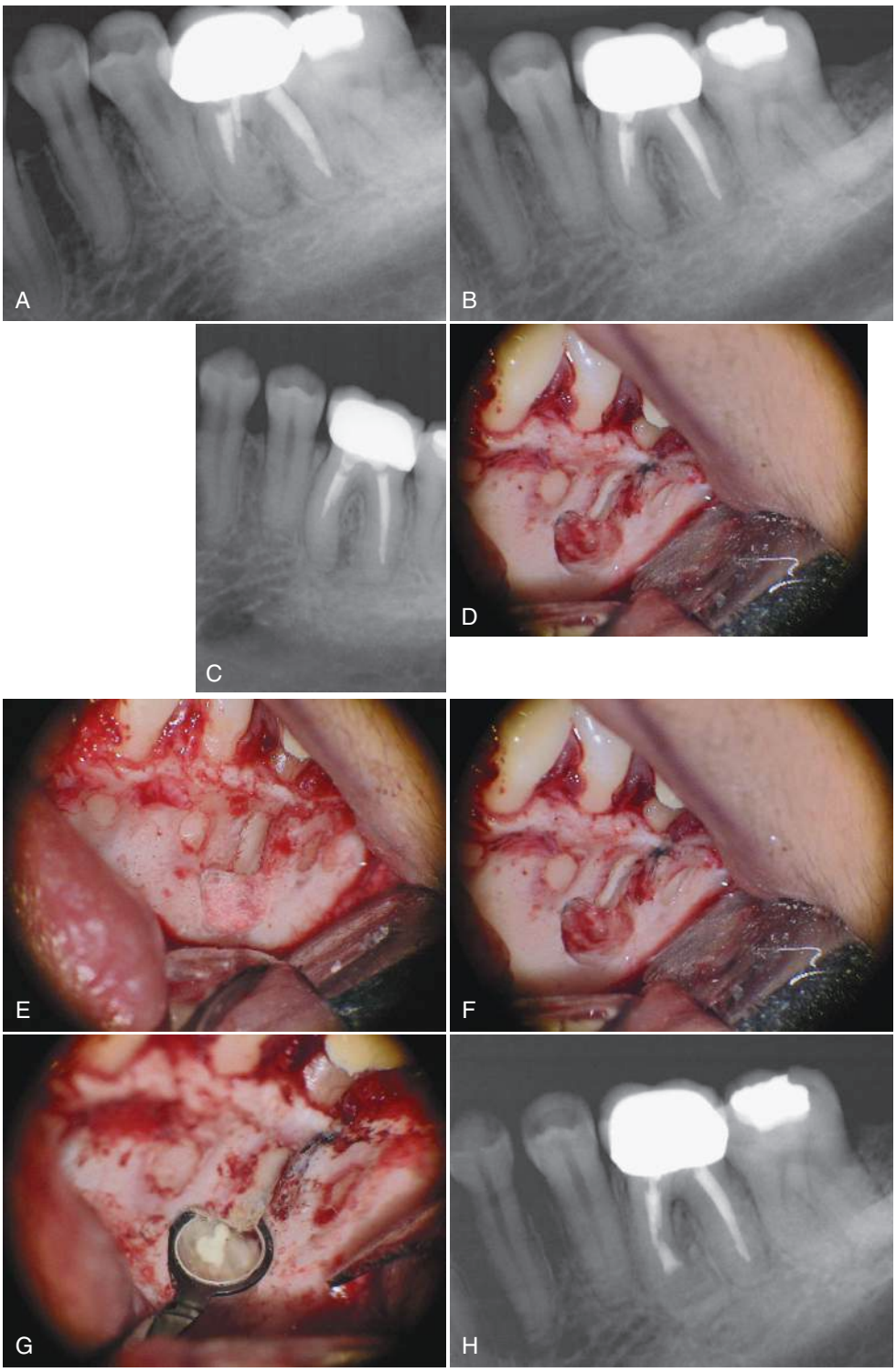



**Fig. 11.47 A,** Preoperative radiograph (mesial angle) of a left mandibular first molar. Periradicular disease and symptoms had persisted after nonsurgical retreatment by an endodontic resident. The mesial canals were completely obstructed at the midroot level. The preoperative evaluation included three periapical radiographs, two different horizontal angulations, and one vertically positioned radiograph. **B,** Straight view preoperative radiograph. **C,** Vertical periapical view preoperative radiograph. **D,** Osteotomy and root resection perpendicular to the long axis of the root were performed; a partial bony dehiscence over the mesial root can be seen. **E,** Racellets were packed into the bony crypt to establish hemostasis. **F,** The Racellets were removed (one usually is left in the deepest part of the crypt during root-end preparation and filling), and the root end was beveled perpendicular to the long axis of the root. Methylene blue dye can be useful for identifying the root outline and locating any cracks. Ultrasonic root-end preparation was completed to a depth of 3 mm, connecting the MB and ML canals. **G,** Mineral trioxide aggregate root-end filling was placed and inspected. The bony crypt was then gently curetted to initiate bleeding and to remove any remnants of hemostatic materials. The flap was repositioned, and a radiograph was taken. **H,** An immediate postoperative radiograph confirmed the depth and density of the root-end filling and the absence of any foreign objects. Note that calcium sulfate and bone grafting material was placed in the osseous defect and over the root because of the large buccal dehiscence, although this is not routinely required in these cases. (Courtesy Dr. Vince Penesis.)

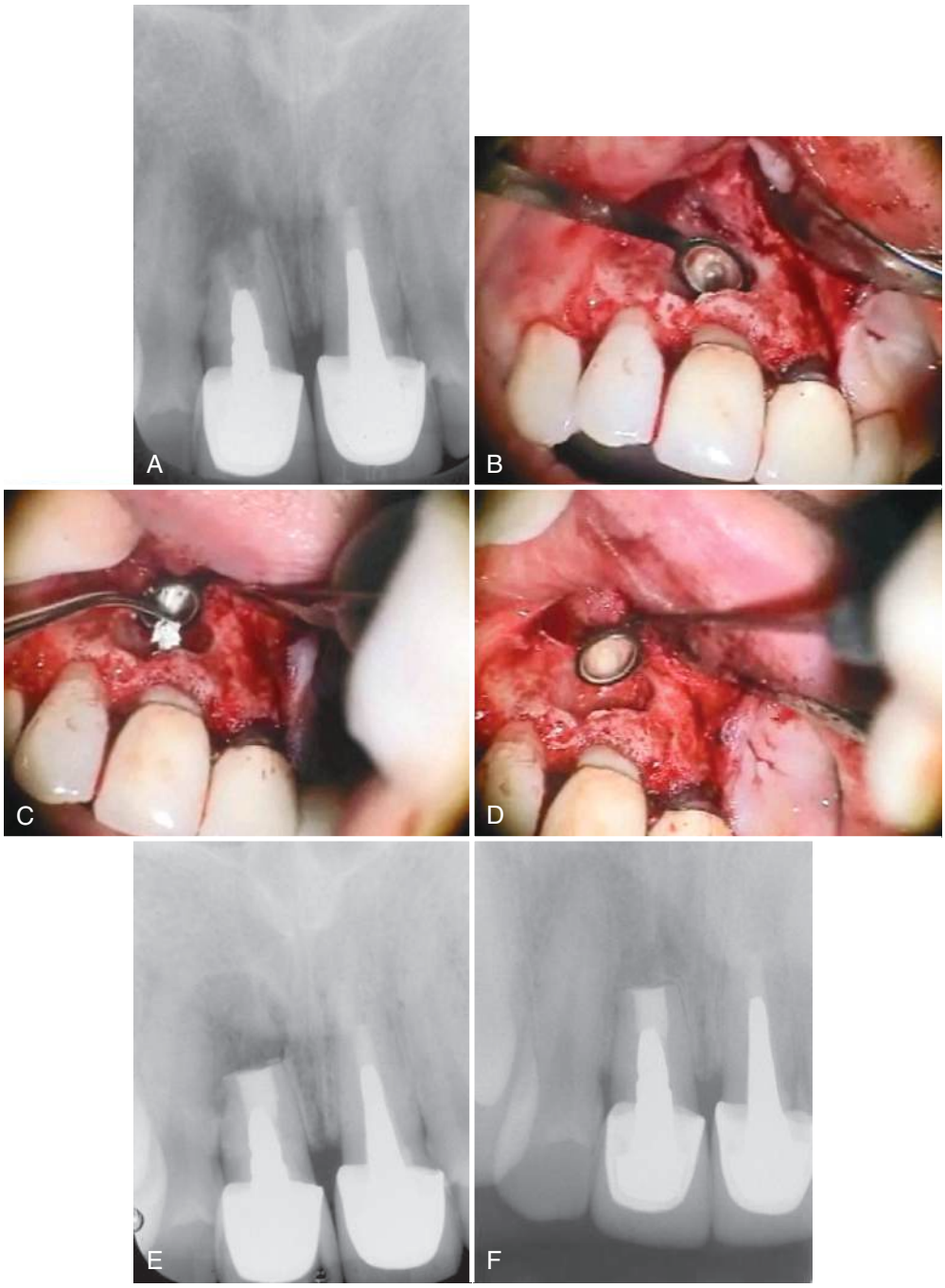



**Fig. 11.48 A,** Preoperative radiograph of a maxillary left central incisor showing evidence of previous surgery but no apparent root-end filling. Because of the short root length and inadequate band of keratinized gingiva, an intrasulcular incision and full-thickness triangular mucoperiosteal flap design were selected. **B,** The root was minimally resected, and the root-end cavity was prepared ultrasonically. **C,** Mineral trioxide aggregate was placed and condensed into the root-end cavity preparation. **D,** The root-end filling was inspected before the flap was repositioned and sutured. **E,** Immediate postoperative radiograph. **F,** The 6-month follow-up radiograph showed good initial periradicular healing. (Courtesy Dr. Shawn Velez.)

soon or, more commonly, may remain in the incision area for longer than desired. Gut suture materials are packed in isopropyl alcohol. The handling properties of gut sutures can be improved by immersion in sterile water for 3 to 5 minutes before use.[430]

Suture materials with a smooth Teflon or polybutilate coating (e.g., Tevdec and Ethibond, respectively) are particularly well suited for use in periradicular surgery. Synthetic monofilament suture materials (e.g., Supramid and Monocryl) are also commonly used. These materials are easy to handle and do not promote bacterial growth or wicking of oral fluids to the same extent as silk. Gortex (expanded PTEE-Teflon) sutures have many desirable properties but are more expensive than the previously mentioned materials.

Tissue adhesives such as cyanoacrylate and fibrin glues may hold promise for wound closure after periradicular surgery.[113,196,403,599] Although the currently available research is insufficient to recommend these adhesives as a routine replacement for more traditional suture materials, future applications in periradicular surgery are possible.

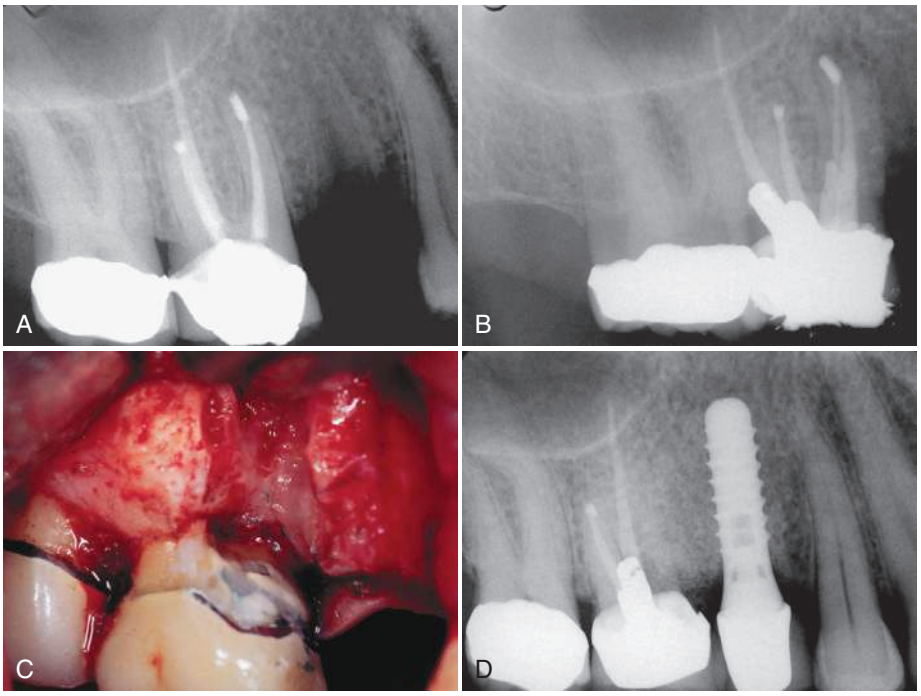



**Fig. 11.49** **A,** A maxillary right first molar that was sensitive to percussion and tender to palpation over the MB root. Periodontal probing revealed a deep, narrow bony defect on the facial aspect of the MB root. The presence of a vertical root fracture was confirmed visually using magnification and methylene blue dye. The maxillary right second premolar had been recently extracted because of a vertical root fracture. **B,** A bonded amalgam core buildup was placed in the DB and P canals, and Geristore was placed in the MB canal system. **C,** The MB root was resected, and methylene blue dye was used to help define the extent of the fracture. **D,** At the 3-year follow-up visit, a new crown had been fabricated for the molar, and the premolar had been replaced with an implant.

**Care of Your Mouth After Endodontic Surgery**

1. Apply an ice pack to your face next to the surgery area (on for 20 minutes and off for 20 minutes) for the next 5 to 6 hours to help decrease postoperative swelling. Swelling is usually greatest the day after surgery and may be at its worst 2 or 3 days after surgery.

2. Take all medications as directed. Approximately 45 minutes should be allowed for you to feel the effect of pain medication.

3. Clean your mouth as usual (brushing, flossing, etc.) in all areas except the surgical site. Modify cleaning procedures of the teeth in the area of the surgical site to keep from disturbing the area. Do not rinse vigorously during the first 24 hours following surgery. Continue using the prescribed mouth rinse twice a day until after you return to have the sutures removed.

4. Minor oozing of blood from the surgical site may occur in the first 24 hours after surgery. This will produce a pink tinge in the saliva and is not a cause for concern. However, if bleeding is excessive, please contact our office. You may apply pressure to the area with a tea bag or moist cotton gauze.

5. Sutures have been placed and will need to be removed at your next appointment. Please do not lift or pull on your lip to examine the surgical site during the first 2 to 3 days because this may disturb the healing process.

6. A soft diet is recommended for the first 2 or 3 days. Try to avoid foods that are hot, spicy, or hard to chew. It is very important that you drink plenty of fluids (nonalcoholic). This will help your mouth heal.

7. Avoid cigarettes and all other tobacco products.

8. A slight increase in body temperature may occur during the first 24 hours following surgery. This is normal. Infection after endodontic surgery is not typical but can occur. If an infection develops, it usually occurs 2 to 3 days after the surgery. Signs of an infection include sudden increase in pain or swelling, feverish feeling, sore glands in the neck area, and a general flulike feeling. If you think an infection has developed, please contact the office immediately.

**If you have any questions, please contact the office during normal office hours at 312-XXX-5555. If you are experiencing problems after office hours, you may contact Dr. _____ at: 312-XXX-1212.**

**Fig. 11.50** Example of postoperative instructions. Written instructions provide an essential reference for the patient, because verbal instructions often are difficult to remember after surgery. The instructions may be modified as needed; it is important to provide instructions that the patient can understand. For example, the readability of these instructions is at approximately the eighth-grade level using the Flesch-Kincaid Grade Level scale.

## Guided Tissue Regeneration and Endodontic Surgery

The amount and location of bone adjacent to the root structures affect the prognosis of periradicular surgery. Kim and Kratchman[282] propose a six-category classification system to assist in predicting surgical prognosis and determining the need for bone grafting and barrier techniques. Class A (no lesion), class B (small periapical lesion), and class C (large periapical lesion without periodontal communication) all represent situations that are favorable for healing without supplemental grafting or barriers. Class D (similar to class C with independent periodontal pocketing), class E (endodontic-periodontal communication to the apex), and class F (apical lesion with complete loss of buccal bone) represent situations with a more guarded prognosis and usually require concurrent use of bone grafting and barrier techniques. Figs. 11.51–11.63 are examples of cases that required GTR.

An apicomarginal defect[141] or a localized bony defect distinguished by a total deficiency of alveolar bone over the entire root length has a significant adverse effect on the outcome, reducing the rate of complete healing by approximately 20% or more when compared to teeth with an isolated endodontic-only lesion.[233,279,499] The presence of a periradicular lesion 15 mm or greater in diameter has also been linked to a poorer prognosis.[233] Advanced periodontitis with deep pocket formation has been associated with chronic periradicular inflammation after endodontic surgery and subsequent failure of the root-end surgery.[450] The cause of failure has been identified as in-growth of non-osteogenic tissues into the periradicular surgical site and down-growth of epithelial tissue along the root surface. Successful treatment may depend more on controlling epithelial proliferation than root-end management. GTR techniques have been advocated for use in such cases.[206,547]

The basic principle of guided tissue and bone regeneration is that different types of cells repopulate a wound at different rates during healing. The soft-tissue cells are considerably more motile than the hard-tissue cells, therefore they tend to migrate into the wound more quickly during healing. A barrier interposed between the gingival tissue and the exposed root surfaces and supporting alveolar bone prevents colonization of the exposed root surface by gingival cells. This encourages selective repopulation of the root surface by PDL cells. The use of an absorbable barrier theoretically would allow PDL cells and other cells with osteogenic potential to repopulate the defect, resulting in new connective tissue attachment and bone formation. Dahlin and coworkers[119,120] demonstrated that in monkeys, a significant increase in osseous healing occurs when membranes are used in through-and-through bone defects in periradicular surgery of the lateral maxillary incisors. The use of resorbable GTR membranes in endodontic surgery with buccal apicomarginal type defects also has been shown to enhance regeneration of the periodontium and surrounding bone in dogs.[149] This type of matrix barrier promoted greater amounts of connective tissue and alveolar bone and minimized the formation of junctional epithelium.

Several case reports have discussed the use of GTR techniques in conjunction with endodontic surgery.[5,32,81,123,154,273,337,414,421,432,515,544,549,607] These studies largely have reported favorable outcomes in cases involving large periradicular lesions, through-and-through bone defects, and repair of a surgical perforation or loss of the buccal cortical plate adjacent to the root.

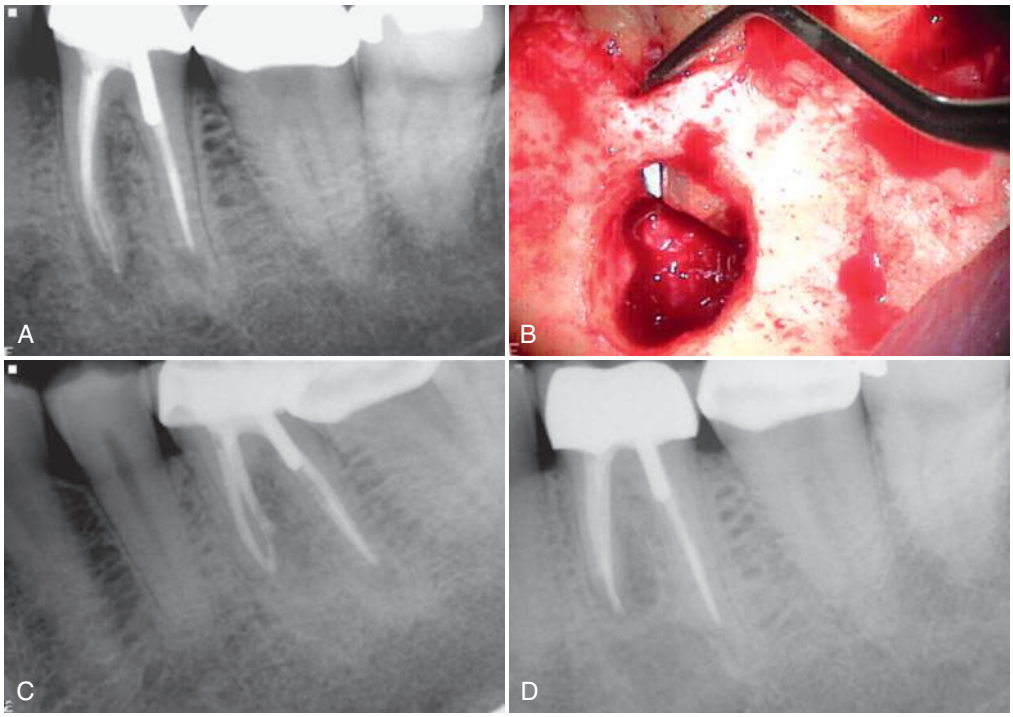



**Fig. 11.51** **A,** Preoperative periapical radiograph of tooth #19. **B,** Clinical image demonstrating periodontal defect along the facial aspect of the mesial root. **C,** Immediate postsurgical radiograph. The mesial root end was prepared with ultrasonics and filled with MTA. A mixture of DFDBA and Capse was placed for guided tissue regeneration. **D,** One-year follow-up radiograph demonstrating good healing. Periodontal probings were WNL.

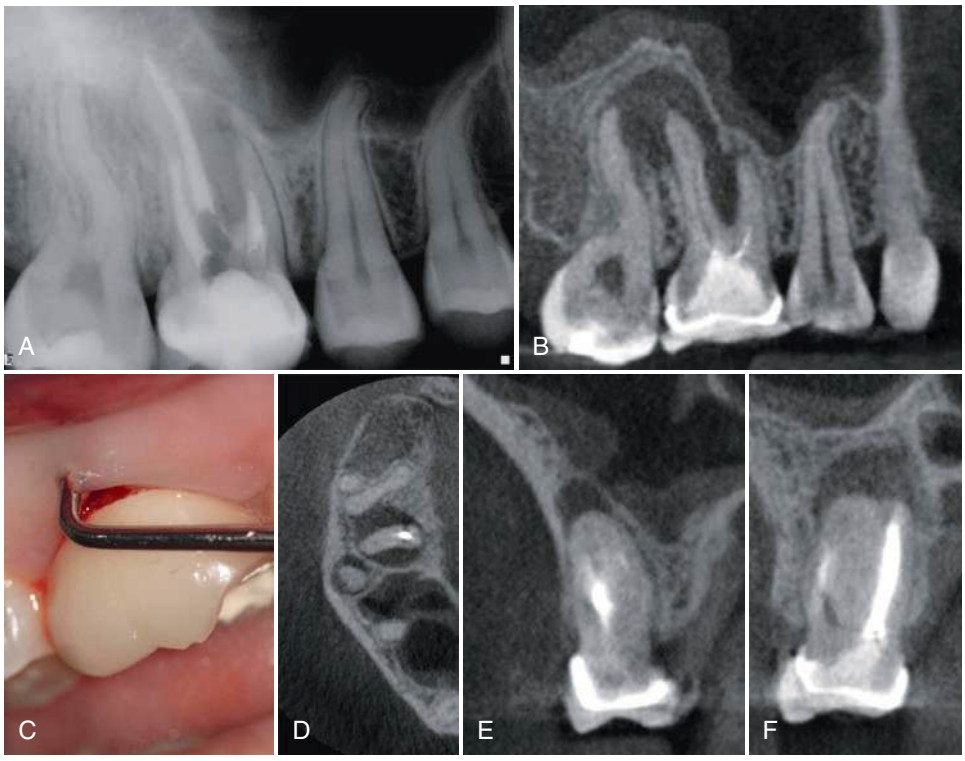



**Fig. 11.52** (Case 1): **A,** Periapical radiograph of tooth #3. Retreatment RCT was attempted, but MB1 and MB2 were blocked. **B,** Sagittal view demonstrating the extent of the periapical lesion. **C,** Clinical view of periodontal probing to the apex. **D,** Axial view of the apical one third demonstrating the close proximity of the periapical defect to the MB root of tooth #2 (#2 responded WNL to pulp vitality testing). **E** and **F,** Coronal views of the MB and DB/P roots, respectively.




**Fig. 11.53** (Case 1, continued): **A,** A 3D reconstruction of the tooth #3 area demonstrating the furcation periodontal defect. **B,** After flap reflection, clinical view demonstrating the periodontal defect. **C,** A 3D reconstruction with the buccal plate cropped to visualize the extent of the periradicular defect. **D** and **E,** DB and P root resection, root-end preparation, root-end filling with mineral trioxide aggregate. **F,** Periradicular defect grafted with Puros allograft (Zimmer Dental, Carlsbad, CA) and CopiOs pericardium membrane (Zimmer Dental).

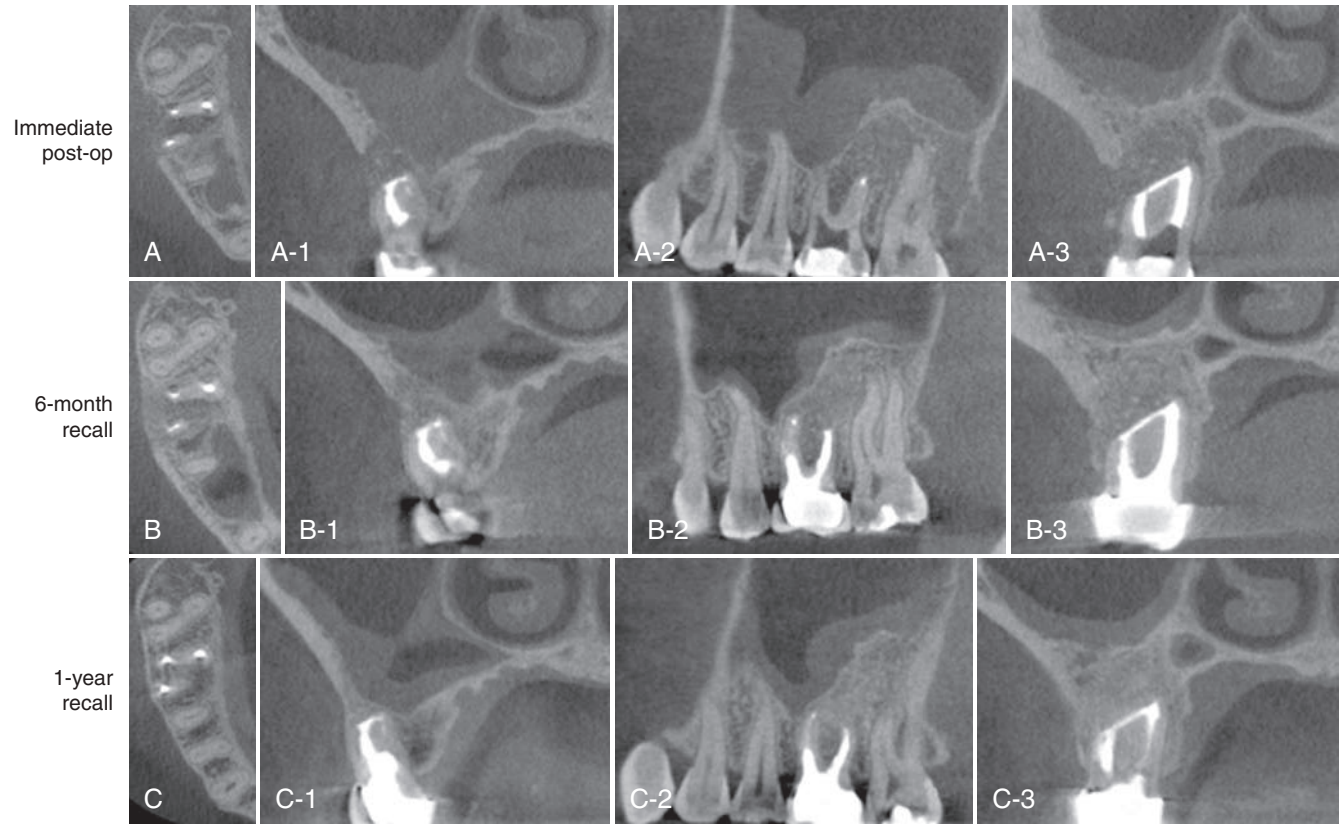



**Fig. 11.54** (Case 1, continued): **A,** Axial view of immediate postoperative CBCT scan, A-1 coronal view of MB root, A-2 sagittal view and A-3 coronal view of DB and palatal roots. **B,** A 6-month recall, axial view, B-1 coronal view of MB root, B-2 sagittal view, and B-3 coronal view of DB and palatal roots. **C,** A 1-year recall, axial view, C-1 coronal view of MB root, C-2 sagittal view, and C-3 coronal view of DB and palatal roots.

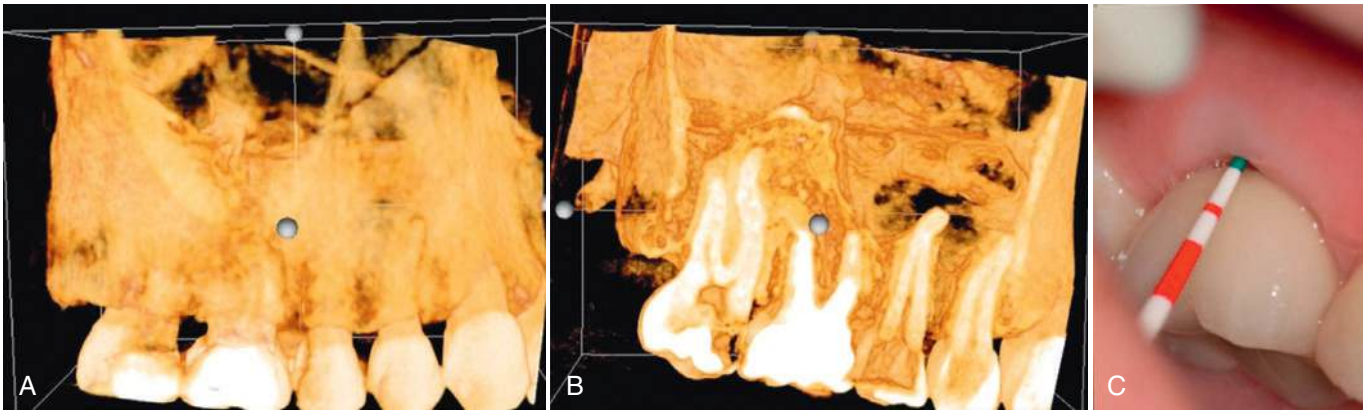



**Fig. 11.55** (Case 1, 1-year recall): **A** and **B,** A 3D reconstruction of 1-year recall. **C,** Clinical picture showing resolution of the periodontal defect.

Pecora and colleagues[400] compared the healing of 20 large periradicular defects (>10 mm diameter) with and without the use of nonresorbable membrane. They reported that at 12 months after surgery, the sites in which membranes had been used had healed more quickly and that the quality and quantity of the regenerated bone was superior. One study evaluated periradicular and periodontal healing in cases involving apicomarginal defects when GTR (Bio-Oss and Bio-Gide membrane; Osteohealth Co., Shirley, NY) was performed in conjunction with periradicular surgery. At 12 months after surgery, 86% were considered healed both clinically and radiographically. It was concluded that GTR should be considered as an adjunct to periradicular surgery in cases of apicomarginal defects.[142] However, use of a resorbable membrane when a standard apical osteotomy is performed and the buccal bone over the remainder of the root is intact has no beneficial effect on healing.[186]

Several different types of membranes are available. They can be grouped into two broad categories, nonresorbable and resorbable (Table 11.1). Resorbable membranes are generally better suited for endodontic uses because a second surgical procedure is not required to remove the membrane.

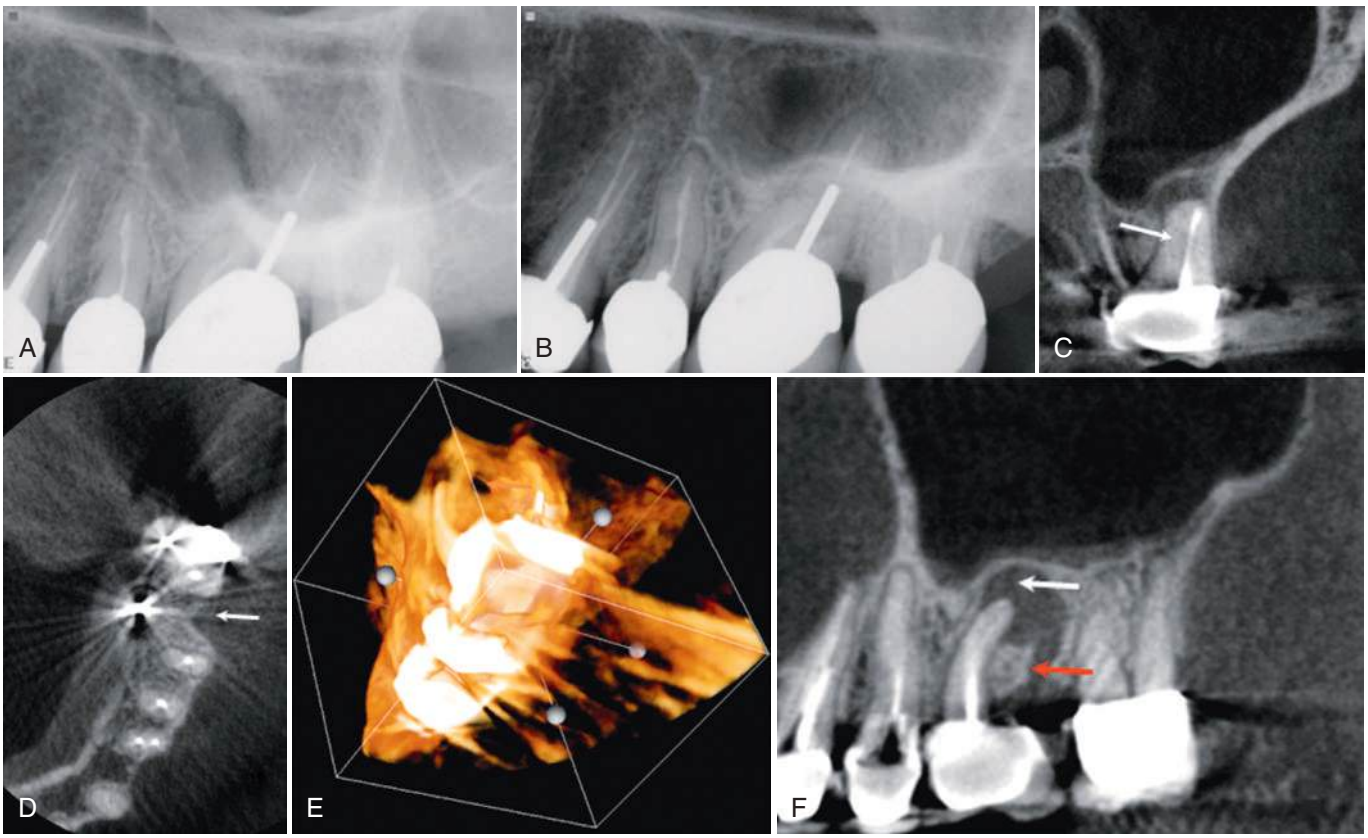



**Fig. 11.56** (Case 2, tooth #14): **A** and **B,** Periapical radiographs of tooth #14, mesial and distal angles, respectively. **C,** Coronal view of the MB root demonstrating a missed MB2 canal *(arrow)*. **D,** Axial view showing a previous DB root amputation site *(arrow)* that was not obvious in the periapical radiograph. **E,** A 3D reconstruction demonstrating a crestal defect. **F,** Sagittal view demonstrating the crestal defect *(red arrow)* communicating with the periapical lesion *(white arrow),* elevation of the floor of the maxillary sinus but no evidence of sinus perforation.

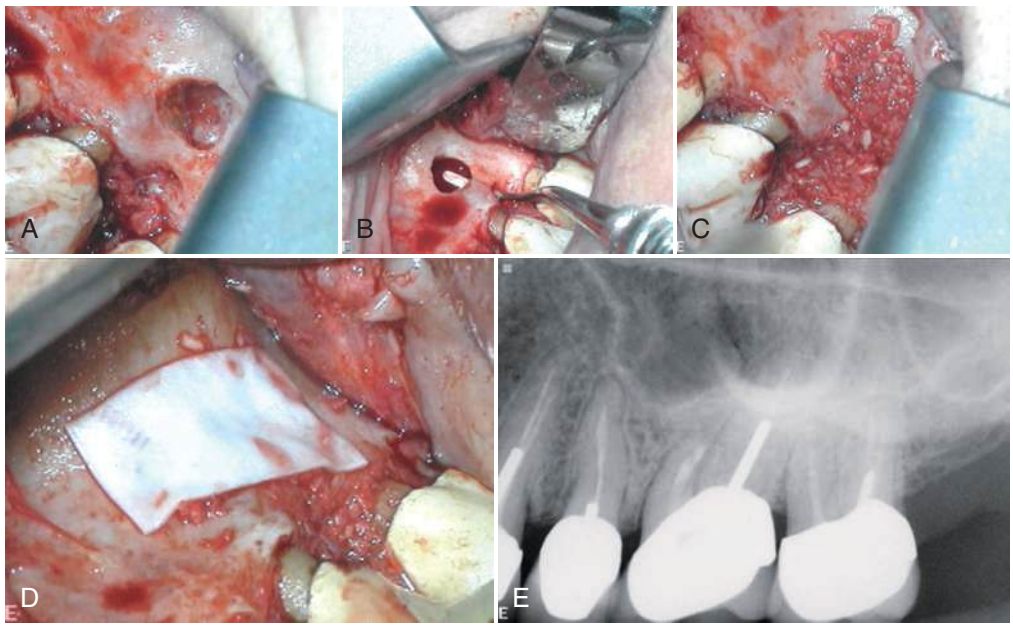



**Fig. 11.57** (Case 2, continued): **A,** Clinical view after flap reflection demonstrating both crestal and periapical lesions. **B,** Communication between both defects. **C,** Both defects grafted with EnCore Combination Allograft (Osteogenics Biomedical, Lubbock, TX). **D,** CopiOs pericardium membrane (Zimmer Dental, Carlsbad, CA). **E,** Immediate postoperative radiograph.

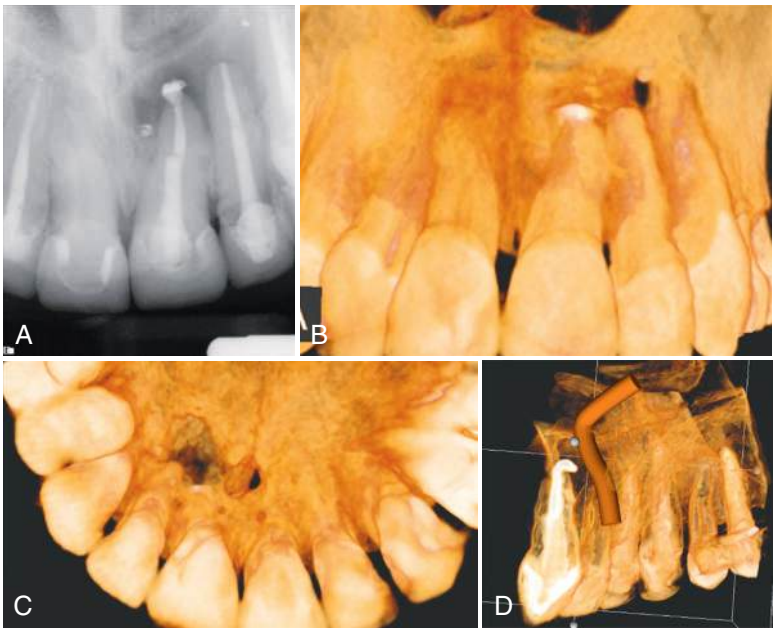



**Fig. 11.58** (Case 3): **A,** Periapical radiograph of maxillary anterior region demonstrating a periapical lesion associated with tooth #9. **B,** CBCT reconstruction demonstrating the intact buccal cortical plate. **C,** Palatal view of the 3D reconstruction demonstrating perforation of the palatal plate. **D,** A 3D reconstruction showing the nasopalatine bundle.

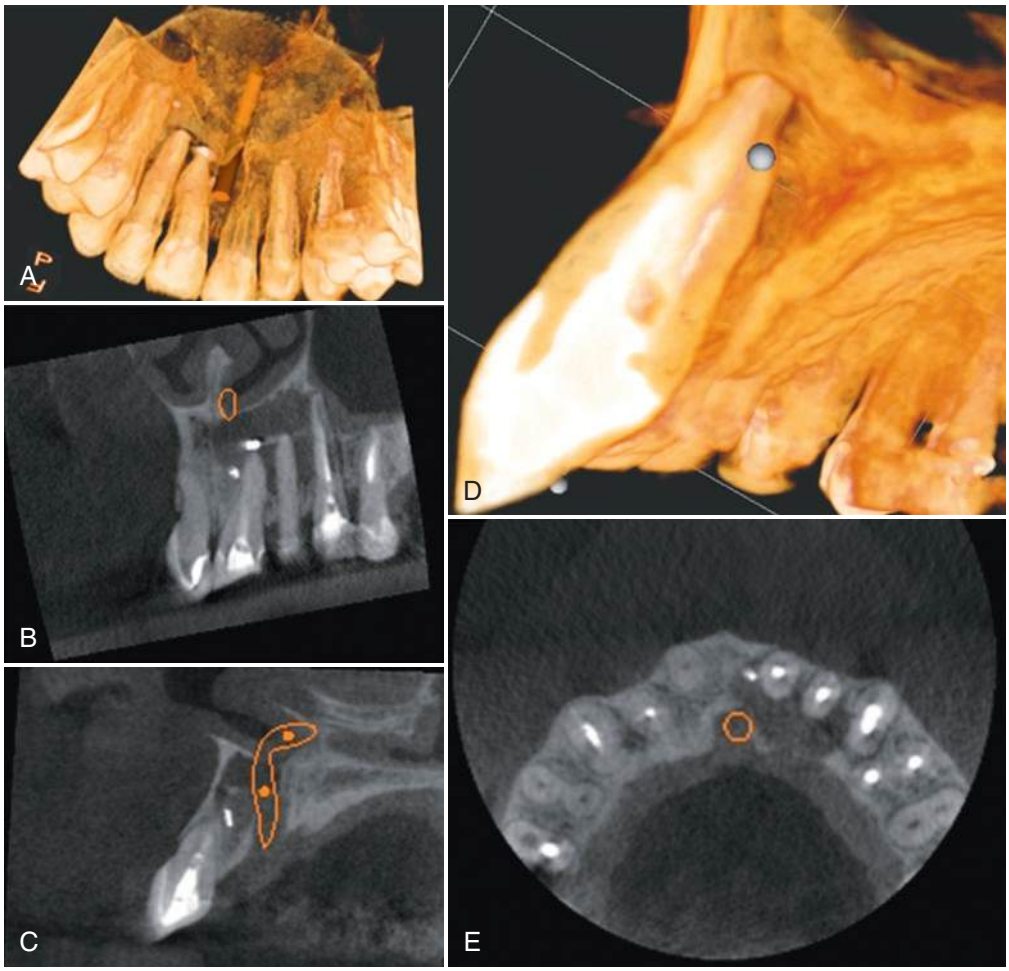



**Fig. 11.59** (Case 3, continued): **A,** Palatal view CBCT reconstruction showing the exit of the nasopalatine neurovascular bundle from the incisive canal. **B,** Sagittal view demonstrating the periapical radiolucency involving teeth #9, 10, and 11. **C,** Coronal view the exit of the nasopalatine bundle from incisive canal. **D,** A 3D reconstruction demonstrating the periapical lesion extension to tooth #11. **E,** Axial view demonstrating the extension of the lesion, palatal plate perforation, and relation of the nasopalatine bundle to the periapical lesion.

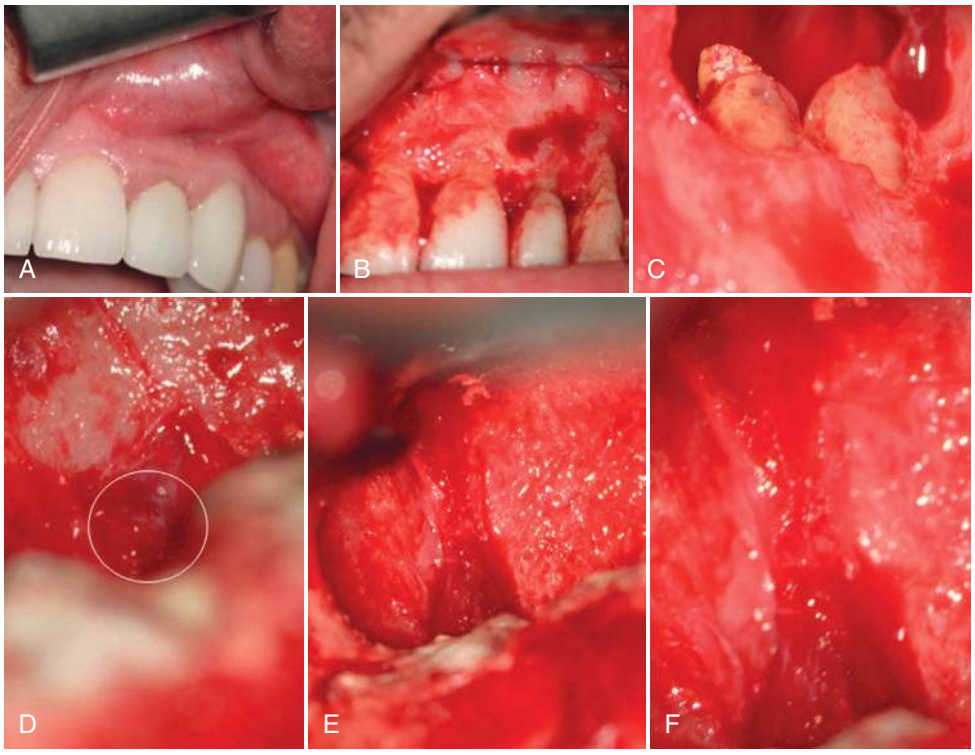



**Fig. 11.60** (Case 3, continued): **A** and **B,** Clinical pictures before and after flap reflection showing the intact buccal plate. **C,** Periapical defect after degranulation showing the apices of teeth #9 and 10 prior to resection. **D,** Palatal bone perforation with palatal mucosa evident *(circle)*. **E** and **F,** Clinical pictures of the nasopalatine bundle intact after degranulation.

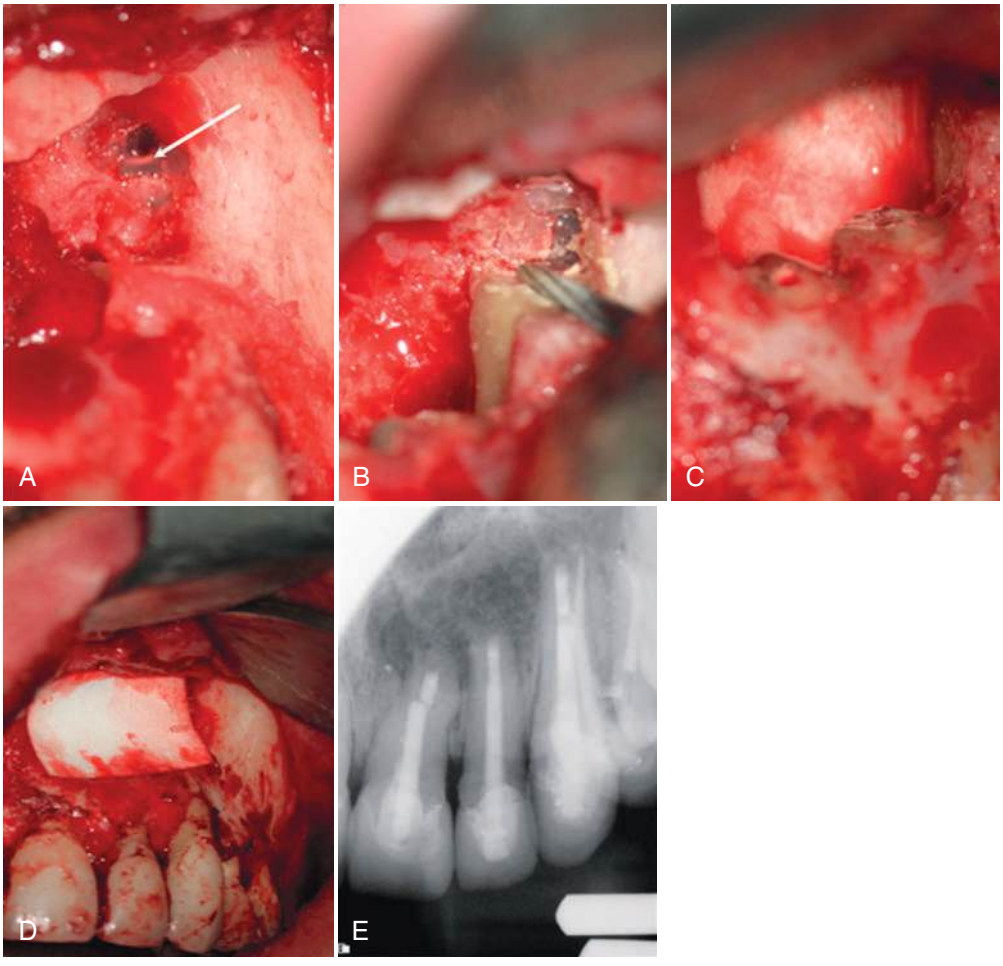



**Fig. 11.61** (Case 3, continued): **A** and **B,** Lateral wall of the maxillary sinus distal to tooth #11. **C,** CopiOs membrane placed palatally to cover the palatal mucosa. **D,** CopiOs membrane covering the Puros allografting material. **E,** Through-and-through defect immediate postoperative radiograph.

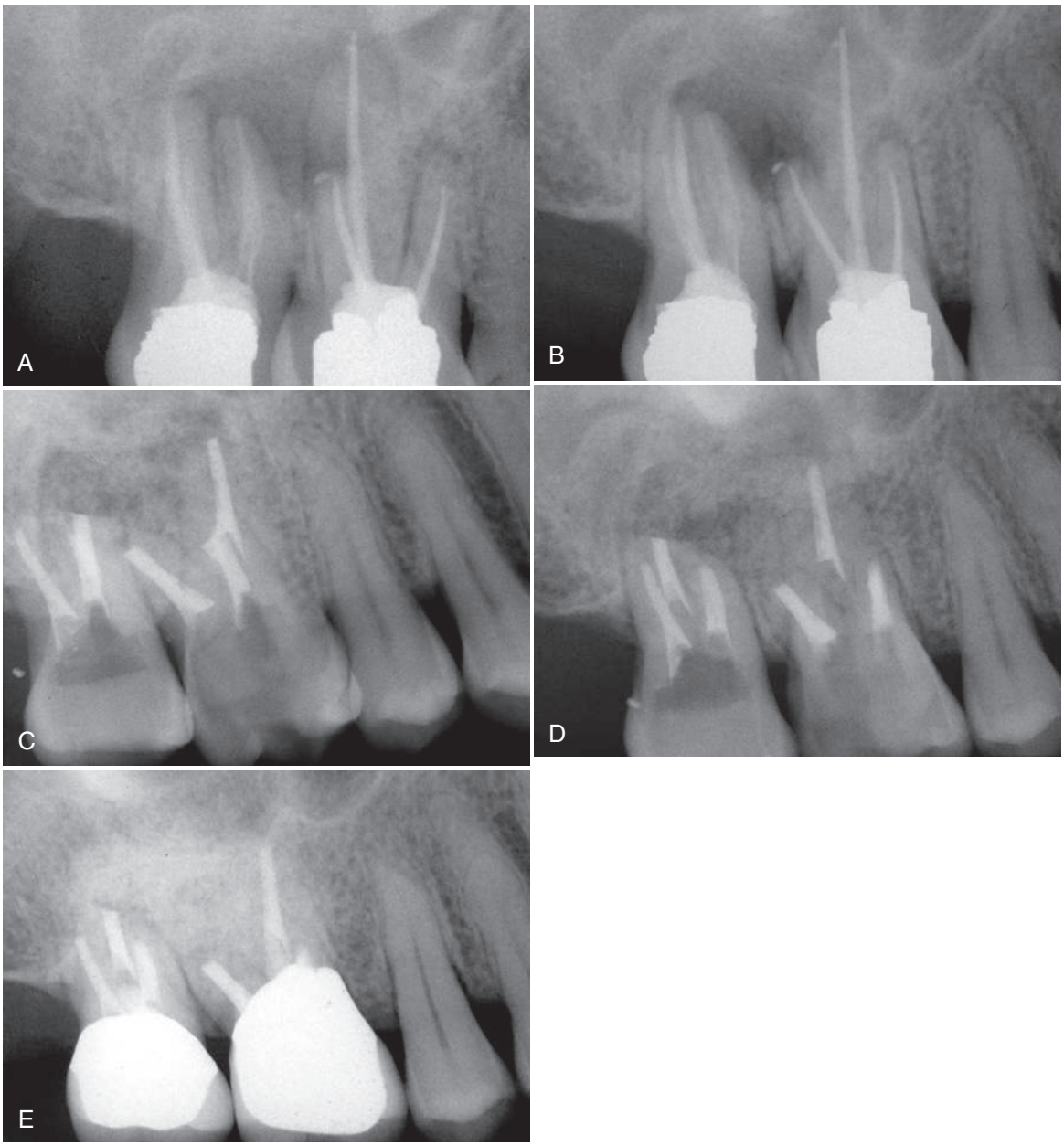



**Fig. 11.62 A,** Preoperative angled radiograph of maxillary right first and second molars. Both teeth had been treated previously, and the patient has reported a history of pain in the area for the past 5 years. The treatment plan included nonsurgical retreatment followed by periradicular surgery with bone grafting and guided tissue regeneration (GTR). **B,** Preoperative straight-on radiograph of the maxillary right first and second molars. **C,** Immediate postoperative radiograph showing root-end resections and fillings. The root ends were prepared with ultrasonics, conditioned with 17% ethylenediaminetetraacetic acid (EDTA), filled with Diaket, and smoothed with a superfine diamond finishing bur. The crypt was packed with BioOss xenograft material, and a Guidor resorbable membrane was placed. **D,** Immediate postoperative radiograph (straight on view). **E,** A 4-year follow-up radiograph. The patient was asymptomatic, and all objective findings were within normal limits. The teeth were restored with porcelain fused to metal crowns.

Membranes frequently require support so that the membrane does not collapse into the defect itself. Support for the membrane may be provided by using either a titanium-tented membrane or a graft material. Graft materials have two main functions: to act as a mechanical substructure that supports the membrane and the overlying soft tissues and to serve as a biologic component that enhances bone formation. Bone graft materials (Table 11.2) can be categorized as osteoconductive or osteoinductive. An osteoconductive material provides a framework into which bone can grow. The pore size of the material is similar to that of normal bone, and the material eventually is absorbed and remodeled. An osteoinductive material stimulates the production of new bone cells such that healing occurs more quickly. The BMP family has been investigated extensively for use in this role. A combination of osteoconductive and osteoinductive materials also can be used for bone grafts.

The use of GTR techniques raises several additional issues that should be discussed with the patient before surgery. These include the cost of the additional material, the origin of the material (synthetic, animal, or human), the need to manage the wound for a longer period, and

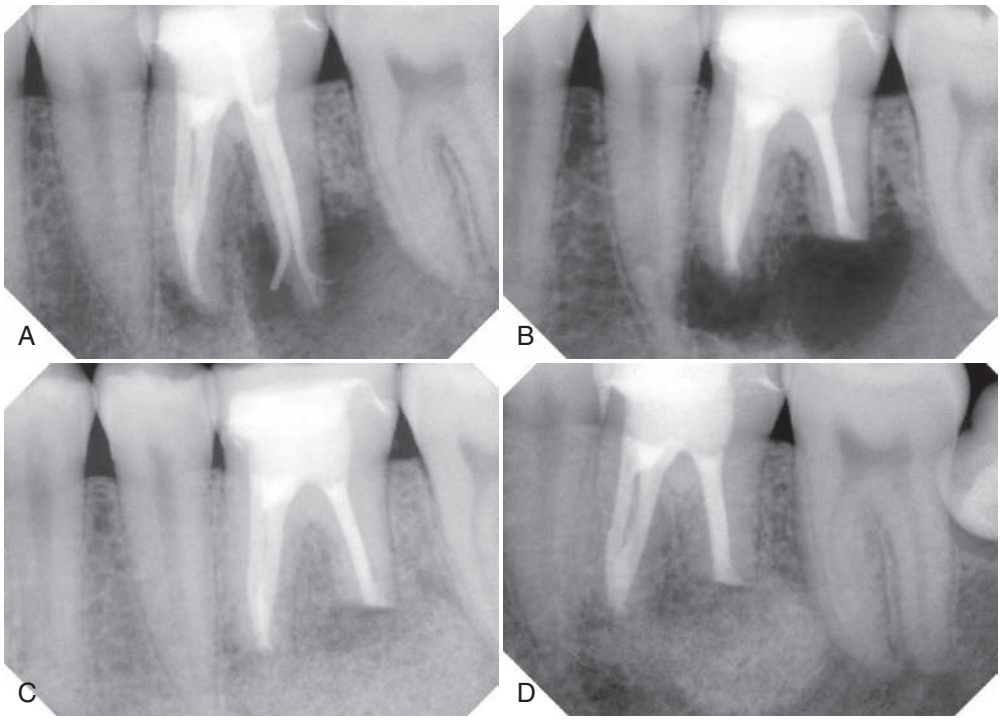



**Fig. 11.63** **A,** Preoperative radiograph of a mandibular left first molar. Gutta-percha was inserted into the buccal sulcus and traced to the apex of the distal root. Nonsurgical root canal treatment had been performed 12 months earlier. **B,** Root-end resection and MTA root-end fillings (M and D roots). **C,** Immediate postoperative radiograph. BioOss xenograft material was placed. **D,** The 19-month follow-up radiograph showed good periradicular healing.

**Table 11.1** Examples of Membrane Materials

| Composition | Trade Name/Manufacturer |
|---|---|
| **NONRESORBABLE** | |
| Polytetrafluoroethylene | Gortex (WL Gore & Associates Inc, Flagstaff, AZ)<br>TefGen FD (Lifecore Biomedical, Chaska, MN)<br>Bicon Barrier Membrane (Bicon, Boston, MA)<br>Cytoflex (Unicare Biomedical, Laguna Hills, CA) |
| **RESORBABLE** | |
| Laminar bone | Lambone (Pacific Coast Tissues Bank, Los Angeles, CA) |
| Polylactic acid | Guidor* This product was used extensively in early research with very favorable results (Guidor USA)<br>Atrisorb (CollaGenex Pharmaceuticals, Newtown, PA) |
| Polyglactic acid | Vicyl Mesh (Ethicon, Somerville, NJ) |
| Polylactic acid, polyglycolic acid, and trimethylene carbonate | Resolut (WL Gore & Associates Inc, Flagstaff, AZ) |
| Collagen | Biomend (Zimmer Dental, Carlsbad, CA)<br>Bio-Guide (Osteohealth, Shirley, NY)<br>Bicon Resorbable Collagen Membrane (Bicon, Boston, MA) |

*No longer available.

potential postoperative complications related specifically to these techniques and materials. Discussion of the composition of the materials to be used is very important, because some patients may have concerns based on religious or ethical grounds. The surgeon must discuss all the ramifications of using these materials with the patient before beginning the procedure, because it is not always possible to predict before surgery when grafting materials may be needed.

If GTR techniques are to be used during periradicular surgery, a resorbable membrane should be chosen and a protocol should be followed (see Figs. 11.62 and 11.63):

1. The membrane is extended to cover 2 to 3 mm of bone peripheral to the margins of the crypt; it should be supported with a bone substitute graft material so that it does not collapse into the crypt or onto underlying tooth structures.
2. Tissue closure techniques should ensure total tissue coverage of the membrane. The traditional postoperative

**Table 11.2** Examples of Bone Graft Materials

| Graft Type | Description | Product/Manufacturer or Source |
|---|---|---|
| Autogenous graft | Obtained from patient's own body | Ramus, chin, iliac crest |
| Allograft | Demineralized freeze-dried human bone (DFDBA) | Osteofil (Regeneration Technologies, Alachua, FL)<br>Grafton (Osteotech, Eatontown, NJ)<br>Dynagraft (GenSci, Toronto, Ontario, Canada)<br>Opteform (Exactech, Gainesville, FL)<br>Puros (Zimmer Dental, Carlsbad CA)<br>MTF DeMin Bone (DENTSPLY Friadent CeraMed, Lakewood, CO) |
| Xenograft | Inorganic bovine/porcine bone particles | BioOss (Osteohealth, Shirley, NY)<br>OsteoGraf (DENTSPLY Friadent CeraMed, Lakewood, CO) |
| Ceramic/synthetic grafts | Calcium sulfate, calcium phosphate/hydroxyapatite, bioactive glass | CapSet (Lifecore Biomedical, Chaska, MN)<br>OsteoSet (Wright Medical Technology, Arlington, TN)<br>HTR (Bioplant HTR, Kerr Corporation, West Collins, CA)<br>Biogran (3i, Palm Beach Gardens, FL)<br>Norian SRS (Synthes, West Chester, PA)<br>NovaBone-C/M (NovaBone Products, LLC, Sales and Manufacturing, Alachua, FL)<br>PerioGlas (NovaBone Products, LLC, Sales and Manufacturing, Alachua, FL) |
| Bioactive proteins | Bone morphogenic proteins (BMPs) | Experimental |
| Combination graft | Allograft, xenograft, or ceramic/synthetic grafts plus bioactive protein | PepGen P15 (DENTSPLY Friadent CeraMed, Lakewood, CO) |

compression is eliminated, because this would collapse the membrane onto the underlying structures.

3. Smoking is contraindicated with GTR techniques because it consistently has been shown to affect the outcome adversely.[73,324,445,525,538,539]

### RIDGE PRESERVATION

With the growing use of dental implants for the replacement of missing teeth, clinicians should be aware of ridge preservation strategies, even if they do not place implants.[308] As an example, ridge preservation should be considered when a tooth is determined to have a vertical root fracture during an exploratory surgical procedure and is extracted. In this situation, there is often a complete absence of the buccal bony plate, and simple extraction of the tooth would predispose the patient to a loss of ridge height and width, thereby complicating future implant placement. GTR with graft and barrier placement (as previously described) at the time of extraction may be indicated to create a more favorable site for future implant placement.[31,251,342,608] An atraumatic extraction technique is desirable, as one of the goals is to preserve the maximum amount of existing bone. Periotomes are particularly useful for this type of bone-preserving extraction technique.

## Intentional Replantation

Intentional replantation may be an option when surgical access is limited or presents unacceptable risks. Mandibular second molars are a common example for this technique because of the typically thick overlying buccal bone, shallow vestibular depth, and proximity of the root apices to the mandibular canal (Fig. 11.64). However, any tooth that can be atraumatically removed in one piece is a potential candidate for intentional replantation. Contraindications include teeth with flared or moderately curved roots and the presence of periodontal disease. Vertical root fracture has often been considered a contraindication,[401] although some investigators have demonstrated moderate success using a dentin-bonded resin and intentional replantation for the treatment of teeth with root fractures.[226,268,506] The prognosis was generally better for incisors and for teeth with fractures less than two thirds of the root length. Clinical success after 1 year was about 89% and decreased to 59% at 5 years.[226]

The tooth should be extracted with minimal trauma to the tooth and socket. Ideally, elevators are not used and the root surface is not engaged with the forceps. All instruments and materials for root-end preparation and filling should be arranged before extraction to minimize extraoral working time. The root surface must be kept moist by wrapping the root with gauze soaked in a physiologic solution, such as Hank's Balanced Salt Solution. After root-end preparation and filling (as described previously in this chapter), the tooth is replanted and the buccal bone is compressed. The patient may be instructed to bite on a cotton roll or other semisolid object to help position the tooth properly in the socket. Occlusal adjustment is indicated to minimize traumatic forces on the tooth during the initial stage of healing. A splint may be applied, but this is often not necessary. The patient should eat a soft diet and avoid sticky foods, candy, and chewing gum for at least 7 to 10 days. Based on clinical observations and several animal model studies, the prognosis for successful healing after replantation is most closely related to avoiding trauma to the PDL and cementum during extraction and minimizing extraoral time.[21,23,391]

## Postoperative Care

As previously noted, NSAIDs are generally the preferred class of drugs for managing postoperative pain.[9,41,54,143] Ibuprofen (400 to 800 mg) or an equivalent NSAID typically is given

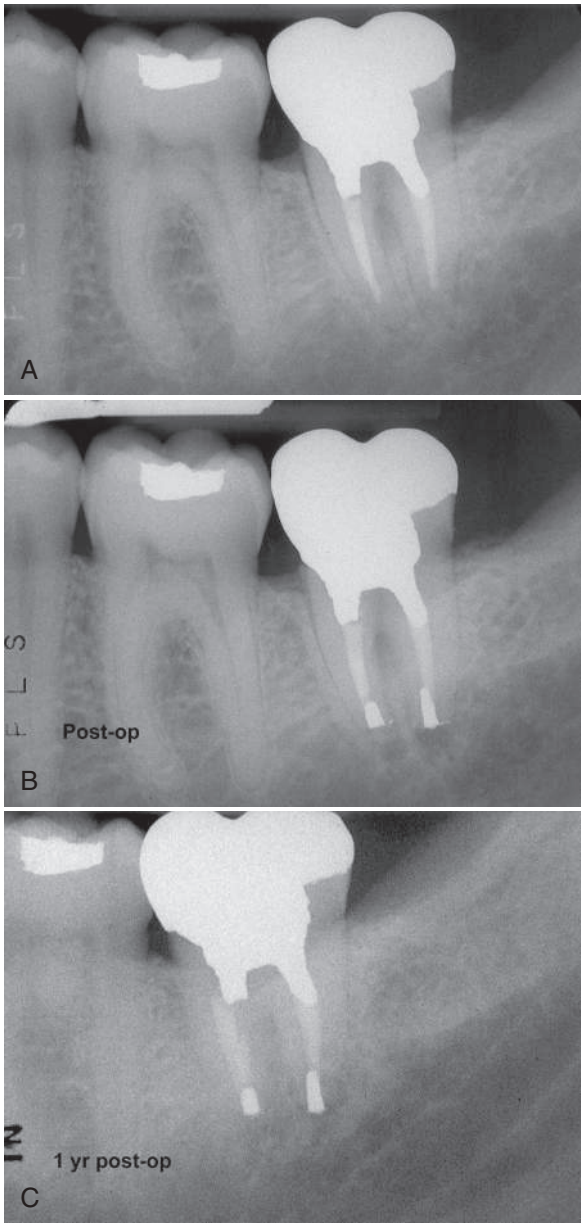



**Fig. 11.64** Intentional replantation. **A,** Preoperative radiograph of a mandibular left second molar. The tooth was persistently sensitive to percussion and biting after nonsurgical retreatment. **B,** Radiograph of the tooth immediately after extraction, root-end preparation and filling, and replantation. **C,** At the 1-year follow-up visit, the tooth was asymptomatic and showed good periradicular healing. (Courtesy Dr. Matt Davis.)

before or immediately after surgery and can be continued for several days postoperatively as needed. When additional pain relief is required, a narcotic such as codeine, hydrocodone, or tramadol may be added to the standard NSAID regimen. This strategy may result in a synergistic effect, and therefore greater pain relief, than would be expected with the separate analgesic value of each drug.[148] A useful short-term approach to the management of moderate to severe pain is a "by the clock" alternating schedule of an NSAID and an acetaminophen/narcotic combination.[250,348] Pain after periradicular surgery is typically only mild to moderate. Postoperative pain usually is managed quite well with NSAIDs only, especially when the previously recommended strategy of preoperative NSAID therapy and a long-acting local anesthetic is combined with a minimally traumatic surgical approach.

Sutures are commonly removed 2 to 4 days after surgery.[92,213] This recommendation is based on the current understanding of wound healing and the desire to remove any potential irritants from the incision area as soon as possible. Local anesthesia is rarely required, although application of a topical anesthetic may be helpful, especially to releasing incisions in nonkeratinized mucosa. Sharp suture scissors or a #12 scalpel blade can be used to cut the sutures before they are removed with cotton pliers or tissue forceps. A transient bacteremia can be expected after suture removal, even when a preprocedural chlorhexidine mouth rinse is used.[79] Antibiotic coverage should be considered only for patients at high risk of developing bacterial endocarditis.

If healing is progressing normally at the suture removal appointment, the patient does not need to be seen again in the office until the first scheduled recall examination, typically 3 to 12 months after surgery. However, phone contact with the patient approximately 7 to 10 days after suture removal is recommended to confirm the absence of problems. Patients with questionable healing at the suture removal appointment should be reevaluated in the office in 7 to 10 days or sooner if necessary.

## Management of Surgical Complications

Although serious postoperative surgical complications are rare, the clinician should be prepared to respond to patient concerns and recognize when additional treatment may be necessary. Careful case evaluation, adherence to a minimally traumatic surgical technique, and proper patient management, as described previously in this chapter, should result in a low incidence of postoperative complications. Even so, some patients experience mild to moderate postoperative pain, swelling, ecchymosis, or infection. In a prospective study of 82 patients undergoing endodontic surgical treatment, Tsesis and coworkers[545] reported that 76.4% were pain free 1 day after surgery and 64.7% did not report any swelling. Only 4% of the patients in this study experienced moderate pain, and this sequela was closely related to the presence of presurgical symptoms. Postoperative pain typically peaks the day of surgery and swelling reaches its maximum 1 to 2 days after surgery.[294] As previously noted, good evidence supports the use of prophylactic NSAID therapy and a long-acting local anesthetic to reduce the magnitude and duration of postoperative pain.

Patients should be advised that some postoperative oozing of blood is normal, but significant bleeding is uncommon and may require attention. Most bleeding can be controlled by applying steady pressure for 20 to 30 minutes, typically with a piece of moist cotton gauze or a tea bag. Bleeding that persists requires attention by the clinician. Pressure to the area and injection of a local anesthetic containing 1:50,000 epinephrine are reasonable first steps. If bleeding continues, it may be necessary to remove the sutures and search for a small severed blood vessel. When

located, the blood vessel can be crushed or cauterized to control bleeding. Cauterization may be performed with a heat source commonly used for warm obturation techniques. Local hemostatic agents, as previously described, may also be used. Occasionally, a patient may require hospitalization and surgical intervention to control bleeding, but this is an extremely rare event. Extraoral ecchymosis (Fig. 11.65) occurs when blood seeps through the interstitial tissues; although it may be alarming to the patient and clinician, this condition is self-limiting and does not affect the prognosis.[281] Moist heat applied to the area may be helpful, although complete resolution of the discoloration may take up to 2 weeks. Heat should not be applied to the face during the first 24 hours after surgery.

Sinus exposure during surgical root canal procedures on maxillary posterior teeth is not uncommon. Postoperative antibiotics and decongestants are often recommended[16,29,281,568]; however, this practice is controversial, and no evidence supports the routine use of antibiotics and decongestants in these cases. Walton[576] makes a persuasive argument that antibiotics are not routinely indicated for the management of sinus exposures during periradicular surgery when primary closure of the oral-antral communication is possible. Further support for this position is provided by other clinicians who have observed excellent healing and minimal complications after sinus exposure during periradicular surgery.[291,456,579] Clinical judgment should guide the use of antibiotics and decongestants on a case-by-case basis until more conclusive evidence on this practice is available.



**Fig. 11.65** Postoperative ecchymosis can be alarming to the patient but resolves spontaneously within 7 to 14 days.

No reliable data are available to provide an accurate estimate of the likelihood of paresthesia after surgical root canal treatment. The incidence of paresthesia after third molar removal is estimated to be 1% to 4.4%[443]; however, most reported cases of paresthesia after third molar extraction involved the lingual nerve, which is rarely encountered in mandibular periradicular surgery. The incidence of damage to the inferior alveolar nerve after third molar surgery is approximately 1.3%, with only about 25% of these cases resulting in permanent injury.[551] Unless the nerve is resected during surgery, most patients can be expected to return to normal sensation within 3 to 4 months. If the paresthesia does not show signs of resolving within 10 to 12 weeks, referral and evaluation for possible neuromicrosurgical repair should be considered.[167,431] Robinson and Williams[443] presented a useful method for charting and documenting paresthesias. The area of altered sensation is determined by pinching the skin or mucosa with cotton pliers; alternatively, a pinprick can be applied with a sharp instrument. The area of paresthesia is noted with a series of marks on a diagram of the face and mouth. This method provides a graphic and chronologic record of the paresthesia.

## Summary

Periradicular surgery today bears little resemblance to the surgical procedures commonly performed as recently as the 1990s. Enhanced magnification and illumination, microsurgical instruments, ultrasonics, new materials for hemostasis, root-end filling, and GTR, and a greater understanding of the biology of wound healing and the etiology of persistent periradicular disease all have contributed to the rapid evolution of periradicular surgery. With proper case selection and operator skill, periradicular surgery can be considered a predictable, cost-effective alternative to extraction and tooth replacement.

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